



GE VERNOVA

ADMS

16.25.1

Outage Assist Installation Guide

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Change History

Date	Change Description
Jun 8, 2023	Outage Assist 16.1.0: • Updated AppMode.json in Outage Assist Customization Files. • Editorial and formatting updates.
Jan 23, 2024	Outage Assist 16.4.0: • Updated section Example sa-manifest.yaml File. • Added chapter Outage Assist 2.X Recommended Deployment
Apr 25, 2024	Outage Assist 16.7.0: • Updated section Deploying the Outage Assist Services Layer • Added section Outage Assist AOR Configuration Updates
May 22, 2024	Outage Assist 16.8.0: • Updated section Updating Outage Assist Customizations • Added chapter Localizing Outage Assist

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1. System Overview

Outage Assist is GE's solution for electric utilities to complement GE's control room distribution platform, Advanced Distribution Management System (ADMS). The Outage Assist solution meets the real-time network management needs of electric distribution utilities, providing a high-quality visualization of distribution and outage management operations.

This chapter provides a high-level overview of the Outage Assist software and hardware. It also introduces key architectural concepts of the Outage Assist design.

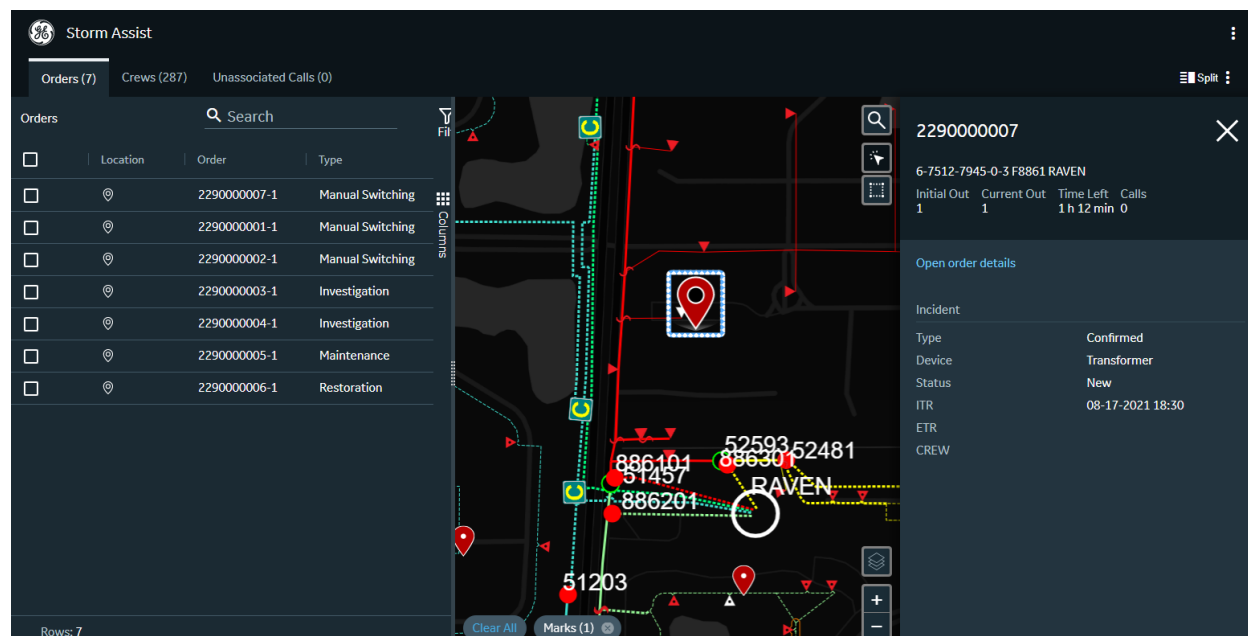


Figure 1. Outage Assist

1.1. Key Architecture and Design Concepts

Outage Assist uses services that are provided by GE's common Kubernetes-based platform (hereinafter "Foundation"), which is used for specific GE products and solutions. In particular, Outage Assist uses the Foundation Base, Foundation OSB, and Foundation UI Server layers of Foundation. The Foundation and Outage Assist deployment, scaling, and management is implemented using Kubernetes.

Outage Assist uses distribution data supplied by GE's Advanced Distribution Management System (ADMS).

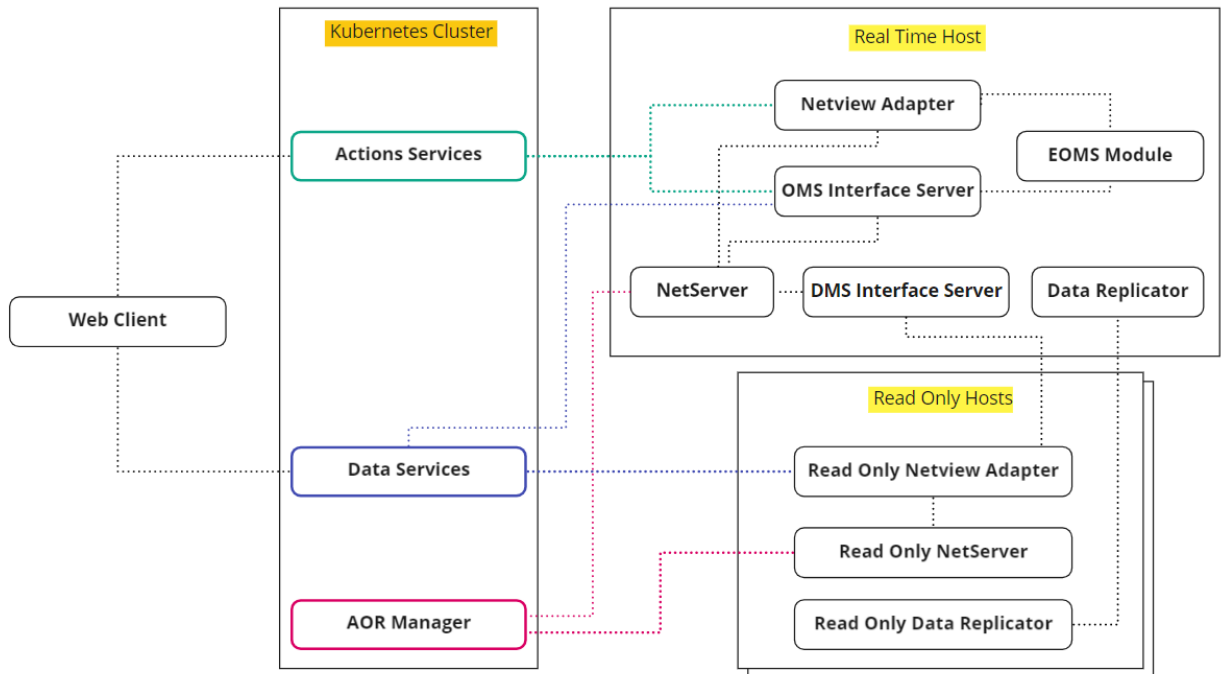


Figure 2. Outage Assist Outage Management Data Flow

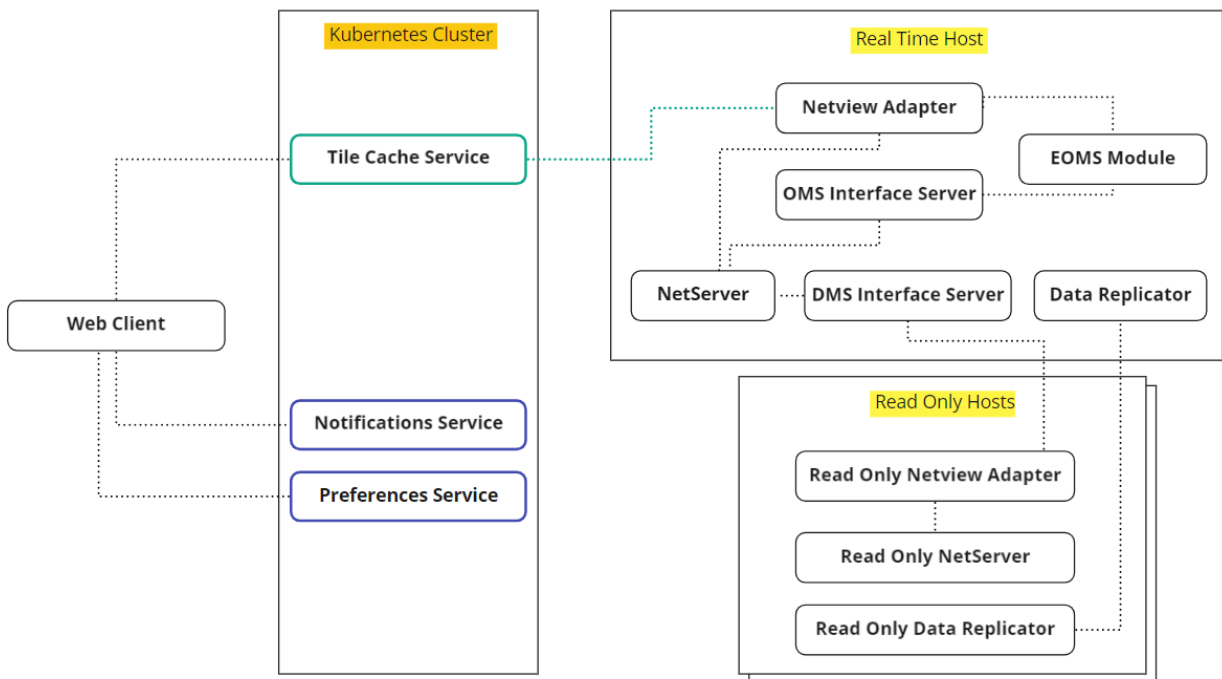


Figure 3. Outage Assist Network Tiles Data Flow

1.1.1. Foundation

The Foundation includes services that support other applications. The key Foundation services used

by Outage Assist are:

- **Foundation Base Services:** This Foundation layer includes common Foundation services:
 - **Security:** Provides security features, including user authentication and authorization. Includes the Zitadel identity management service and AOR Manager..
 - **Elasticsearch:** Stores application logs in a persistent Elasticsearch database.Foundation Base Services layer also includes alerting, monitoring, and routing features.
- **Foundation OSB Services:** This Foundation layer includes integration services:
 - **activemq-artemis:** Supports the Notification service.
- **Foundation UI Server Services:** This Foundation layer includes services that are required for Outage Assist:
 - **preferences-service:** Service for saving user preferences and configurations in the PostgreSQL database.
 - **notification-service:** Service for sending notification events to clients via a web socket.
 - **tile-service:** Receives tiles and configurations from the ADMS NetViewer Adapter and provides the data to the Outage Assist user interface (UI).
 - **graphql-server:** Connects to NetViewer Adapter and communicates DMS and OMS data in GraphQL format to the Outage Assist user interface (UI).

For more information about Foundation, refer to the following:

- **Foundation Software Installation:** Information about installing, configuring, and maintaining the Foundation.

1.1.2. Outage Assist

Outage Assist includes the following key applications, services, and custom resources:

- **storm-assist:** Supports user interface functions.
- **adms-external-gateway:** Supports failover capabilities and secure communication to ADMS servers.
- **nva-readonly-eg-route:** Represents the External Gateway routing configuration for connection to the GraphQL datapoint of the NetViewer Adapter, to provide a data interface.
- **ois-eg-route:** Represents the External Gateway routing configuration for connection to the OMS Interface Server, to provide a data and actions interface.
- **nva-main-eg-route:** Represents the External Gateway routing configuration for connection to the OMS datapoint of the NetViewer Adapter, to provide a data and actions interface.

The Outage Assist UI supports a thin client, where the client is simply a web browser that connects to the Outage Assist web server. The Outage Assist UI is implemented using Angular.

1.1.3. ADMS

The following ADMS servers support ADMS communication with Outage Assist:

- **Main Server Host:** Contains applications that support the production system functions. Located in the most secure zone.
 - **NetServer (OMS Server):** Supports the Outage Management System functionality. Receives the Area of Responsibility (AOR) and Functional Permissions data from the AOR Manager application.
 - **Net Data Replicator:** Replicates OMS data to the read-only hosts.
 - **NetViewer Adapter (Map and Query Modes):** Provides map data to Outage Assist UI clients. Supports the GraphQL interface for modifying OMS data from Outage Assist. Only one instance of NetViewer Adapter in the Map mode can connect to NetServer. Therefore, the Failover Broker must be used if multiple instances of NetServer are running across a failover system.
 - **Failover Broker:** Supports redundancy for the NetViewer Adapter.
 - **Net Data Server:** Supports reviewing and correcting cleared and canceled incidents in the Outage Management System. Net Data Server also supports read-only access to the current OMS data and recent history for users inside and outside of the control room environment via web services. Generates daily snapshots for data replication to read-only hosts.
 - **OMS Adapter:** Records ADMS customization data in the Preference service.
 - **DMS Interface Server:** A standard web server interface to allow third-party applications to communicate with the ADMS DMS server (NetSrv).
 - **OMS Interface Server (with Zitadel authentication):** A standard web server interface to allow third party applications to communicate with the ADMS OMS server (NetServer). Supports communication between NetServer and Outage Assist.
 - **OMS Interface Server (without Zitadel authentication):** A standard web server interface to allow third party applications to communicate with the ADMS OMS server (NetServer). Supports communication between NetServer and EOMS services.
 - **Enterprise OMS Applications:** Supports the Enterprise OMS system functionality. For more information, refer to the *Enterprise OMS Software Installation and Maintenance Guide*.
- **Enterprise OMS Server Host:** The host for Enterprise OMS services (on Windows or Kubernetes).
 - **Enterprise OMS Services:** Supports the Enterprise OMS system functionality. For more information, refer to the *Enterprise OMS Software Installation and Maintenance Guide*.
- **Read-Only Server Host:** The read-only system configuration replicates a copy of the production DMS/OMS system data so that users who require read-only access to the data can view it without interfacing directly with the production system. Located in the demilitarized zone (DMZ) outside the production system's network firewall. Multiple read-only systems can be used for scaling and performance purposes.
 - **NetServerRO (OMS Server Read-Only):** Supports the read-only Outage Management

System functionality. Receives DMS and OMS data from the main server host. Receives EOMS data from the Enterprise OMS server host. Receives the Area of Responsibility (AOR) and Functional Permissions data from the AOR Manager application.

- **Net Data Replicator:** Receives OMS data from the main server host.
- **NetViewer Adapter (Query mode):** Supports the GraphQL and other interfaces. Receives DMS and OMS data from NetServer Read-Only. Provides DMS and OMS data to the GraphQL service for Outage Assist UI. One NetServer Read-Only instance can serve data to up to 5 instances of NetViewer Adapter. Multiple NetViewer Adapter instances in the Query mode can connect to NetServer Read-Only. Therefore, this configuration does not require Failover Broker.

1.2. Connection Security

A secure connection among the Outage Assist and ADMS systems is established using an HTTPS connection with TLS termination.

For HTTPS with TLS, you need to obtain SSL certificates from your certificate authority. The certification authority is based on Active Directory or other services. Therefore, you need to have two certificates: one for the Kubernetes Ingress Controller and another for ADMS interface servers.

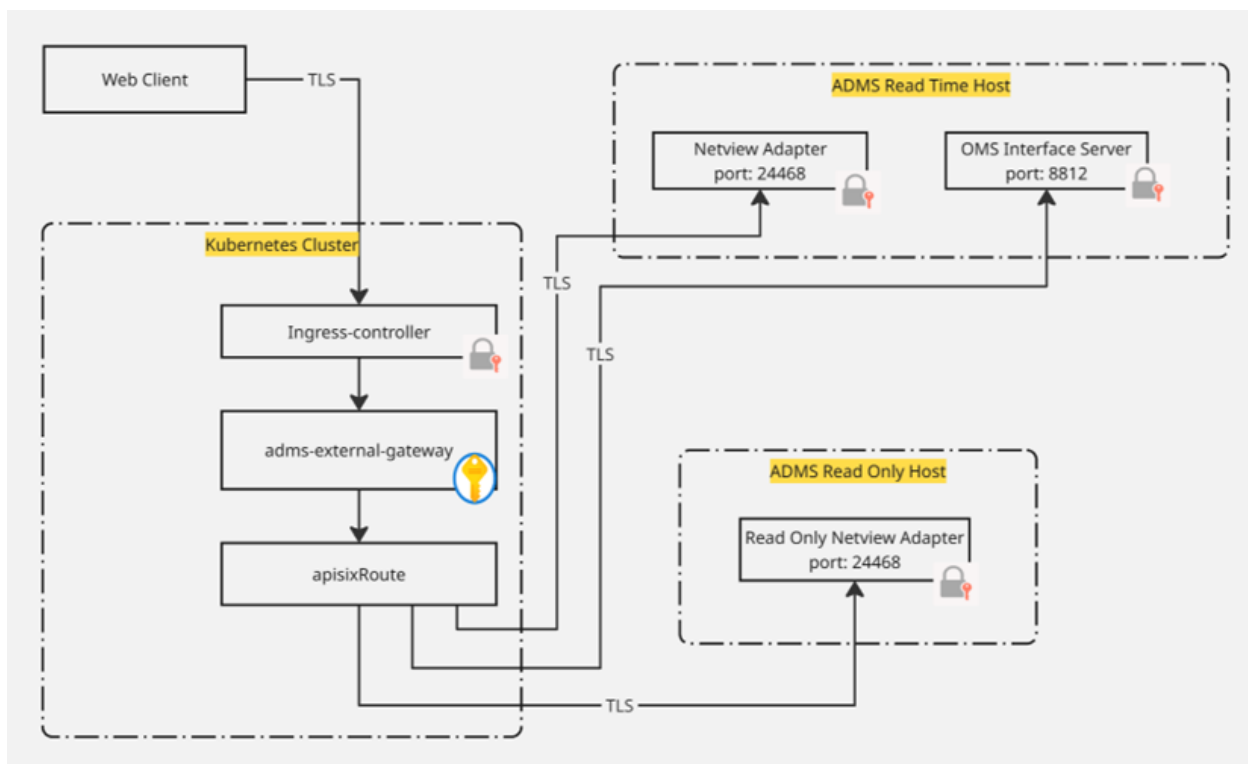


Figure 4. HTTPS Connection with TLS Termination between Outage Assist and ADMS

2. Installation Overview

For a list of certified and supported operating systems, as well as the required third-party applications, refer to the "Operating System" chapter in the *Outage Assist Release Notes*.

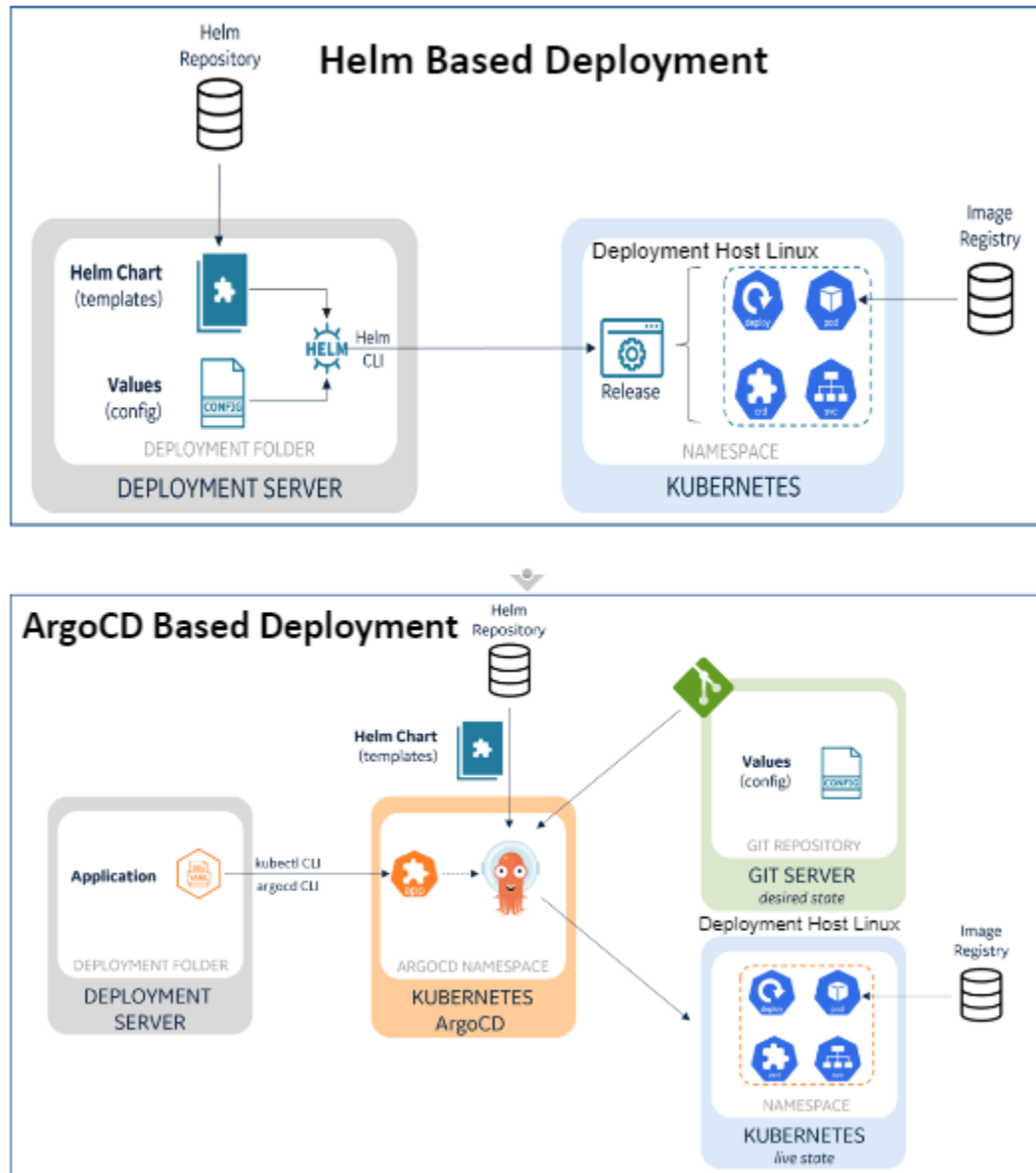


Figure 5. Deployment Overview

Outage Assist components (containers) are orchestrated by Kubernetes. Therefore, you must first create a Kubernetes environment before deploying Outage Assist. The application deployment is certified for the Kubernetes RKE2 cluster that is deployed using GE's Kubernetes PDI. For details about installing the Kubernetes RKE2 cluster and deploying the Foundation solution, consult your GE project team.

Outage Assist uses GE's Deployment Model 2.0 for deployment. This model is based on Helm-base or Argocd with Helm to the Kubernetes cluster.

Outage Assist uses Helm for templating deployments. Helm helps you define, install, and upgrade Kubernetes applications and manage Helm charts.

A simplified deployment includes the following steps:

1. Creating a Kubernetes environment using GE's Kubernetes PDI.
2. Deploying the Foundation Base, Foundation OSB, and Foundation UI Server Helm charts.
3. Performing pre-deployment steps for Outage Assist.
4. Deploying the Outage Assist Helm Chart.
5. Performing post-deployment steps for Outage Assist.
6. Configuring ADMS.
7. Validating the Outage Assist setup.

Detailed deployment information is provided in this document.

Note

Prerequisites for ADMS are outside the scope of this document. For more information, refer to the *ADMS Release Notes*.

2.1. Hardware and Operating System Prerequisites

For a list of supported hardware and operating systems, refer to the "Hardware Requirements" and "Operating System" sections in the *Outage Assist Release Notes*.

2.2. Software Prerequisites

2.2.1. Installation Artifacts

Before installation, obtain the following artifacts from your GE project team:

- Cache containers with the required Docker containers (sa-cache-16.18.0.tar.gz)
- Outage Assist Helm chart (outage-assist-16.18.0.tgz)
- Sample input manifest files for deployment (k8s-base-manifest.yaml, foundation-base-manifest.yaml, foundation-osb-manifest.yaml, foundation-ui-manifest.yaml, and sa-manifest.yaml)

2.2.2. Server

Outage Assist server has the following software requirements:

- RKE2 Kubernetes (installed using GE's Kubernetes PDI)
- Foundation Base, Foundation OSB, and Foundation UI Services

For more information about setting up the RKE2 Kubernetes cluster and deploying Foundation services, refer to your GE project team.

2.2.3. Deployment Server

Outage Assist deployment server has the following software requirements:

- Docker version 18.09.3 or later
- Connection to a Docker Trusted Registry or a similar supported API/v2 registry (for example, Nexus)

2.2.4. Clients

Outage Assist clients have the following software requirements:

- Google Chrome x64 version 100 or later with an enabled JavaScript engine.

3. Deploying the Outage Assist Services Layer

Important

It is assumed that you already have the RKE2 Kubernetes cluster running and Foundation services deployed. For more information, consult your GE project team.

3.1. Pre-Deployment

This section describes the steps to perform before deploying the Outage Assist solution to the Kubernetes cluster.

Note

Refer to the Foundation documentation for more details on Helm-based and ArgoCD-Based deployment and maintenance of YAML files in Nexus, Git or Local repo.

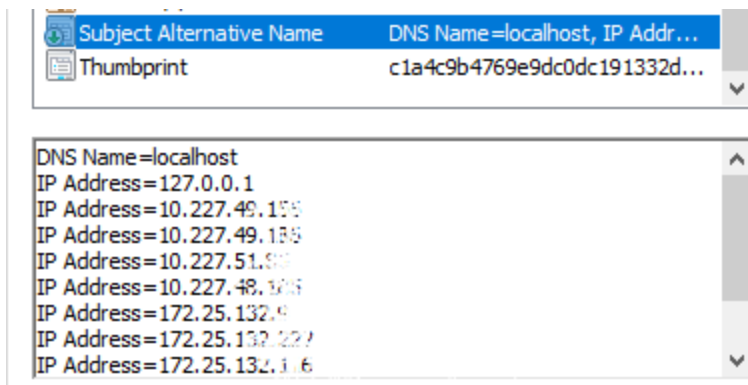
3.1.1. Generating Certificates for SSL/TLS Termination with ADMS

For production, it is recommended to use certificates issued by a certificate authority. For testing and demo purposes, you can use self-signed certificates.

For Outage Assist, you need the following certification files:

- Root certificate authority certificate. This certificate is used to issue and sign certificates. (For example, file name rootCA.crt; CN Storm Assist Root CA)
- Ingress controller certificate and key. These files are used for establishing connections with Active Directory. (For example, file name ingress-local-tls.crt and ingress-local-tls.key; ingress.local)
- ADMS server CRT certificate file signed by the Root certificate authority certificate. This file is used for secure communication from the Kubernetes cluster to ADMS interface servers. (For example, file name adms-server.crt; CN ADMS; O GE).

The certificate Subject Alternative Name must include all DNS host names and IPs as referenced in Outage Assist helm chart to enable the certificate for all ADMS IPs and host names. For example:



- ADMS server PFX file with a certificate (CRT file) and the associated private key. This file is used for certificate hosting by ADMS interface servers. (For example, file name adms-server.pfx; CN ADMS; O GE). The installed certificate has the same properties as the CRT file used in the Kubernetes cluster.

The example below shows how to generate OpenSSL certificates using the openssl tool. This is a standard tool for most Linux distributions. For more information, see <https://www.openssl.org/source/>.

To generate self-signed certificates:

1. Generate a root certificate authority certificate and key:

```
openssl genrsa -out rootCA.key 2048
openssl req -x509 -new -key rootCA.key -days 10000 -out rootCA.crt
```

2. Generate a certificate and key for Ingress Controller:

Note

It is recommended to set CN to the actual ingress controller DNS host name (web address you access when connecting to a secure site).

```
openssl genrsa -out ingress-local-tls.key 2048
openssl req -new -key ingress-local-tls.key -out ingress-local-tls.csr
openssl x509 -req -in ingress-local-tls.csr -CA rootCA.crt -CAkey rootCA.key -CAcreateserial -out ingress-local-tls.crt -days 5000
```

3. Generate a certificate and key for ADMS servers using the openssl tool:

- Create the csr.conf file in the cert subfolder of the deployment folder (see [Preparing a Deployment Folder](#)) with the IPs or DNS names of the ADMS machines. All DNS names can be listed in the file as DNS.1, DNS.2, DMS.N:

```
[v3_ext]
subjectAltName = @alt_names
[alt_names]
```

```
DNS.1 = localhost
DNS.2 = admshost1
DNS.3 = admshost2
DNS.4 = admshost3
DNS.5 = admshost4
IP.1 = 10.227.49.11
IP.2 = 10.227.49.12
IP.3 = 10.227.49.13
IP.4 = 10.227.49.14
```

- Execute the following commands using the Apache openssl tool supplied with the ADMS UX Prerequisites kit, where NetViewer Adapter uses O=GE value in <CertificateSubject>GE</CertificateSubject>.

```
"C:\Eterra\Distribution\PrerequisitesServices\Apache\bin\openssl.exe" genrsa -out adms-server.key 2048
```

This generates the ADMS Server key file, adms-server.key.

```
"C:\Eterra\Distribution\PrerequisitesServices\Apache\bin\openssl.exe" req -new -key adms-server.key -config "C:\Eterra\Distribution\PrerequisitesServices\Apache\conf\openssl.cnf" -out adms-server.csr -subj "/C=US/ST=WA/L=Bothell/O=GE/OU=Digital/CN=ADMS Server"
```

This generates the ADMS Server certificate signing request file, adms-server.csr, using adms-server.key.

```
"C:\Eterra\Distribution\PrerequisitesServices\Apache\bin\openssl.exe" x509 -req -in adms-server.csr -CA rootCA.crt -CAkey rootCA.key -CAcreateserial -out adms-server.crt -extensions v3_ext -extfile csr.conf -days 3650
```

This generates the ADMS Server certificate file, adms-server.crt, using csr.conf, adms-server.csr, rootCA.crt, and rootCA.key.

```
"C:\Eterra\Distribution\PrerequisitesServices\Apache\bin\openssl.exe" pkcs12 -export -out adms-server.pfx -inkey adms-server.key -in adms-server.crt
```

This generates the ADMS Server PFX file, which includes a certificate and key to use on Windows certificate installation / import. (During the certificate installation, set an empty password value and press Enter to disable the password, or use a non-empty password to enable the password).

i Note

It is recommended that you set CN to the actual ADMS DNS host name (web address you access when connecting to a secure site).

3.1.2. Setting Up Firewall Rules

Open the firewall ports as shown in the table below to enable communication among the Kubernetes cluster, ADMS servers, and LDAP/AD server.

Source	Destination	Port	Protocol
Kubernetes Cluster (sws-openresty)	OMS Interface Server (ADMS) - with Zitadel/Apisix	8812	HTTPS
Kubernetes Cluster (sws-openresty)	NetViewer Adapter (ADMS)	24468	HTTPS
Kubernetes Cluster (sws-openresty)	NetViewer Adapter (ADMS) on Read-Only Host	24468	HTTPS
Kubernetes Cluster	LDAP/AD Server	Project-specific	LDAPS
ADMS Interface Server (DMS Interface Server, OMS Interface Server, and NetViewer Adapter)	Kubernetes Cluster	443	HTTPS/WSS
ADMS Main Server (NetServer)	Kubernetes Cluster	443	HTTPS
ADMS Read-Only Server (NetServerRO)	Kubernetes Cluster	443	HTTPS
OMS Interface Server	DMS Interface Server	8801*	HTTPS
NetViewer Adapter on Read-Only Host	DMS Interface Server	8801	HTTPS
ADMS Main Server (NetServer)	EOMS Event Handler	8083	HTTP(S)
ADMS Read-Only Server (NetServerRO)			
NetViewer Adapter			
ADMS Main Server (NetServer)	EOMS Service	8080	HTTP(S)
ADMS Read-Only Server (NetServerRO)			
NetViewer Adapter			
ADMS Read-Only Server (NetServerRO)	ADMS Main Server (NetSrv, NetServer)	5025, 5026, 5050	TCP
ADMS NetData Replicator	ADMS NetData Replicator Read-Only	8200	TCP
ADMS NetData Replicator Read-Only	ADMS NetData Replicator	8201	TCP
ADMS Read-Only Server (NetServerRO)	Habitat Permit Service	5345	TCP
EOMS Event Handler	OMS Interface Server (ADMS) - without Zitadel/Apisix	8802	HTTP(S)

Source	Destination	Port	Protocol
--------	-------------	------	----------

Note

Open this port if DMS Interface Server and OMS Interface Server are running on different hosts.

Note

Port numbers and protocols can be configured in configuration files for the associated applications.

3.1.3. Preparing Helm deployment for Outage Assist Deployment

Note

Outage Assist Namespaces needs to create during deployment of Foundation and Create required certificate secrets in target kubernetes cluster.

This section describes the Helm files that are needed to deploy the RKE2 Kubernetes cluster, Foundation services, and Outage Assist components.

3.1.3.1. Preparing Outage Assist helm chart

Outage Assist deployment is defined using the Outage Assist helm chart. Helm charts are package and deliver to customer as tar file along with App and App Config yaml file

To prepare an YAML files to deploy Outage Assist:

1. Prepare the App file and App config to do the deployment of Outage Assist and Foundation
2. Configure IP addresses according to your current setup.
3. Specify the address of the Docker Registry and GitHub [GE's Docker Trusted Registry (DTR) or a Nexus registry on a deployment server].
4. To integrate with a multiple standby host (MSH) ADMS configuration (4 hosts: 1 enabled and 3 standby), define all ADMS interface hosts for the OMS Interface Server (oisExternalGatewayRoute), NetViewer Adapter Main (nvaMainExternalGatewayRoute), and NetViewer Adapter Read-Only (nvaReadOnlyExternalGatewayRoute) elements in the etd-services configuration.
5. Provide the Public key details in App config yaml to load the admsCertSecret element in cluster.
6. Set the application mode using the omsMode flag: eoms for Enterprise OMS mode or soms for Standard OMS mode.
7. Make other changes, as appropriate.

3.1.3.1.1. Outage Assist Deploy YAML Parameters

This section describes parameters in YAML file of the helm chart values file, optional overlay customization values file, and a cluster-customized global values file.

The below table lists the param_groups parameters that you can override with an input manifest file.

Table 1. Configurable storm-assist-deploy param_groups

param_group	Interpreter	Description
Outage-Assist-values	yaml	Configurations: • repoURL for the github repo • targetRevision , branch of github repo • Cluster external Url • OMS Interface server connection configuration • NetViewer Adapter Main connection configuration • NetViewer Adapter Read-Only connection configuration • admsCertSecret certificate in YAML file

3.1.3.1.2. Example Outage Assist helm chart File

The following is an example of the Outage Assist helm chart file for deploying Outage Assist.

Note

Backend services are still using naming convention as UAA but they all have been upgraded to use Zitadel token. Below example of manifest have 'uaaClient' which is just namet to align the correct roles.

App Yaml file for the outage Assist

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: outage-assist
  namespace: argocd
  annotations:
    argocd.argoproj.io/sync-wave: "1"
  finalizers:
    - resources-finalizer.argocd.argoproj.io
spec:
  destination:
    namespace: oms
    server: https://kubernetes.default.svc
  project: default
  source:
    helm:
```

```

# parameters:
# - name: global.registry
#   value: dig-grid-artifactory.apps.ge.com
valueFiles:
- ../../apps-config/outage-assist-values.yaml
path: charts/outage-assist
repoURL: https://github.build.ge.com/IDMS/outage-assist-charts.git
targetRevision: FRA-VERSION
syncPolicy:
  automated:
    prune: true
    selfHeal: true
  syncOptions:
    - CreateNamespace=true
    - RespectIgnoreDifferences=true

```

Outage-Assist-values YAML file

```

# Default values for breaker summary deployment.
# This is a YAML-formatted file.
# Declare variables to be passed into your templates.

global:
  platformNamespace: foundation-env-default
  clusterExternalUrl: http://ingress.local
  fqdnSuffix: .svc.cluster.local
  netviewAdapterClientId: netview-adapter-client-id
  netviewAdapterClientSecret: netview-adapter-client-secret
  omsInterfaceServiceClientId: oms-interface-service-client-id
  omsInterfaceServiceSecret: oms-interface-service-secret
  netServerMainClientId: net-server-main-client-id
  netServerMainSecret: net-server-main-secret
  netServerReadOnlyClientId: net-server-readonly-client-id
  netServerReadOnlySecret: net-server-readonly-secret
  dcInterfaceServiceClientId: dc-interface-service-client-id
  dcInterfaceServiceSecret: dc-interface-service-secret
  zitadel:
    enabled: true
  zitadelProxyDns: ""
  serviceProxyDns: ""
  foundation:
    zeroTrustNamespace: foundation-cluster-zerotrust

etd-services:
  oisExternalGatewayRoute:
    endpoints:
      - https://{WIN_SERVER}:8812
      # - https://172.227.51.85:8802
      # - https://172.227.51.85:8803
      # - https://172.227.51.85:8804
  nvaMainExternalGatewayRoute:
    endpoints:
      - https://{WIN_SERVER}:24468

```

```

# - https://172.227.51.85:24468
# - https://172.227.51.85:24469
# - https://172.227.51.85:24470
nvaReadOnlyExternalGatewayRoute:
  endpoints:
    - http://{WIN_SERVER}:24461
    #- https://172.227.51.85:24468
    #- https://172.227.51.85:24469
    #- https://172.227.51.85:24470
dciExternalGatewayRoute:
  endpoints:
    - http://{WIN_SERVER}:50001
admsExternalGateway:
  loggingLevel: ERROR
  replicas: 1
  annotations: {}
  maxHttpRequestHeaderSize: "1MB"
admsCertSecret:
  cert: |
    -----BEGIN CERTIFICATE-----
    MIIDpDCCAoygAwIBAgIUPqxi/b5xxFAyNlRFjW/OTewfxlYwDQYJKoZIhvcNAQEL
    BQAwYTELMAkGA1UEBhMCQVUxEzARBgNVBAgMClnvbWUtU3RhdGUxEzARBgNVBAoM
    CkdFIERpZ2l0YWwxKDAwBgNVBAMMH0RFMi4wIEZvdW5kYXRpb24gQmFzZSAuIFJv
    b3QgQ0EwHhcNMjExMjA4MTkzNDQ0WhcNMzExMjA4MTkzNDQ0WjBhMQswCQYDVQQL
    EwJVUzELMAkGA1UECAwCV0ExEDAOBgNVBACMB0JvdGh1bGwxCzAJBgNVBAoMAkdF
    MRAwDgYDVQQLDAdEaWdpdGFsMRQwEgYDVQQDDAtBRE1TIFNlcnZlcjCCASIwDQYJ
    KoZIhvcNAQEBBQADggEPADCCAQoCggEBALlPUvH4acN1rf1zcNvwvXV7uEZIPwNO
    Wo/3rcf+yqWH/7Etdn3jOBtipSdzpRsXIcXE/NA9J7S5UzLwE8EqHjRPQsyVE0q
    wtaG6GhoFOKAQ13dNEkcED59jeY1e8hJs/2Q7dsjVJXFMGGhkD0L6muco9G7zQj9k
    7a7ciLisyCU83o8u57eLI9/8g4EHK8LWOFARkABFWRKMipHeJTtstjX4IVCy4hGk
    lL2/cMXfgjk1BkMjh9ZQVnmR36lKJQD0cdiKJgchtt1D9UT3VNbQu0XkURb/b4xw
    hC/xfHauswAe/yihBZqGZ+Nde5E6ar1JL18dzPBjwKid002L+OTz9I0CAwEAAaNU
    MFiIwUAYDVRR0RBEkwr4IJbG9jYWxob3N0hwR/AAABhwQK4zGchwQK4zG6hwQK4zNV
    hwQK4zC5hwSsGYQJhwSsGYTehwSsGYR0hwSsGYTHhwQK4zMhMA0GCSqGSIb3DQEB
    CwUAA4IBAQAAPVpuR0fzYp00SUGXwHYQ39WiyEGNN8Kb3p3+owk09g7B0mV0KscZc
    D+pU7IYL8FIHilbeiJklfDMfcrCaxHc/HqvB9CaqWdSnSGbwGNqbPAYcrCXFheY0
    HeFFKLqsJmhmMotRpcG7eNcEeo/OTehfeqhBbOP7tdYEmFaQnrVpkBQ3zrJbZ/wg
    AeXRDZnpkx+z0v9Jqnc052eDNutKlohG5VxP3YtzBpz+enVubLnixaMD1ropMgGf
    0zE6HGUZEnvJhR/PJgN6TuL/AykuFHxEq0pFwUHuso/uhB3vMLhXaMAMJSw9+pG
    +TkVuQyQadPWEh7Du6Rse8p6QNFCoJYM
    -----END CERTIFICATE-----
uaaClient:
  nvaUaaClient:
    authorities: # Needs to match etd group defined in roles-config YAML
      - service.account.group5
  oisUaaClient:
    authorities: # Needs to match etd group defined in roles-config YAML
      - service.account.group5
  mainUaaClient:
    authorities: # Needs to match etd group defined in roles-config YAML
      - service.account.group5
  roUaaClient:
    authorities: # Needs to match etd group defined in roles-config YAML

```

```

    - service.account.group5
  dcIUaaClient:
    authorities: # Needs to match etd group defined in roles-config YAML
    - service.account.group5
  pref-init:
    omsMode: eoms

```

3.1.3.2. Preparing k8s-base-manifest.yaml

The RKE2 Kubernetes cluster deployment is defined using the k8s-base-manifest.yaml file.

3.1.3.2.1. Example k8s-base-manifest.yaml File

The following is an example of the k8s-base-manifest.yaml file for deploying Outage Assist.

```

id: k8s-base-stack-{{ env['BRANCH_NAME'] }}
api_version: '5.2'
targets:
  - id: AWS
    products:
      - id: k8s-infra
        pdi:
          image: dig-grid-artifactory.apps.ge.com/foundation-docker/k8s-infra-pdi
          tag: 'latest'
          local_dir_mount_path: './local/'
          param_groups:
            - id: environment
              params:
                TF_VAR_operating_system: 'CENTOS7'
                TF_VAR_number_of_masters: '1'
                TF_VAR_number_of_agents: '4'

      - id: k8s
        pdi:
          image: dig-grid-artifactory.apps.ge.com/foundation-docker/pdi-k8s
          tag: '1.0.0Uv1.24.4'
          local_dir_mount_path: './local/'
          param_groups:
            - id: environment
              params:
                INVENTORY_FILE_PATH: '/data/local/k8s_inventory.yaml'
            - id: k8s_all_group_vars
              params:
                ansible_user: gecloud # expected user for the image above
                ansible_ssh_private_key_file: '/data/local/ssh.pem'
                monitoring_stack: true
                install_rke2_version: 'v1.23.12+rke2r1'
                install_rke2_channel: stable

      - id: k8s-validate

```

```

pdi:
  image: dig-grid-artifactory.apps.ge.com/foundation-docker/pdi-k8s-validate
  tag: 'latest'
  depends_on_product: 'k8s'
  local_dir_mount_path: './local/'

```

3.1.3.3. Preparing foundation-base-manifest.yaml

The Foundation Base deployment is defined using the yaml file. To support Outage Assist, the following sa-verrides yaml file needs to present while doing the deployment of foundation-ui

sa-overrides YAMI file to install graphql-server

```

# Tabular GraphQL Server
graphql-server:
  netview:
    namespace: oms
    name: adms-external-gateway
    prefix: nva-ro

```

3.1.3.3.1. Example foundation-base.yaml File

The following is an example of the foundation-base-manifest.yaml file for deploying Outage Assist.

```

apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: foundation-base
  namespace: argocd
  finalizers:
    - resources-finalizer.argocd.argoproj.io
spec:
  destination:
    namespace: "argocd"
    server: https://kubernetes.default.svc
  project: default
  sources:
    - chart: foundation-base-argocd-wrapper
      repoURL: https://dig-grid-artifactory.apps.ge.com/artifactory/api/helm/foundation-helm
      targetRevision: BASE-VERSION
      helm:
        valueFiles:
          - $valuesRepo/2024-04/foundation-base-argocd/foundation-base-app-values.yaml
    - repoURL: https://github.build.ge.com/IDMS/outage-assist-charts.git
      targetRevision: FRA-VERSION
      ref: valuesRepo
  syncPolicy:
    automated:
      prune: false
      selfHeal: true

```

```
allowEmpty: true
syncOptions:
- CreateNamespace=false
- RespectIgnoreDifferences=true
```

Foundation Base App yaml file

```
spec:
  destination:
    server: "https://kubernetes.default.svc"
  sources:
    helmRepo: "https://dig-grid-artifactory.apps.ge.com/artifactory/api/helm/foundation-helm"
    values:
      repo: "https://github.build.ge.com/grid-foundation/foundation-reference-app"
      targetRevision: "2024-04"
      foundationProfile:
        prefixPath: "release/foundation-helm-profiles"
        name: "default"

#### sync-wave: 10
kyvernoRancherCrds:
  chartName: "kyverno-rancher-crds"
  #overrideTargetRevision: 1.3.0

#### sync-wave: 20
foundationPoliciesCerts:
  chartName: "foundation-policies-certs"
  #overrideTargetRevision: 1.3.0

#### sync-wave: 30
zeroTrustOperator:
  chartName: "zero-trust-operator"
  #overrideTargetRevision: 1.3.0

#### sync-wave: 40
istioPrep:
  chartName: "istio-prep"
  #overrideTargetRevision: 1.3.0

#### sync-wave: 40
istiod:
  chartName: "istiod"
  #overrideTargetRevision: 1.3.0

#### sync-wave: 50
foundationOperators:
  chartName: "foundation-operators"
  #overrideTargetRevision: 1.3.0

#### sync-wave: 60
monitoringApps:
  chartName: "monitoring-apps"
  #overrideTargetRevision: 1.3.0
```

```

#### sync-wave: 70
k8sIngressNginx:
  chartName: "k8s-ingress-nginx"
  #overrideTargetRevision: 1.3.0

#### sync-wave: 80
zerotrustApps:
  chartName: "zerotrust-apps"
  #overrideTargetRevision: 1.3.0

#### sync-wave: 81
foundationAuthWS:
  chartName: "foundation-auth-ws"
  #overrideTargetRevision: 1.0.0

#### sync-wave: 90
velero:
  chartName: "velero"
  #overrideTargetRevision: 1.3.0

#### sync-wave: 100
foundationClusterTools:
  chartName: "foundation-cluster-tools"
  #overrideTargetRevision: 1.3.0

```

3.1.3.4. Preparing foundation-osb.yaml

The Foundation OSB deployment is defined using the foundation-osbyaml file.

3.1.3.4.1. Example foundation-osb.yaml File

The following is an example of the foundation-osb.yaml file for deploying Outage Assist.

```

apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: foundation-osb
  namespace: argocd
  finalizers:
    - resources-finalizer.argocd.argoproj.io
spec:
  destination:
    namespace: "foundation-env-default"
    server: https://kubernetes.default.svc
  project: default
  sources:
    - chart: foundation-osb
      repoURL: https://dig-grid-artifactory.apps.ge.com/artifactory/api/helm/foundation-helm
      targetRevision: OSB-VERSION

```



```

helm:
  valueFiles:
    - $valuesRepo/2024-04/foundation-helm-profiles/PROFILE/foundation-umbrella-values.yaml
    - $valuesRepo/2024-04/foundation-helm-profiles/PROFILE/global-values.yaml
  - repoURL: https://github.build.ge.com/IDMS/outage-assist-charts.git
    targetRevision: FRA-VERSION
    ref: valuesRepo
  syncPolicy:
    automated:
      prune: false
      selfHeal: true
      allowEmpty: true
    syncOptions:
      - CreateNamespace=false
      - RespectIgnoreDifferences=true

```

3.1.3.5. Preparing foundation-ui.yaml

The Foundation UI deployment is defined using the foundation-ui.yaml file.

3.1.3.5.1. Example foundation-ui-manifest.yaml File

The following is an example of the foundation-ui-manifest.yaml file for deploying Outage Assist.

```

apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: foundation-ui-server
  namespace: argocd
  finalizers:
    - resources-finalizer.argocd.argoproj.io
spec:
  destination:
    namespace: "foundation-env-default"
    server: https://kubernetes.default.svc
  project: default
  sources:
    - chart: foundation-ui-server
      repoURL: https://dig-grid-artifactory.apps.ge.com/artifactory/api/helm/foundation-helm
      targetRevision: UI-SERVER-VERSION
      helm:
        releasenName: foundation-ui-server
        valueFiles:
          - $valuesRepo/2024-04/foundation-helm-profiles/PROFILE/foundation-umbrella-values.yaml
          - $valuesRepo/apps/sa-overrides.yaml
          - $valuesRepo/2024-04/foundation-helm-profiles/PROFILE/global-values.yaml
    - repoURL: https://github.build.ge.com/IDMS/outage-assist-charts.git
      targetRevision: FRA-VERSION
      ref: valuesRepo
  syncPolicy:

```

```

automated:
  prune: false
  selfHeal: true
  allowEmpty: true
syncOptions:
- CreateNamespace=false
- RespectIgnoreDifferences=true

```

3.1.3.6. Preparing Foundation Data Yaml file

```

apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: foundation-data
  namespace: argocd
  finalizers:
  - resources-finalizer.argocd.argoproj.io
spec:
  destination:
    namespace: "foundation-cluster-zerotrust"
    server: https://kubernetes.default.svc
  project: default
  source:
    helm:
    # valueFiles:
    # - ../../apps-config/global-config/00-foundation-global-values.yaml
    path: 2024-04/foundation-data
    repoURL: https://github.build.ge.com/IDMS/outage-assist-charts.git
    targetRevision: FRA-VERSION
  syncPolicy:
    automated:
      prune: false
      selfHeal: true
    syncOptions:
    - CreateNamespace=false
    - Replace=true

```

3.1.4. Preparing AOR Data

Configure the following files:

In this version of Foundation, users can use version 2 or version 3 of the config files related to AOR.

To switch between V2 and V3, use the following config setting in outage-assist-charts/2024-04/foundation-data/values.yaml

aorsManager:

- For v3 aors set below to true

```
importYamlFileContent: true
useV3Config: true
useV3Import: true
```

- For v2 aors set below to true

```
importFileContent: false
useV2Config: false
useAorsXML: false
```

- V2: aors.xml, roles-config.yaml
- V3: aors-config-v3.yaml, roles-config.yaml, aors-import-v3.yaml
- **aors.xml**: Contains a list of AORs exported from Permit. Used by the data-loader for V2.
- **aors-config-v3.yaml**: Contains the AORs-to-Group mapping. Used by the data-loader for V3.
- **roles-config.yaml**: Contains the Roles-to-Group mapping. Used by the data-loader.

The AOR model for Outage Assist must include the DISPATCH_FUNCTION_AOR_NAME and OUTAGE_INCIDENT_FUNCTIONAL_PERMISSION_NAME parameters values defined in NetServerOmsConfig.txt (by default, OUTAGE and DISPATCH) as the lowest AORs. The model must also include AORs specified using the FunctionalPermission parameters in the OrdersPermissionsTable section of OmsServerConfig.xml (by default, ARBOR and EDISON). The AOR model must also include the FUNC AOR for functional updates.

For example, add the following entries in aors.xml:

```
<AORRef>
  <Children>
    <AORRef>
      <Children />
      <CheckedState>Unchecked</CheckedState>
      <CanDetach>true</CanDetach>
      <IsAOR>true</IsAOR>
      <DescriptiveName>Outage Permission</DescriptiveName>
      <FunctionalAOR />
      <NameAOR>DEFPAREA</NameAOR>
    </AORRef>
    <AORRef>
      <Children />
      <CheckedState>Unchecked</CheckedState>
      <CanDetach>true</CanDetach>
      <IsAOR>true</IsAOR>
      <DescriptiveName>Outage Permission</DescriptiveName>
      <FunctionalAOR />
      <NameAOR>OUTAGE</NameAOR>
    </AORRef>
  </Children>
  <CheckedState>Unchecked</CheckedState>
  <CanDetach>true</CanDetach>
```

```

    <IsAOR>true</IsAOR>
    <DescriptiveName>Dispatch Permission</DescriptiveName>
    <FunctionalAOR />
    <NameAOR>DISPATCH</NameAOR>
  </AORRef>
  <AORRef>
    <Children />
    <CheckedState>Unchecked</CheckedState>
    <CanDetach>true</CanDetach>
    <IsAOR>true</IsAOR>
    <DescriptiveName>EDISON Quick Order Permission</DescriptiveName>
    <FunctionalAOR />
    <NameAOR>EDISON</NameAOR>
  </AORRef>
  <AORRef>
    <Children />
    <CheckedState>Unchecked</CheckedState>
    <CanDetach>true</CanDetach>
    <IsAOR>true</IsAOR>
    <DescriptiveName>ARBOR Quick Order Permission</DescriptiveName>
    <FunctionalAOR />
    <NameAOR>ARBOR</NameAOR>
  </AORRef>
</Children>
<CheckedState>Unchecked</CheckedState>
<CanDetach>true</CanDetach>
<IsAOR>true</IsAOR>
<DescriptiveName>Functional</DescriptiveName>
<FunctionalAOR />
<NameAOR>FUNC</NameAOR>
</AORRef>
<AORRef>
  <Children />
  <CheckedState>Unchecked</CheckedState>
  <CanDetach>true</CanDetach>
  <IsAOR>true</IsAOR>
  <DescriptiveName>Default Area</DescriptiveName>
  <FunctionalAOR />
  <NameAOR>DEFPAREA</NameAOR>
</AORRef>

```

Add the following entry in aors-config-v3.yaml:

```
defaultAors: "ROOT.GE.FUNC"
```

For more information about configuring AORs, consult your GE project team.

3.1.4.1. Outage Assist AOR Configuration Updates

aors-config-v3.yaml

In the `aors-config-v3.yaml` file, you'll find configurations specific to AORs for Outage Assist.

- Different types of AORs are supported in v3 like Geographic, DeviceType, Distribution and so on. Each AOR should start with ROOT followed by AOR Type. For example, is ROOT.GEOGRAPHIC....
- AORs are defined under each usergroupName. You can assign access levels such as Read or Write independently for each AOR. In v2, the equivalent values were oms.Read and oms.Write.
- Each AOR can be enabled or disabled using the `isDefault` flag set to true or false. If `isDefault` is not provided, the default value is false.

For example:

```
- usergroupName: Supervisors
  selectableAor:
    aors:
      - aorName: ROOT.GEOGRAPHIC.GE.WA
        permissionLevel: ['Read','Write']
        isDefault: true
      - aorName: ROOT.GEOGRAPHIC.GE.OR
        permissionLevel: ['Read','Write']
        isDefault: true
      - aorName: ROOT.GEOGRAPHIC.GE.ORG
        permissionLevel: ['Read','Write']
        isDefault: true
      - aorName: ROOT.GEOGRAPHIC.GE.FUNC
        permissionLevel: ['Read','Write']
        isDefault: true
      - aorName: ROOT.GEOGRAPHIC.GE.DEFPAREA
        permissionLevel: ['Read','Write']
        isDefault: true

- usergroupName: Readonly Viewers
  selectableAor:
    aors:
      - aorName: ROOT.GEOGRAPHIC.GE.WA
        permissionLevel: ['Read']
        isDefault: true
      - aorName: ROOT.GEOGRAPHIC.GE.OR
        permissionLevel: ['Read']
        isDefault: true
      - aorName: ROOT.GEOGRAPHIC.GE.ORG
        permissionLevel: ['Read']
        isDefault: true
```

aors-import-v3.yaml

In the `aors-import-v3.yaml` file, you'll define the AOR hierarchy information for Outage Assist in v3. Here are the key points:

- **AOR Hierarchy Configuration:** Unlike v2 version, v3 version draws the AOR hierarchy information from this file instead of `aors.xml`. Ensure all parent and child level AORs are defined in this file.

- **ShortName Property:** The shortName property in v3 is equivalent to the NameAOR property in aors.xml of v2. If shortName is not provided, the default value would be the last part of an AOR (after the last dot in the name property).

For example:

```
- name: ROOT
  description: Root
  shortName: ROOT
- name: ROOT.GEOGRAPHIC
  description: GEOGRAPHIC TYPE
  shortName: GEOGRAPHIC
- name: ROOT.GEOGRAPHIC.GE.WA.GORDON.ROARKE.3LIONSSC.RAVEN
  description: RAVEN
  shortName: RAVEN
- name: ROOT.GEOGRAPHIC.GE.ORG
  description: ORG
  shortName: ORG
- name: ROOT.GEOGRAPHIC.GE.ORG.SO_HIGH
  description: SO_HIGH
  shortName: SO_HIGH
- name: ROOT.GEOGRAPHIC.GE.FUNC
  description: FUNC
  shortName: FUNC
- name: ROOT.GEOGRAPHIC.GE.FUNC.OUTAGE
  description: OUTAGE
  shortName: OUTAGE
```

roles-config-v2.yaml

In the roles-config-v2.yaml file, which is used by both v2 version and v3 version, you'll find role configurations.

- There's no change to the existing file unless a role or permission needs to be added or updated.
- When deleting roles in an existing environment and re-deploying, ensure existing AORs are deleted as well because both AORs and roles share the same usergroups.

3.1.5. Preparing a Deployment Folder

Note

Before running deployment procedure below, make sure Foundation and Outage Assist deployment prerequisites are met (Namespaces, root certificates, external credentials secrets, and so on).

On the machine where you deploy the Outage Assist solution, prepare a folder for deployment:

1. Create a folder named "App and App-Config" to store the yaml file to deploy using Helm or ArgoCD

2. Add the Helm chart of Outage Assist by unpacking the chart, outage-assist-16.8.0.tgz. If doing the deployment using the git repository and reading the chart
3. Copy the Kubernetes cluster's admin.conf file to the "deployment" folder. By default, the admin.conf file is stored in the /etc/kubernetes/ folder.
4. Add the adms Public certificate Authority certificate (ADMS) in YAML file. This certificate is used to sign certificates on the ADMS side. (See [Generating Certificates for SSL/TLS Termination with ADMS.](#))
5. Copy the certificate and key for the Ingress controller and for establishing connections with Active Directory to the "cert" folder. (See [Generating Certificates for SSL/TLS Termination with ADMS.](#))

i Note

Ingress certificates are not required for Outage Assist deployment if you already deployed Ingress Controller with enabled TLS termination during the Foundation deployment.

6. Copy the AOR files to the "data-loader" folder.

A sample folder structure that is used for deployment is shown below. This structure can be used as a reference to create the folder structure.

```
<folder to run Bsfdeploy>
├── k8s-base-manifest.yaml
├── Foundation helm chart
├── Outage Assist helm chart
├── apps
│   ├── Custom-namespace.yaml
│   ├── foundation-base.yaml
│   ├── foundation-data.yaml
│   ├── foundation-osb.yaml
│   ├── foundation-ui-server.yaml
│   ├── outage-assist.yaml
│   └── sa-overrides.yaml
├── apps-config
└── outage-assist-values.yaml
```

In the example above, the Helm chart folder contains all the required files for deploying the Foundation Base, UI, OSB, and sa-deploy Gateway Helm chart.

3.2. Helm Based Deployment

- add helm repository

The example command below assumes digital grid artifactory is used as the helm repository, and that the corresponding user and password are stored in respective environment variables \${GRIDART_USR} and \${GRIDART_PSW}

In airgap environments, replace <https://dig-grid-artifactory.apps.ge.com/artifactory/api/helm/adms-helm> with the url to your local helm repository. And pass in the local helm repo's username and password accordingly.

```
helm repo add adms-helm https://dig-grid-artifactory.apps.ge.com/artifactory/api/helm/adms-helm
--username ${GRIDART_USR} --password ${GRIDART_PSW}

# create ImagePullSecret secret
./kubectl create secret docker-registry ${GRIDART_SECRET_NAME} --docker-server=${DGA_REGISTRY}
--docker-username=${GRID_ARTIFACTORY_CREDS_USR} --docker-password=${GRID_ARTIFACTORY_CREDS_PSW} -n
"${OA_DEPLOYMENT_NAMESPACE}" || true
./kubectl patch serviceaccount default -n "${OA_DEPLOYMENT_NAMESPACE}" -p '{"imagePullSecrets\":[{"name\":"docker-registry"}]}'

# create PG secret
./kubectl get secret postgres-db-pg-cluster-postgres-secret -n "${PLATFORM_NAMESPACE}" -o yaml |
grep -v '^\\s*namespace:\\s' | kubectl apply -n "${OA_DEPLOYMENT_NAMESPACE}" -f - || true

# Export Variables
export VALUES_ARGS="--values outage-assist-values.yaml
--set global.zitadel.enabled=true
--set global.clusterExternalUrl=https://<Cluster_External_DNS>"
helm upgrade -i <release-name> adms-helm/<umbrella-chart-name> --version <umbrella-chart-version>
--wait $VALUES_ARGS
```

3.3. ArgoCD Based Deployment

i Note

Unlike the helm-only use case, ArgoCD deployment requires pushing the configuration manifests to a Git repository additionally. In either case, make a note of the location of each configuration manifest in Git, as they will be used to configure the outage-assist ArgoCD application.

The command below assumes that your kubectl client is configured with the argocd cluster kubeconfig file, meaning you have access to it. If not, please contact the argocd admin user/team to obtain the argocd credentials and log in using the classical way.

```
kubectl config set-context --current --namespace=argocd
```

```
argocd login --core
```

- add helm repository

The example command below assumes digital grid artifactory is used as the helm repository, and that the corresponding user and password are stored in respective environment variables `${GRIDART_USR}` and `${GRIDART_PSW}`

In airgap environments, replace <https://dig-grid-artifactory.apps.ge.com/artifactory/api/helm/adms-helm> with the url to your local helm repository. And pass in the local helm repo's username and password accordingly.

```
helm repo add adms-helm https://dig-grid-artifactory.apps.ge.com/artifactory/api/helm/adms-helm
--username ${GRIDART_USR} --password ${GRIDART_PSW}
./kubectl create secret docker-registry ${GRIDART_SECRET_NAME} --docker-server=${DGA_REGISTRY}
--docker-username=${GRID_ARTIFACTORY_CREDS_USR} --docker-password=${GRID_ARTIFACTORY_CREDS_PSW} -n
"${OA_DEPLOYMENT_NAMESPACE}" || true
./kubectl patch serviceaccount default -n "${OA_DEPLOYMENT_NAMESPACE}" -p '{"imagePullSecrets\":[{"name\":"docker-registry"}]}'
# create PG secret
./kubectl get secret postgres-db-pg-cluster-postgres-secret -n "${PLATFORM_NAMESPACE}" -o yaml |
grep -v '^\\s*namespace:\\s' | kubectl apply -n "${OA_DEPLOYMENT_NAMESPACE}" -f - || true
argocd repo add https://dig-grid-artifactory.apps.ge.com/artifactory/api/helm/adms-helm \
--type helm \
--name adms-helm \
--username ${GRIDART_USR} \
--password ${GRIDART_PSW} \
--upsert

argocd repo add https://github.build.ge.com/IDMS/outage-assist-charts.git \
--username ${GIT_USR} \
--password ${GIT_PSW} \
--upsert

argocd app create -f outage-assist.yaml --repo https://github.build.ge.com/IDMS/outage-assist-
charts.git \
--helm-set-string
spec.sources.values.adhocGlobals.global.clusterExternalUrl=https://${CLUSTER_EXTERNAL_DNS} \
--helm-set-string spec.sources.values.adhocGlobals.global.zitadel.enabled=true \
--upsert
argocd app wait outage-assist --sync --timeout 600
argocd app wait outage-assist --operation --timeout 600
argocd app wait outage-assist --health --timeout 600
```

Important

After deploying a Foundation , restart the ADMS interface servers (DMS Interface Server, OMS Interface Server, and NetViewer Adapter).

4. Outage Assist Recommended Deployment

This chapter describes the standardized deployment method for Outage Assist. This method ensures consistency, streamlining the installation and support process to enhance overall system efficiency. Although variations in deployment may have been used by customers in the past, GE strongly encourages the use of this standard deployment in order to effectively support active installations.

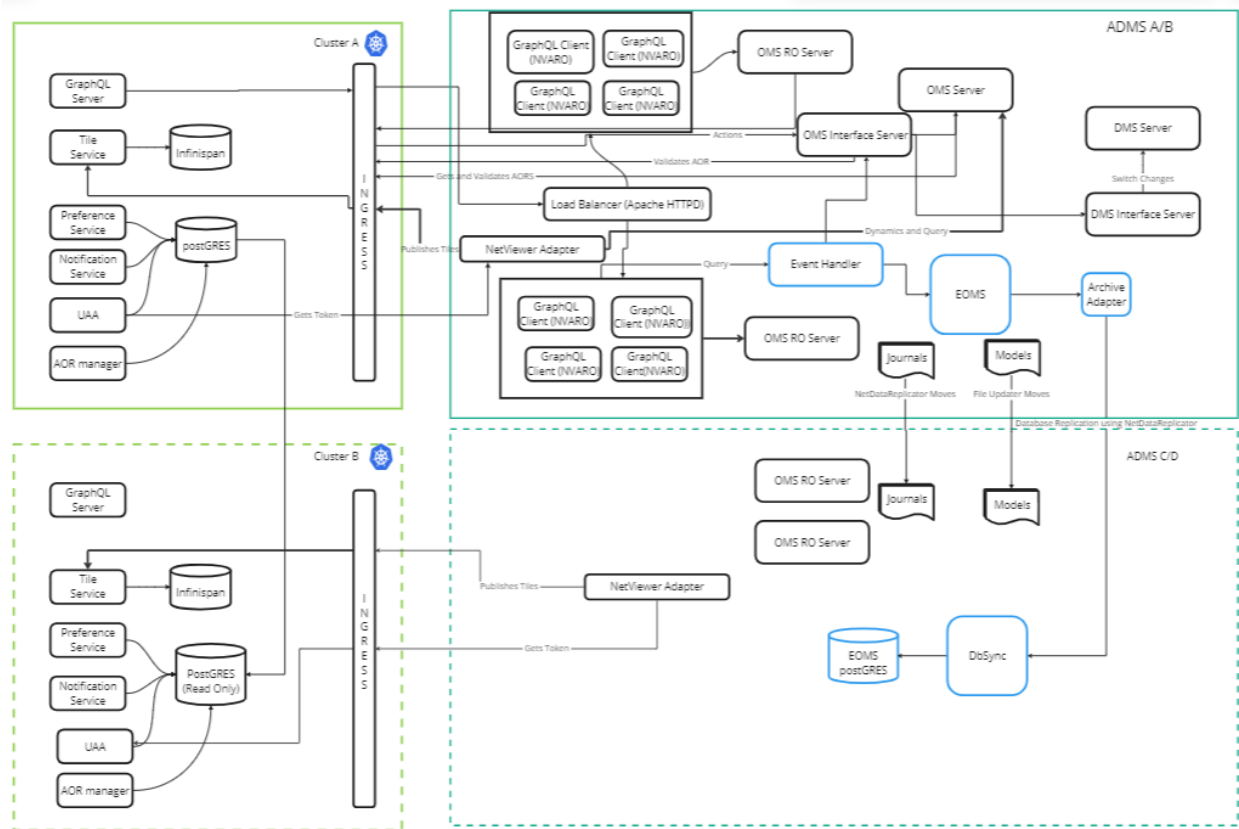


Figure 6. Outage Assist/ADMS Deployment Architecture

The figure above shows the recommended deployment architecture, mainly designed for Outage Assist. Only the relevant components for Outage Assist are shown in the ADMS section. The current foundation architecture supports two RKE2 clusters in an active/standby mode, while ADMS can support four nodes. In this setup, ADMS nodes A and B run on the same site as Cluster A, and nodes C and D operate where Cluster B is located. The main highlights of this deployment architecture are:

1. Planned Failover Coordination:
 - Crucial for coordinated planned failovers between ADMS and Outage Assist clusters.
 - In the case of unplanned failovers, the system continues operation with limited performance. Relocating to the same cluster is highly advised.
2. GTM (Global Traffic Manager) Limitation:

- Currently, the foundation does not support GTM.
- In case of failovers, users need an alternate shortcut to log in to the new cluster.
- If a project chooses to implement its own GTM, it is crucial to replicate session information stored in Infinispan to the secondary cluster for a smooth transition. Note that this support is not anticipated in the foundation's future development roadmap.

3. LTM (Local Traffic Manager) Limitation:

- The cluster currently does not support LTM.
- To achieve scalability and availability within the cluster, projects should deploy the recommended architecture outlined in the Kubernetes documents provided here. The foundation plans to include this support in the roadmap for future development.

4. Database Replication using PostGRES:

- The Foundation supports database replication between sites using PostGRES.
- The secondary cluster's database instance is set to read-only.
- No user authentication is possible via the secondary cluster, even when all services are running.

5. ADMS Node Tasks in Secondary Site:

The following tasks are run in active mode to avoid any delay after site transfer:

- **NetviewerAdapter:** Publishes tiles to the secondary cluster.
- **Read-Only NetServer:** Resolves queries from client sessions.
- **NetDataReplicator:** Replicates journals to the secondary site.
- **File Updater Client:** Moves all model files to the secondary site.
- **DbSync** (EOMS Procman deployment mode): Copies EOMS database from the Primary site to the Secondary site.

All other ADMS tasks are fast to start or transition from Standby to Enabled.

Note

For systems without Outage Assist 2.X deployed, `OMS_SPECIAL_LOCK_PROCESSING` defaults to `TRUE`, reducing redundant entries in the Incident Journal. However, when deploying Outage Assist 2.X, ensure to set this parameter to `FALSE`. This is essential for proper handling of lock information in NetServerRO's in-memory database when incidents are locked.

4.1. For Dual or Multi-Site Deployments:

Each cluster will have its own set of credential values for external system to access. There are three elements: `clientID`, `clientSecret`, and `projectID`. `ProjectID` is unique per cluster. At any given time, only one cluster is active; and the other is standby and must shutdown services to not accept connections. The external system should be configured to use the credentials of the active cluster.

In case of a failover, the external system must switch to using the credentials of the newly active cluster.

1. OMS Interface Server:

- Old configuration: `<UaaEndpoints>https://endpoint0/uaa/oauth/token;https://endpoint1/uaa/oauth/token </UaaEndpoints>`
- New configuration: `<UaaEndpoints> <UaaEndpoint> <!-- <Name>ENDPOINT0</Name> -→ <Url><a href="https://endpoint0/uaa/oauth/token</Url>" class="bare">https://endpoint0/uaa/oauth/token</Url> <!-- <IdKey></IdKey> <SecretKey></SecretKey> -→ <UaaEndpoint> <UaaEndpoint> <!-- <Name>ENDPOINT1</Name> -→ <Url><a href="https://endpoint1/uaa/oauth/token</Url>" class="bare">https://endpoint1/uaa/oauth/token</Url> <!-- <IdKey></IdKey> <SecretKey></SecretKey> -→ <UaaEndpoint>`

`</UaaEndpoints>`

If the clusters have same client IDs and Secrets, don't need to specify `<Name></Name>` tags, only token endpoint Urls need to be specified. It's backward compatible and the same following environment variables are used for authentication:

OMSINTERFACESERVER_UAA_AUTH_CLIENTID
OMSINTERFACESERVER_UAA_AUTH_CLIENTSECRET
OMSINTERFACESERVER_UAA_AUTH_PROJECTID

If the clusters have different client IDs and Secrets (Zitadel authentication clusters), use `<Name></Name>` tag to differentiate the environment variables used for authentication. The endpoint name is the namespace prefix of the environment variable, and the following environment variables need to be defined:

ENDPOINT0_OMSINTERFACESERVER_UAA_AUTH_CLIENTID
ENDPOINT0_OMSINTERFACESERVER_UAA_AUTH_CLIENTSECRET
ENDPOINT0_OMSINTERFACESERVER_UAA_AUTH_PROJECTID
ENDPOINT1_OMSINTERFACESERVER_UAA_AUTH_CLIENTID
ENDPOINT1_OMSINTERFACESERVER_UAA_AUTH_CLIENTSECRET
ENDPOINT1_OMSINTERFACESERVER_UAA_AUTH_PROJECTID

`<IdKey></IdKey>` and `<SecretKey></SecretKey>` tags need to be specified if the client IDs and Secrets are stored in CyberArk secret store, not the environment variables.

2. Net Viewer Adapter:

- Old configuration: `<UaaTokenServer><a href="http://insert_server_name/uaa/oauth/token</UaaTokenServer>" class="bare">http://insert_server_name/uaa/oauth/token</UaaTokenServer></a>`
- New configuration: `<UaaTokenServer> <!-- <Name>OA</Name> -→ <Url><a href="http://insert_server_name/uaa/oauth/token</Url>" class="bare">http://insert_server_name/uaa/oauth/token</Url></a> </UaaTokenServer>` If `<Name></Name>` tag is not specified, the same following environment variables are used for authentication:
NETVIEW_UAA_AUTH_CLIENTID NETVIEW_UAA_AUTH_CLIENTSECRET
NETVIEW_UAA_AUTH_PROJECTID If running multiple Net Viewer Adapter instances

connecting to different clusters, use `<Name>` tag to differentiate the environment variables used for authentication. The token server name is the namespace prefix of the environment variable, and the following environment variables need to be defined: `OA_NETVIEW_UAA_AUTH_CLIENTID` `OA_NETVIEW_UAA_AUTH_CLIENTSECRET` `OA_NETVIEW_UAA_AUTH_PROJECTID` "OA" is defined in name tag. For example, users can run two NVA instances – one for outage assist, one for Hazzard Management, and name tag is defined as "OA" and "HM" accordingly.

3. OMS NetServer:

If clusters have different client IDs, Secrets, and ProjectIDs, the configuration for `PERMISSION_UAA_END_POINT` is same: `PERMISSION_UAA_END_POINT = https://ingress0.local/uaa/oauth/token; https://ingress1.local/uaa/oauth/token` Same environment variables are used for authentication: `NETSERVER_UAA_AUTH_CLIENTID` `NETSERVER_UAA_AUTH_CLIENTSECRET` `NETSERVER_UAA_AUTH_PROJECTID` Users need to specify the endpoint name if clusters have different client IDs, Secrets, and ProjectIDs as following: `PERMISSION_UAA_END_POINT = ENDPOINT0=https://ingress0.local/uaa/oauth/token; ENDPOINT1=https://ingress1.local/uaa/oauth/token` The following environment variables need to be defined: `ENDPOINT0_NETSERVER_UAA_AUTH_CLIENTID` `ENDPOINT0_NETSERVER_UAA_AUTH_CLIENTSECRET` `ENDPOINT0_NETSERVER_UAA_AUTH_PROJECTID` `ENDPOINT1_NETSERVER_UAA_AUTH_CLIENTID` `ENDPOINT1_NETSERVER_UAA_AUTH_CLIENTSECRET` `ENDPOINT1_NETSERVER_UAA_AUTH_PROJECTID`

5. Customizing Outage Assist

You can customize Outage Assist for your needs using configuration files. In addition, you must export project-specific configuration data from ADMS.

By default, sample configuration files are stored in the %IDMS_ROOT%\config\interfaces\StormAssist folder on the ADMS host.

5.1. Outage Assist Customization Files

i Note

The OMS Client Configuration Editor exports certain files with the default Fantasy Island configuration to be used as an example (template) for customization.

The following table lists configuration files that can be used to customize the Outage Assist application for your needs.

File	Export Tool	Exported Data	Description	Editing Tool
AppMode.json	OMS Client Configuration Editor	Default configuration	Used to switch the Outage Assist application mode between SOMS and EOMS. Two files are delivered in ADMS kit: AppMode_EOMS.json and AppMode_SOMS.json Used to customize the application name. Used to show an incident or work order icon.	Text Editor
CallMapQuery.json	OMS Client Configuration Editor	Default configuration	Specifies unassociated call fields that are queried from ADMS for Outage Assist.	Text Editor
CrewMapQuery.json	OMS Client Configuration Editor	Default configuration	Specifies crew fields that are queried from ADMS for Outage Assist.	Text Editor
GridServiceLayerRefreshConfig.json	OMS Client Configuration Editor	Default configuration	Specifies the refresh rates for incidents, crews, and unassociated calls.	Text Editor

File	Export Tool	Exported Data	Description	Editing Tool
IncidentMapQuery.json	OMS Client Configuration Editor	Default configuration	Specifies incident fields that are queried from ADMS for Outage Assist.	Text Editor
NetworkViewer.json	OMS Client Configuration Editor	Default configuration	Lists customer layers on the network view.	Text Editor
PrintedIncidentDetail.json	OMS Client Configuration Editor	Default configuration	Customize Incident Print view	Text Editor
PrintDateFormat.json	OMS Client Configuration Editor	Default configuration	Customized date format to print	Text Editor

The following table lists files that contain project-specific configuration data from ADMS.

File	Export Tool	Exported Data	Description	Editing Tool
AMIIIn.json	OMS Client Configuration Editor	Project-specific configuration	AMIIIn coded translation item	Coded Translation Editor
AMIOOut.json	OMS Client Configuration Editor	Project-specific configuration	AMIOOut coded translation item	Coded Translation Editor
AreaNote.json	OMS Client Configuration Editor	Project-specific configuration	AreaNote coded translation item	Coded Translation Editor
CallType.json	OMS Client Configuration Editor	Project-specific configuration	CallType coded translation item	Coded Translation Editor
Condition_1.json	OMS Client Configuration Editor	Project-specific configuration	Condition_1 coded translation item	Coded Translation Editor
Condition_2.json	OMS Client Configuration Editor	Project-specific configuration	Condition_2 coded translation item	Coded Translation Editor
CrewStatusCode.json	OMS Client Configuration Editor	Default configuration	Data from the ARMSStatus coded translation item and color codes to define the crew color on the network view.	Text Editor
CriticalType.json	OMS Client Configuration Editor	Project-specific configuration	CriticalType coded translation item	Coded Translation Editor
CriticalType2.json	OMS Client Configuration Editor	Project-specific configuration	CriticalType2coded translation item	Coded Translation Editor
CriticalType3.json	OMS Client Configuration Editor	Project-specific configuration	CriticalType3coded translation item	Coded Translation Editor
CriticalType4.json	OMS Client Configuration Editor	Project-specific configuration	CriticalType4coded translation item	Coded Translation Editor
CriticalType5.json	OMS Client Configuration Editor	Project-specific configuration	CriticalType5coded translation item	Coded Translation Editor
Description.json	OMS Client Configuration Editor	Project-specific configuration	Description coded translation item	Coded Translation Editor

File	Export Tool	Exported Data	Description	Editing Tool
DeviceType.json	OMS Client Configuration Editor	Project-specific configuration	DeviceType coded translation item	Coded Translation Editor
ARMSStatus.json	OMS Client Configuration Editor	Project-specific configuration	DeviceType coded translation item	Coded Translation Editor
IncidentSymbol.json	OMS Client Configuration Editor	Project-specific configuration	IncidentSymbol coded translation item	Coded Translation Editor
OrderSymbol.json	OMS Client Configuration Editor	Project-specific configuration	OrderSymbol coded translation item	Coded Translation Editor
NonOutageType.json	OMS Client Configuration Editor	Project-specific configuration	NonOutageType coded translation item	Coded Translation Editor
OutageDeviceTypes.json	OMS Client Configuration Editor	Project-specific configuration	OutageDeviceTypes coded translation item	Coded Translation Editor
OutageIncidentType.json	OMS Client Configuration Editor	Project-specific configuration	OutageIncidentType coded translation item	Coded Translation Editor
PhaseOut.json	OMS Client Configuration Editor	Project-specific configuration	PhaseOut coded translation item	Coded Translation Editor
Priority.json	OMS Client Configuration Editor	Project-specific configuration	Priority coded translation item	Coded Translation Editor
PriorityCode.json	OMS Client Configuration Editor	Project-specific configuration	PriorityCode coded translation item	Coded Translation Editor
SuppressReason.json	OMS Client Configuration Editor	Project-specific configuration	SuppressReason coded translation item	Coded Translation Editor
SwitchType.json	OMS Client Configuration Editor	Project-specific configuration	SwitchType coded translation item	Coded Translation Editor
SystemAffected.json	OMS Client Configuration Editor	Project-specific configuration	SystemAffected coded translation item	Coded Translation Editor
SystemMode.json	OMS Client Configuration Editor	Project-specific configuration	SystemMode coded translation item	Coded Translation Editor
VehicleType.json	OMS Client Configuration Editor	Project-specific configuration	VehicleType coded translation item	Coded Translation Editor
WorkModelOrderStatus.json	OMS Client Configuration Editor	Project-specific configuration	WorkModelOrderStatus coded translation item	Coded Translation Editor
WorkModelOrderType.json	OMS Client Configuration Editor	Project-specific configuration	WorkModelOrderType coded translation item	Coded Translation Editor
WorkOrderCrewSummaryStatus.json	OMS Client Configuration Editor	Project-specific configuration	WorkOrderCrewSummaryStatus coded translation item	Coded Translation Editor
YesNo.json	OMS Client Configuration Editor	Default configuration	Customize 'Yes', 'No'	
CustomerStatus.json	OMS Client Configuration Editor	Default configuration	Customize Customer Energization Status	Coded Translation Editor

The following table lists configuration files that can be used to customize Outage Assist tabular

displays for your needs.

File	Export Tool	Exported Data	Description	Editing Tool
AffectedCustomersTableConfig.json	OMS Client Configuration Editor	Default configuration	Configuration for the Affected Customers table	Text Editor
CalledCustomersTableConfig.json	OMS Client Configuration Editor	Default configuration	Configuration for the Called Customers table	Text Editor
CriticalCustomersTableConfig.json	OMS Client Configuration Editor	Default configuration	Configuration for the Critical Customers table	Text Editor
IncidentsTableConfig.json	OMS Client Configuration Editor	Default configuration	Configuration for the Incidents table	Text Editor
OrdersTableConfig.json	OMS Client Configuration Editor	Default configuration	Configuration for the Orders table	Text Editor
PriorityCallsTableConfig.json	OMS Client Configuration Editor	Default configuration	Configuration for the Priority Calls table	Text Editor
SACrewAssignmentTableConfig.json	OMS Client Configuration Editor	Default configuration	Configuration for the Crew Assignment table	Text Editor
SACrewTableConfig.json	OMS Client Configuration Editor	Default configuration	Configuration for the Crew table	Text Editor
SAUnassociatedCallsTableConfig.json	OMS Client Configuration Editor	Default configuration	Configuration for the Unassociated Calls table	Text Editor
SAChronologyLogsTableConfig.json	OMS Client Configuration Editor	Default configuration	Configuration for the chronology table	Text Editor

The following table lists configuration files that can be used to customize Outage Assist entities for your needs.

File	Export Tool	Exported Data	Description	Editing Tool
EntityDataConfig.json	OMS Client Configuration Editor	Default configuration	Outage Assist entity types and query definitions.	Text Editor
EntityPanelConfig.json	OMS Client Configuration Editor	Default configuration	Definitions for component attribute panels.	Text Editor
IncidentDataFetchConfig.json	OMS Client Configuration Editor	Default configuration	Incident types and query definitions.	Text Editor
IncidentPanelConfig.json	OMS Client Configuration Editor	Default configuration	Definitions for the incident attribute panel.	Text Editor
OrdersMenuConfig.json	OMS Client Configuration Editor	Default configuration	Definitions for the Orders table context menu.	Text Editor
OrderPanelConfig.json	OMS Client Configuration Editor	Default configuration	Definitions for the order attribute panel.	Text Editor

File	Export Tool	Exported Data	Description	Editing Tool
PanelUpdaterConfiguration	OMS Client Configuration Editor	Default configuration	Time period for updating data on attribute panels.	Text Editor
PrintedOrderSummary.json	OMS Client Configuration Editor	Default configuration	Order summary print layout.	Text Editor
StandardEntityMenuHandlerConfiguration.json	OMS Client Configuration Editor	Default configuration	Definitions for entity context menus.	Text Editor

The following table lists configuration files that can be used to customize Outage Assist network view tooltips for your needs.

File	Export Tool	Exported Data	Description	Editing Tool
CrewLayerTooltipConfiguration.json	OMS Client Configuration Editor	Default configuration	Crew tooltip	Text Editor
EntityTooltipConfiguration.json	OMS Client Configuration Editor	Default configuration	Component tooltips	Text Editor
IncidentLayerTooltipConfiguration.json	OMS Client Configuration Editor	Default configuration	Incident symbol color depending on the working status	Text Editor
UnassociatedCallLayerTooltipConfiguration.json	OMS Client Configuration Editor	Default configuration	Unassociated call tooltip	Text Editor

The following table lists configuration files that can be used to customize miscellaneous Outage Assist items for your needs.

File	Export Tool	Exported Data	Description	Editing Tool
ContextMenus.json	OMS Client Configuration Editor	Default configuration	Definitions for entity context menus	Text Editor
CrewLayerConfiguration.json	OMS Client Configuration Editor	Default configuration	Specifies the name of the geographic geometry that is used in the display call-up URLs. The default name is "detail".	Text Editor
CrewSVGS.json	OMS Client Configuration Editor	Default configuration	Specifies images for crew working statuses in the base64 encoding format.	Text Editor
CrewStatus.json	OMS Client Configuration Editor	Default configuration	Data from the ARMSStatus coded translation item and color codes to define the crew color on the network view.	Text Editor

File	Export Tool	Exported Data	Description	Editing Tool
GridOperatorConfig.json	OMS Client Configuration Editor	Default configuration	Specifies the name of the geographic geometry that is used in the display call-up URLs. The default name is "detail".	Text Editor
IncidentLayerConfig.json	OMS Client Configuration Editor	Default configuration	Specifies the name of the geographic geometry that is used in the display call-up URLs. The default name is "detail". Also specifies the zoom level at which incident clustering is performed. This configuration is required to show incidents on the map.	Text Editor
IncidentSVGs.json	OMS Client Configuration Editor	Default configuration	Specifies images for incident working statuses in the base64 encoding format.	Text Editor
IncidentWorkingStatus.json	OMS Client Configuration Editor	Default configuration	Data from the IncidentWorkingStatus coded translation item and color codes to define the incident color on the network view.	Text Editor
LayersDetail.json	OMS Client Configuration Editor	Default configuration	Configures the layout of the Layers panel (includes several background and overlay map layers).	Text Editor
Reason.json	OMS Client Configuration Editor	Project-specific configuration	Reason coded translation item	Coded Translation Editor
StandardMenuHandlerConfig.json	OMS Client Configuration Editor	Default configuration	Table menu handler definitions	Text Editor
UnassociatedCallsLayerConfig.json	OMS Client Configuration Editor	Default configuration	Specifies the name of the geographic geometry that is used in the display call-up URLs. The default name is "detail".	Text Editor
SADDataCorrectionTableConfig.json	OMS Client Configuration Editor	Default configuration	Configuration for Data Correction table	Text Editor

5.2. Customization Examples

5.2.1. Configuring Incident Symbols for Outage Assist

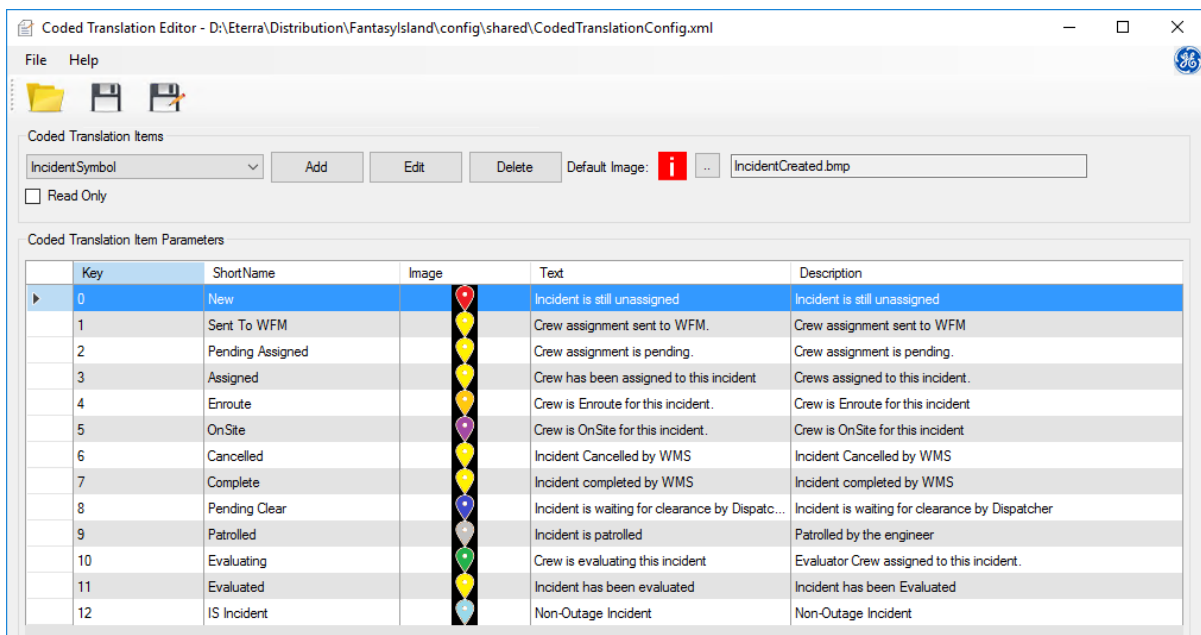
Incident symbols on the Outage Assist network view are defined by the IncidentSymbol.json file. This file includes incident working status configurations from the IncidentSymbol coded translation item and images that correspond to the working statuses.

To configure incident symbols for Outage Assist:

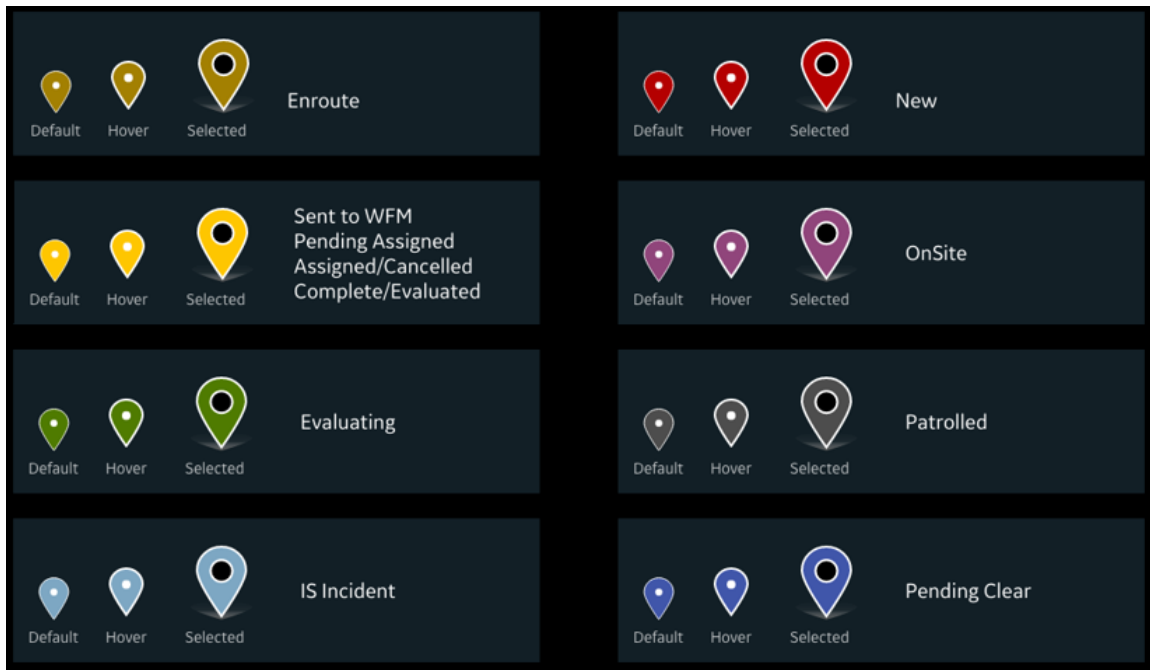
1. Open the Coded Translation Editor.
2. Specify images for incident working statuses using the IncidentSymbol coded translation item.

Note

The image size must not exceed 24x24 px.



3. Save the configuration.
4. Open the ADMS bitmaps folder (by default, %IDMS_ROOT%/bitmaps).
5. Make sure that there are three types of images (default, hovered, and selected) for each incident working status. The image name format is <default_image_name>-<type>.



Note

For Outage Assist, only PNG and SVG image formats are supported.

For example, in ADMS, new incidents use the incident-new2.bmp image. For Outage Assist, the following three images are required for new incidents: incident-new2.svg, incident-new2-hovered.svg, and incident-new2-selected.svg (or the same images in PNG format).

- Export OMS configuration data from ADMS using the OMS Client Configuration Editor.

The incident symbol configurations are saved to the IncidentSymbol.json file.

Note

You can manually define images in the IncidentSymbol.json file using the base64 encoding format.

5.3. Exporting Project-Specific Configuration Data from ADMS

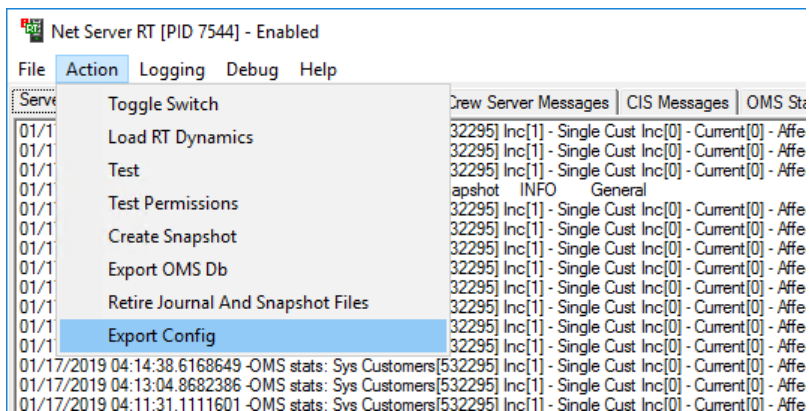
You can export project-specific configuration data using the OMS Client Configuration Editor and NetServer applications.

Important

Before exporting preferences, verify that boilerplates (samples) of the preferences are present in the %IDMS_ROOT%\config\interfaces\StormAssist folder.

To export files using NetServer (OMS):

1. Open NetServer.
2. From the Action menu, select Export Config.



The CrewAreas.json and CrewAreaServiceCenterMap.json files with crew area and crew area service center map data are saved to the %IDMS_ROOT%\config\output folder.

To export files using the OMS Client Configuration Editor:

1. Open the OMS Client Configuration Editor.
2. From the File menu, select Export to JSON Data.

Outage Assist customization files are saved to the %IDMS_ROOT%\config\output folder.

To export files using the OMS Client Configuration Editor command line operations:

1. Open command line.
2. Run the following command:

```
d:\Eterra\Distribution\Tools\bin\OmsConfigClient.exe /EXPORT /BPATH <Path_to_Bitmaps_folder> /CONFIG <Path_to_OMS_Client_Config>
```

For example:

```
d:\Eterra\Distribution\Tools\bin\OmsConfigClient.exe /EXPORT /BPATH %IDMS_ROOT%\bitmaps\ /CONFIG d:\Eterra\Distribution\FantasyIsland\config\client\OmsClientConfig.xml
```

5.4. Updating Outage Assist Customizations

You can update Outage Assist customizations at run time using the OMS Adapter command line.

i Note

Backend services are still using naming convention as UAA but they all have been upgraded to use Zitadel token. Below we are setting up environment variable have 'UAA' but its align to use

Zitadel token

For the steps below, use SAPreferencesSOMS.json for SOMS and SAPreferencesEOMS.json for EOMS.

1. Place Outage Assist customization files in the %IDMS_ROOT%\config\output folder.

For more details, see [Exporting Project-Specific Configuration Data from ADMS](#).

2. Modify the File parameters in the SAPreferencesEOMS.json file to specify files that store Outage Assist preferences for running in the Standard OMS or Enterprise OMS mode. For example:

```
{
  "path": "/grid/sa/appConfig",
  "file": "AppMode.json"
},
```

Where:

- path: The path for the preference to be stored in the Preference service.
 - file: The name of the file that stores preferences for Outage Assist.
3. Make sure that the following environment variables are set so that the OMS Adapter can authenticate with the Outage Assist cluster. Alternatively, set environment variables from command line as follows:

```
set OMSINTERFACESERVER_UAA_AUTH_CLIENTID=oms-interface-service-client-id
set OMSINTERFACESERVER_UAA_AUTH_CLIENTSECRET=oms-interface-service-secret
```

4. Push the preference data to the Outage Assist cluster using the OMS Adapter command-line operations. The format is:

```
OmsAdapter.exe /CONFIG <Path to the OMS Adapter configuration file> /IMPORT /PREFERENCE_SERVICE
<Preference Service URL> /UAA_SERVICE <UAA Service URL> /PREFERENCES_CONFIG <Path to
SAPreferencesEOMS.json>
```

For example:

```
OmsAdapter.exe /CONFIG OMS_ADAPTER_CONFIG /IMPORT /PREFERENCE_SERVICE
https://10.227.51.59/preferences /UAA_SERVICE https://10.227.51.59/uaa/oauth/token
/PREFERENCES_CONFIG D:\Eterra\Distribution\FantasyIsland\config\server\SAPreferencesEOMS.json
```

The customizations are exported to Outage Assist's Preference Service and applied to the Outage Assist user interface at run time.

Note

Outage Assist supports either EOMS or SOMS mode. So use the SA preferences json file accordingly to the mode you are using.

6. Localizing Outage Assist

You can update the labels on the UI or show them in a different language. Here are the steps:

6.1. Configure OA Headers

- Edit the provided file `gx-sa-ui.en-US.xlf` and update the `<target state="new">` value for each of the header fields you want to edit.

```
<trans-unit id="gx-sa-ui-incident-detail-overview-failed-equipment" datatype="html">
  <source>FAILED EQUIPMENT</source>
  <target state="new">EQUIPMENT CODE</target>
```

- Save the file.
- Upload the file to your deploy server.
- Ensure you have exported the correct kubeconfig for the cluster you want to update:

```
export KUBECONFIG=<path to admin.conf file>
```

- Create configmap storing the localization file:

```
kubectl create configmap oa-localization -n oms --from-file ./gx-sa-ui.en-US.xlf
```

```
[root@duke-predeploy Localizing OA]# kubectl create configmap oa-localization -n oms --from-file ./gx-sa-ui.en-US.xlf
configmap/oa-localization created
```

- Mount the `oa-localization` configmap to the Outage Assist container by adding the following to the storm-assist deployment yaml file:


```
spec:
  volumes:
    - name: oa-localization
      configMap:
        name: oa-localization
        defaultMode: 420
  containers:
    - name: storm-assist
      image: >=
        dig-grid-artifactory.apps.ge.com/foundation-docker/deui-storm-assist:16.7.0-rc.3
      ports:
        - name: http
          containerPort: 8080
          protocol: TCP
      resources: {}
      volumeMounts:
        - name: oa-localization
          readOnly: true
          mountPath: /etc/locale
      livenessProbe:
        httpGet:
          path: /
          port: http
          scheme: HTTP
          timeoutSeconds: 1
          periodSeconds: 10
          successThreshold: 1
          failureThreshold: 3
      readinessProbe:
        httpGet:
          path: /
          port: http
          scheme: HTTP
          timeoutSeconds: 1
          periodSeconds: 10
          successThreshold: 1
          failureThreshold: 3
      terminationMessagePath: /dev/termination-log
      terminationMessagePolicy: File
      imagePullPolicy: Always
```

- Save deployment.yaml.
- Delete the storm assist pod.

^ Responsible System & Cause

CAUSE	RESPONSIBLE SYSTEM	EQUIPMENT CODE
Add cause	None	▼ None

^ Restoration / Completion

INCIDENT TYPE	POWER QUALITY TYPE	WORK QUEUES
Not Selected	▼ None	▼ None

WEATHER	ACTION TAKEN	FAILURE MODE
Calm	▼ None	▼ None

PLANNED OUTAGE	MANUFACTURER
None	▼ None

- The localizing labels are persistent.

```

Terminal x Pod storm-assist-6c7676dd9-n7j9v x +
Namespace oms Owner ReplicaSet storm-assist-6c7676dd9 Pod storm-assist-6c... Container storm-assist
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/localize.sh
Localization files copied successfully!
/docker-entrypoint.sh: Configuration complete; ready for start up

```

6.2. Update OA Headers

- Edit the provided file `gx-sa-ui.en-US.xlf` and update the `<target state="new">` value for each of the header fields you want to edit:

```

[root@duke-predeploy localizing-oa]# kubectl create configmap oa-localization -n oms --from-file ./gx-sa-ui.en-US.xlf -o yaml --dry-run=client | kubectl apply -f -
Warning: resource configmaps/oa-localization is missing the kubect1.kubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply should only be used on resources created declaratively by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.
configmap/oa-localization configured

```

-

Save the file.

- Upload the file to your deploy server.
- Ensure you have exported the correct kubeconfig for the cluster you want to update:

```
export KUBECONFIG=<path to admin.conf file>
```

- Execute:

```
kubectl create configmap oa-localization -n oms --from-file ./gx-sa-ui.en-US.xlf -o yaml --dry-run=client | kubectl apply -f -
```

- Delete the storm assist pod.
- Confirm changes applied to app.

7. Configuring ADMS

! Important

It is assumed that you already have the ADMS system with the Enterprise OMS module up and running. For more information, refer to the *ADMS Software Installation and Maintenance Guide* and *Enterprise OMS Software Installation and Maintenance Guide*.

7.1. Configuring the ingress.local Path

i Note

Skip this step if you have already performed the configuration on the DNS server and your hosts already have DNS names.

To enable access to your Kubernetes master IP address using the ingress.local alias:

1. Open the hosts file (C:\Windows\System32\drivers\etc\hosts) using a text editor that is run with Administrator rights.
2. Add the following entry to the hosts file:

```
<Kubernetes master IP address> ingress.local service.ingress.local zitadel.ingress.local  
admin.ingress.local a6dashboard.ingress.local
```

After this configuration, you can use ingress.local instead of your Kubernetes host name in all configurations. After migrating to Zitadel UAA, the URL changes to <https://zitadel.ingress.local/oauth/v2/token> and the service is accessed via <https://service.ingress.local>.

7.2. Importing Certificates to the ADMS Hosts

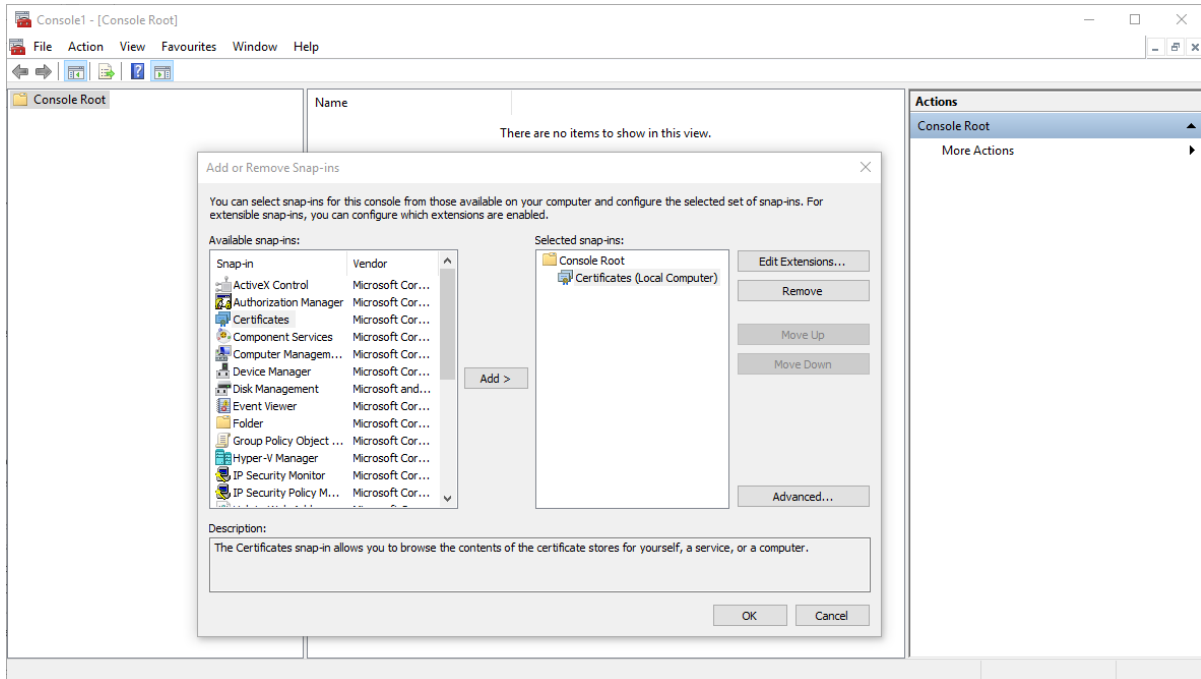
! Important

Perform the following steps on each host where DMS Interface Server, OMS Interface Server, and NetViewer Adapter are running.

To import certificates to an ADMS interface server host:

1. Copy the Root certificate authority certificate (rootCA.crt) and the ADMS server PFX file with a certificate and the associated private key (adms-server.pfx) to the ADMS machine.
2. Open the Microsoft Management console.
3. From the File menu, select Add/Remove Snap-in.
4. From the available snap-ins, select Certificates and click Add.

5. Select the Computer Account option and click Next.
6. Select the Local Computer option and click Finish.

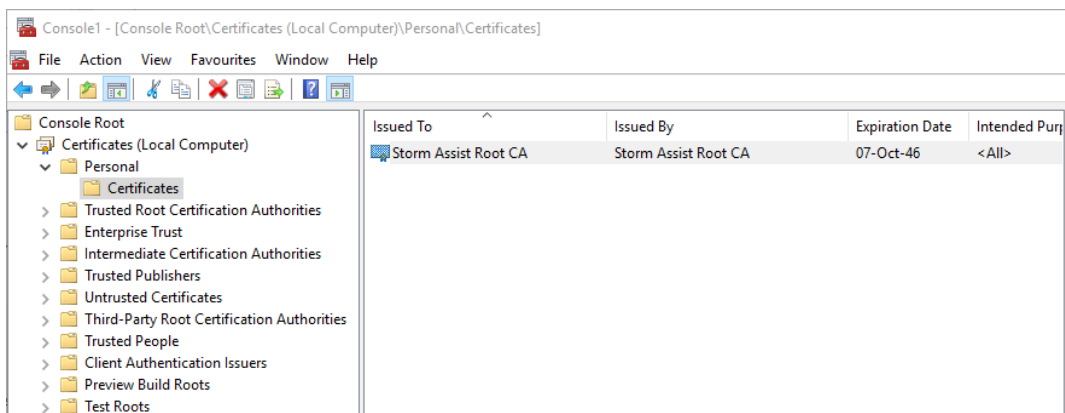


7. Click OK.
8. In Console Root, expand the Certificates (Local Computer) list.
9. Right-click Personal, select All Tasks, and click Import.

The Certificate Import wizard appears.

10. Verify that the Store location is Local Machine and click Next.
11. Click Browse, select the rootCA.crt file to import, and click Next.
12. Select the Place All Certificates in the Following Store option, select Personal, and click Next.
13. Click Finish.

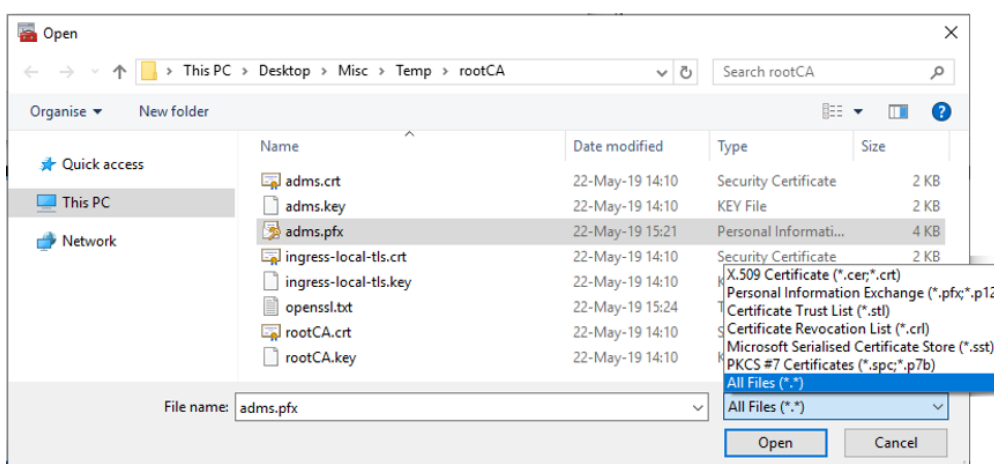
The root certificate appears in the Certificates list.



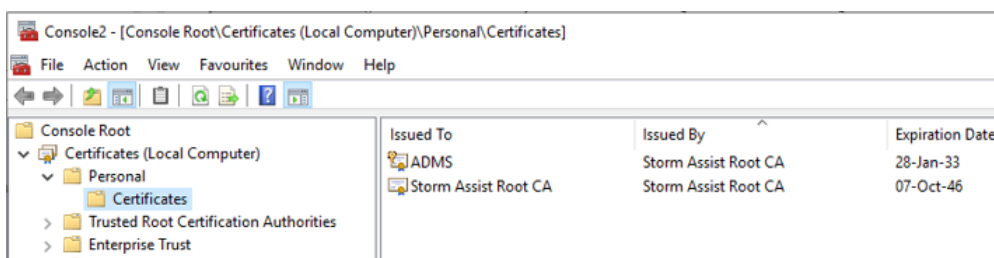
14. Perform the same operations to import the ADMS PFX file with a certificate and the associated private key (adms-server.pfx). Enter the password that you used when generating the certificate (see [Generating Certificates for SSL/TLS Termination with ADMS](#)).

Note

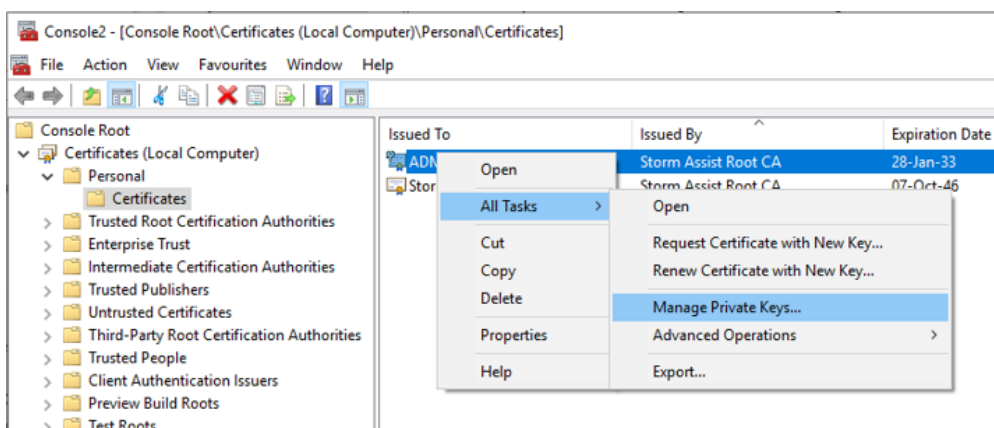
To browse for a .pfx file, select the Personal Information Exchange or All Files filter in the Open dialog box.



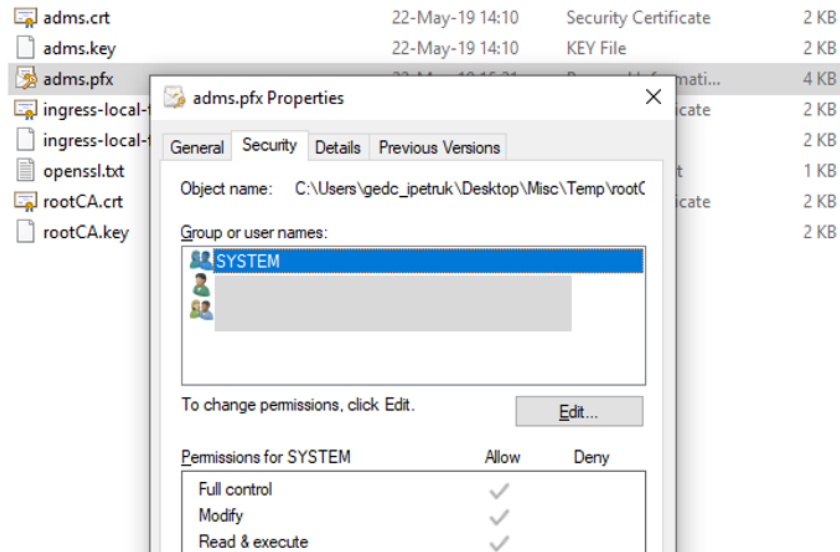
15. Verify that the Root certificate authority certificate (Storm Assist Root CA) and the ADMS server certificate (ADMS) are listed in the Personal folder under Certificates (Local Computer).



16. Right-click the ADMS server certificate (ADMS), point to All Tasks, and click Manage Private Keys.



Alternatively, open the Security tab of the certificate Properties display.

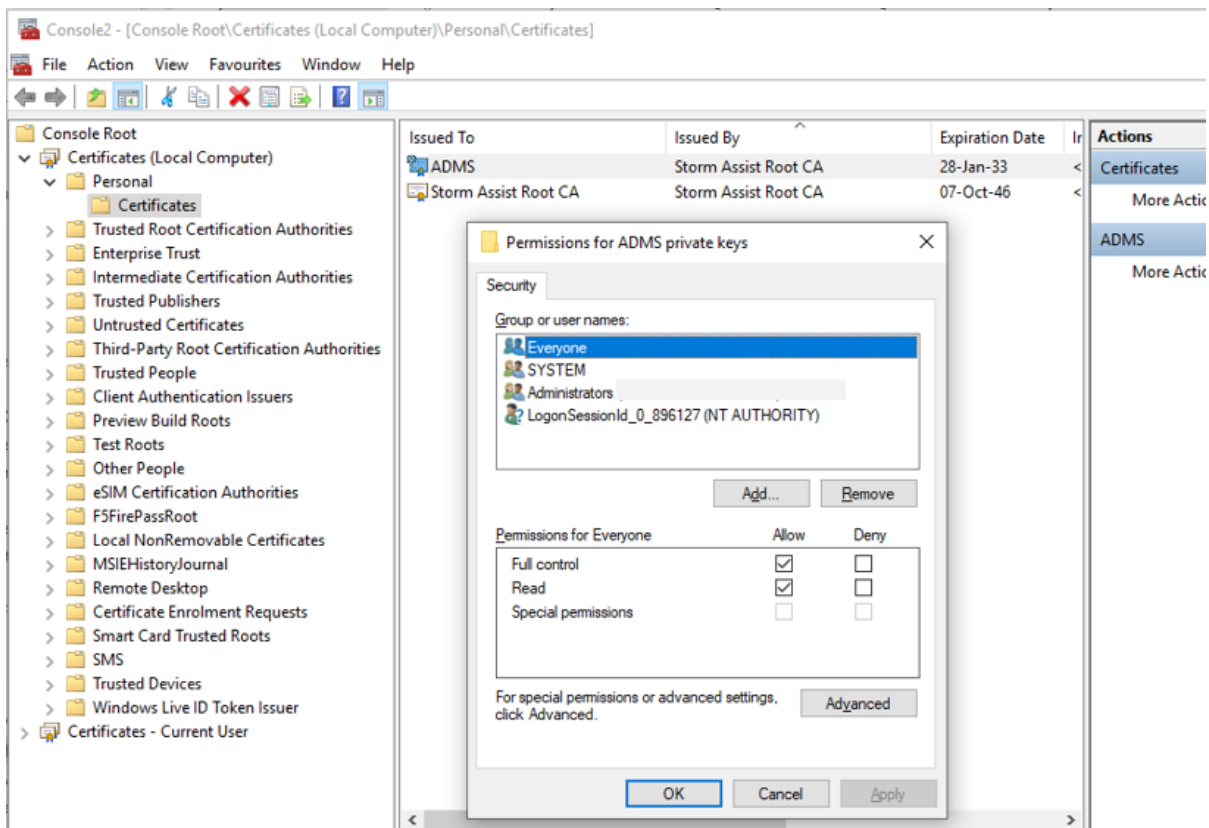


17. Add the user or service account that is used to run NetViewer Adapter (or select Everyone, which is less secure) and give Read permissions.

To read from Trusted Root Certification Authorities, permission must be granted for users that do not have read permissions.

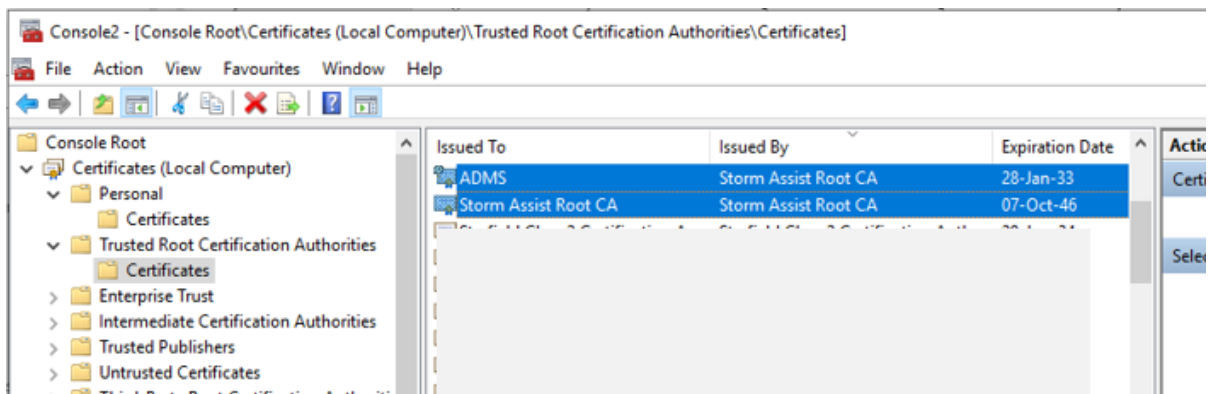
! Important

Make sure you are giving permissions to the private key, not the public key.



This allows access to the private key of the imported certificate to the user (service account) that is used to run the NetViewer Adapter. Otherwise, the NetViewer Adapter is not able to serve data in HTTPS mode.

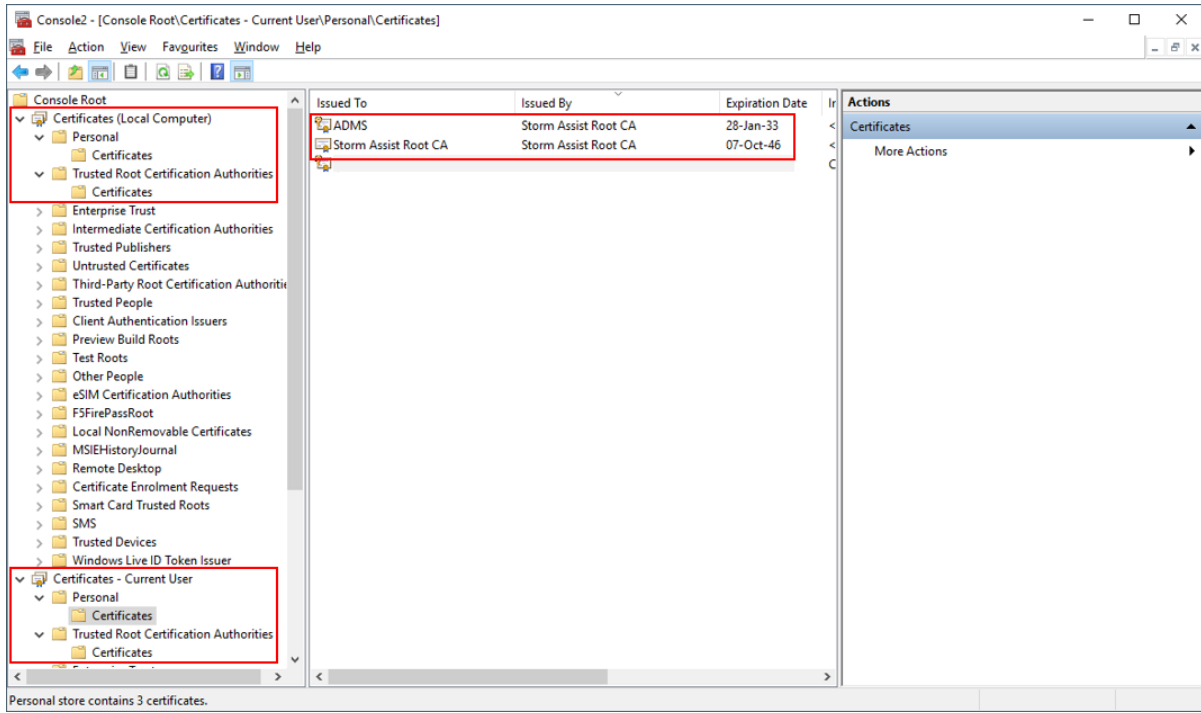
18. Restart the NetViewer Adapter if it was already running.
19. In the Microsoft Management Console, copy the Root certificate authority certificate (Storm Assist Root CA) and the ADMS server certificate (ADMS) certificate using the context menu.
20. Navigate to the Certificates > Trusted Root Certification Authorities > Certificates folder.
21. Paste the Storm Assist Root CA and ADMS certificates using the context menu.



22. Add or copy the Storm Assist Root CA and ADMS certificates to the Certificates - Current User

store.

This step is optional, but should be done if you do not bind a certificate to a port as described further.



7.3. Binding Certificates to the ADMS Interface Servers

! Important

Perform the following steps on each server where OMS Interface Server and DMS Interface Server are running.

To bind certificates to the ADMS interface servers:

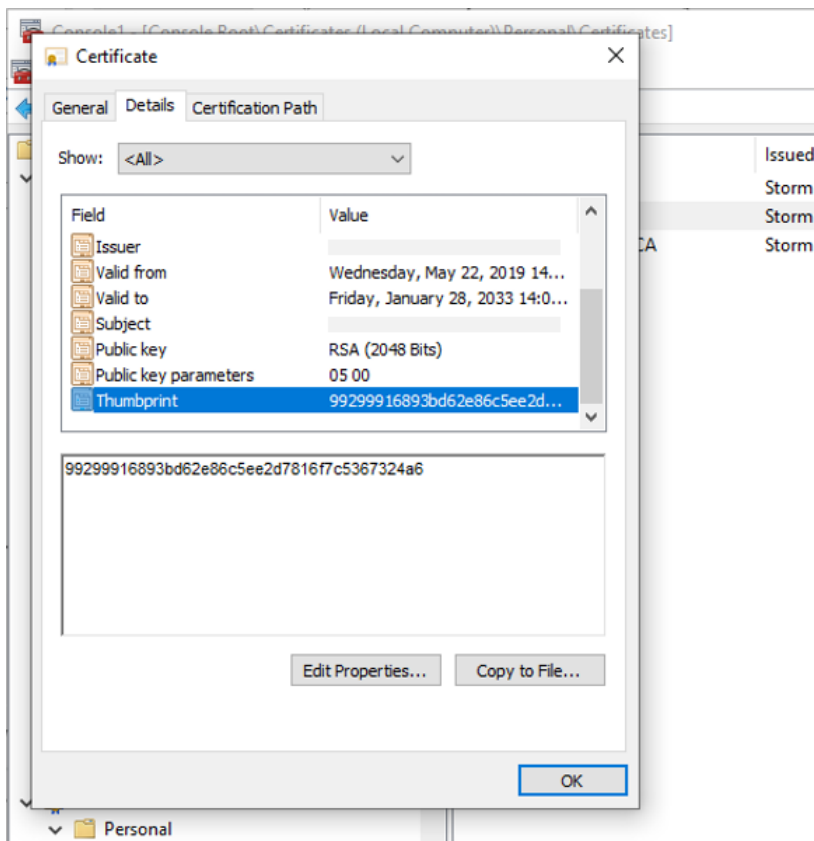
1. In the Microsoft Management Console, double-click the ADMS certificate.

The certificate properties window appears.

2. On the Details tab, copy the Thumbprint value.

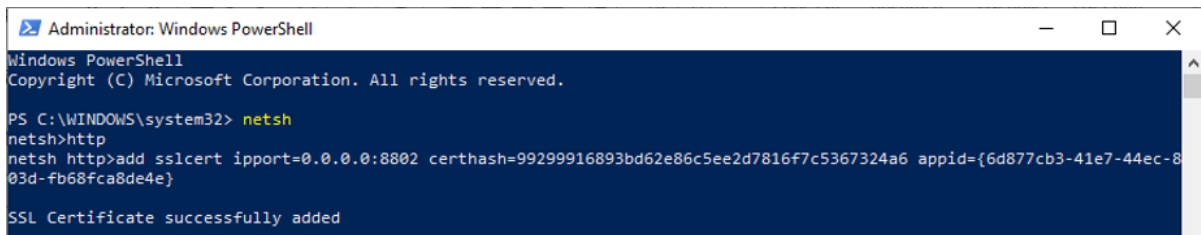
! Important

Delete all spaces in the copied string, if any.



3. Run Windows PowerShell as an administrator.
4. Run the netsh command.
5. Run the http command.
6. Run the following command to bind the ADMS SSL certificate to the OMS Interface Server web service ports (8802 and 8812):

```
netsh http>add sslcert ipport=0.0.0.0:8802 certhash=<Cert_Thumbprint> appid=<Application_GUID>
netsh http>add sslcert ipport=0.0.0.0:8812 certhash=<Cert_Thumbprint> appid=<Application_GUID>
```



Note

This action fails if a certificate is already bound to the specified port.

7. Run the following command to bind the ADMS SSL certificate to the DMS Interface Server web service port (8801):

```
netsh http>add sslcert ipport=0.0.0.0:8801 certhash=<Cert_Thumbprint> appid=<Application_GUID>
```

8. Verify the certificate port binding configuration using the following command:

```
netsh http> show sslcert
```

7.4. Configuring Main Servers

! Important

Perform the following steps on each server where the referenced applications are running.

i Note

It is assumed that the ADMS Server and ADMS Interfaces kits are already deployed, and the Outage Assist and Foundation ports are open in Windows firewall.

i Note

Backend services are still using naming convention as UAA but they all have been upgraded to use Zitadel token. Below configuration files still have 'UAA' but its actually using Zitadel token/secret.

Configuring the NetServerOmsConfig.txt file

NetServer needs to access the Zitadel server to communicate with the AOR Manager application in the Outage Assist's Kubernetes cluster to get AOR data for Outage Assist.

To configure the %IDMS_ROOT%\config\server\NetServerOmsConfig.txt file:

1. Specify the Zitadel server endpoint.

```
PERMISSION_UAA_END_POINT = https://zitadel.ingress.local/oauth/v2/token
```

2. Specify the value that defines the name of the user permission to write data to the NetServer.

```
PERMISSION_READ_VALUE = Write
```

3. Specify the value that defines the name of the user permission to read data from the NetServer.

```
PERMISSION_READ_VALUE = Read
```

4. Specify the AOR Manager API version.

```
PERMISSION_API_VERSION = v3
```

5. Specify the values that define the functional permissions that Outage Assist users must have for managing outage incidents and operating protective devices.

```
DISPATCH_FUNCTION_AOR_NAME = DISPATCH  
OUTAGE_INCIDENT_FUNCTIONAL_PERMISSION_NAME = OUTAGE
```

6. Enable writing the incident locking information to the Incident journal.

```
OMS_SPECIAL_LOCK_PROCESSING = false
```

Configuring the NetViewerAdapterConfig.xml file (Main Server, Map and Query Modes)

To configure the %IDMS_ROOT%\config\interfaces\NetViewerAdapterConfig.xml file:

1. Enable a secure web service:

```
<SecureWS>true</SecureWS>
```

2. Set certificate subject matching O=GE of the ADMS CRT certificate (adms-server.crt/adms-server.pfx):

```
<CertificateSubject>GE</CertificateSubject>
```

3. Specify the listening port.

```
<Port>24468</Port>
```

4. Disable Common Notification Framework:

```
<DisableCnf>true</DisableCnf>
```

5. Enable pushing tiles to the Tile service.

```
<EnablePublishingToTileService>true</EnablePublishingToTileService>
```

6. Set the Tile service compression to Brotli.

```
<TileServiceCompression>br</TileServiceCompression>
```

7. Set a secured connection to the Tile service:

```
<TileServiceUrl>wss://service.ingress.local/tilecacheservice</TileServiceUrl>
```

8. Set a zitadel endpoint:

```
<UaaTokenServer>https://zitadel.ingress.local/oauth/v2/token</UaaTokenServer>
```

9. Create the following system environment variables to provide the Zitadel access token required by the NetViewer Adapter. You can obtain these values from the Kubernetes deployment:

```
NETVIEW_UAA_AUTH_PROJECTID
NETVIEW_UAA_AUTH_CLIENTID
NETVIEW_UAA_AUTH_CLIENTSECRET

# Get the Zitadel Project ID
kubectl get configmap -n foundation-cluster-zerotrust zitadel-projects -o
jsonpath='{.data.foundation\.foundation\.id}'

# Get NetView UAA Client ID and Secret
kubectl get secret netview-adapter-client-secret -n oms -o jsonpath='{.data.clientId}' | base64
--decode
kubectl get secret netview-adapter-client-secret -n oms -o jsonpath='{.data.clientSecret}' |
base64 --decode
```

10. Configure NetViewer Adapter running modes:

```
<ConfigMode>Normal</ConfigMode>
<CustomerMode>Normal</CustomerMode>
<DnomMode>Normal</DnomMode>
<DocumentMode>Normal</DocumentMode>
<LandbaseMode>Normal</LandbaseMode>
<OverviewMode>Normal</OverviewMode>
```

11. Configure connection properties:

```
<DmsInterfaceServerUrl>https://<Main_Server_IP>:8801</DmsInterfaceServerUrl>
<OmsServerIPs><Main_Server_IP></OmsServerIPs>
<OmsServerPort>5050</OmsServerPort>
<HostName><Main_Server_IP></HostName>
<HostPort>5025</HostPort>
```

12. Configure connection properties:

```
<WorkModelWebDbEnabled>1</WorkModelWebDbEnabled>
<WorkModelServiceApi>http://<EOMS_Server_IP>:8080</WorkModelServiceApi>
<WorkModelEventApi>ws://<EOMS_Server_IP>:8083/oms/api/v1/events</WorkModelEventApi>
```

13. Enable DNAF result queries.

```
<EnabledDNAFQueries>true</EnabledDNAFQueries>
```

14. Disable DNOM device searching and data querying.

```
DNOMQueriesInterfaceServer>>false</DNOMQueriesInterfaceServer>
```

15. Save the file.

For more details, see appendix 9.1 NetViewer Adapter Configuration File.

Configuring the NetDataReplicatorConfig.xml file (Main Server)

To configure the %IDMS_ROOT%\config\server\NetDataReplicatorConfig.xml file:

1. Specify the connection properties:

```
<LocalListenerAddress>localhost:8200</LocalListenerAddress>  
<RemoteListenerAddresses><Read_Only_Server_IP>:8201</RemoteListenerAddresses>
```

2. Make sure that the OMS data is replicated:

```
<ReplicationQueryIntervalSeconds>5</ReplicationQueryIntervalSeconds>  
<ReplicateIncidentJournal>true</ReplicateIncidentJournal>  
<ReplicateOrderJournal>true</ReplicateOrderJournal>  
<ReplicateRealTimeIncidentJournal>true</ReplicateRealTimeIncidentJournal>  
<ReplicateCrewJournal>true</ReplicateCrewJournal>  
<ReplicateDailySnapshots>true</ReplicateDailySnapshots>  
<ReplicationDailySnapshotDaysRequired>7</ReplicationDailySnapshotDaysRequired>
```

Configuring the NetDataServerConfig.txt file

To configure the %IDMS_ROOT%\config\server\NetDataServerConfig.txt file:

- Specify the number of daily snapshots required by the system:

```
OMS_NUMBER_OF_DAILY_SNAPSHOTS_REQUIRED = 7
```

Note that daily snapshots are required for Net Data Replicator.

Configuring credentials to access the Kubernetes cluster

For NetServer to access the UAA server in the Kubernetes cluster, add environment variables that define service connection credentials:

1. Open a command prompt.
2. Get the Netserver UAA credentials by executing kubernetes command and add environment variable for NetServer.

```
SETX NETSERVER_UAA_AUTH_CLIENTID net-server-main-client-id  
SETX NETSERVER_UAA_AUTH_CLIENTSECRET net-server-main-secret  
SETX NETSERVER_UAA_AUTH_PROJECTID {VALUE_of_Cluster_Project_ID}  
  
# Get NetServer UAA Client ID, Secret, and Project ID  
kubectl get secret net-server-main-secret -n oms -o jsonpath='{.data.clientId}' | base64 --decode  
kubectl get secret net-server-main-secret -n oms -o jsonpath='{.data.clientSecret}' | base64  
--decode  
kubectl get configmap -n foundation-cluster-zerotrust zitadel-projects -o  
"jsonpath={.data.foundation\.foundation\.id}"
```

Configuring the interface_OmsInterfaceServerConfig_UAA.xml file

To configure the %IDMS_ROOT%\config\interfaces\interface_OmsInterfaceServerConfig_UAA.xml file:

1. Specify the DMS Interface Server connection URL:

```
<DMSInterfaceServerURL>https://<Main_Server_Host>:8801/dms/services/</DMSInterfaceServerURL>
```

2. Enable the secure web service:

```
<SecureWebService>true</SecureWebService>
```

3. Enable UAA and set a UAA endpoint:

```
<UseUAAToken>true</UseUAAToken>  
<UaaEndpoints>https://zitadel.ingress.local/oauth/v2/token</UaaEndpoints>
```

4. Enable a web socket security web service:

```
<WSSecurityEnabled>true</WSSecurityEnabled>
```

5. Specify the AOR Manager API version.

```
<AorApiVersion>v3</AorApiVersion>
```

6. Save the file.

For more details, see appendix 9.2 OMS Interface Server Configuration.

Configuring the interface_OmsInterfaceServerConfig.xml file:

This configuration only applies when EOMS services are run using PROCMAN.

To configure the %IDMS_ROOT%\config\interfaces\interface_OmsInterfaceServerConfig.xml file:

1. Specify the DMS Interface Server connection URL:

```
<DMSInterfaceServerURL>https://<Main_Server_Host>:8801/dms/services/</DMSInterfaceServerURL>
```

2. Edit the default port values:

```
<WebServiceBaseURL>http://<Main_Server_Host>:8802/</WebServiceBaseURL>  
<UriPort>18802</UriPort>
```

3. Disable UAA:

```
<UseUAAToken>false</UseUAAToken>
```

4. Save the file.

For more details, see appendix 9.2 OMS Interface Server Configuration.

Configuring the work-model-eventhandler.properties file

When using two instances of the OMS Interface Server (with and without UAA support), modify the work-model-eventhandler.properties file to reference the two OMS Interface Servers. This configuration only applies when EOMS services are run using PROCMAN.

To configure the d:\Eterra\Distribution\WorkModelServices\config\work-model-eventhandler.properties file:

1. Modify the "ois" parameter value:

```
ois=http://localhost:8802
```

2. Add the "oisuaa" parameter:

```
oisuaa=https://localhost:8812
```

Configuring credentials to access the Kubernetes cluster

For the OMS Interface Server and NetViewer Adapter to access the Kubernetes cluster, add environment variables that define service connection credentials:

1. Open a command prompt.
2. Execute the following commands to add environment variables for the applications.

```
SETX OMSINTERFACESERVER_UAA_AUTH_CLIENTID oms-interface-service-client-id
SETX OMSINTERFACESERVER_UAA_AUTH_CLIENTSECRET oms-interface-service-secret
SETX OMSINTERFACESERVER_UAA_AUTH_PROJECTID {VALUE_of_Cluster_Project_ID}

SETX NETVIEW_UAA_AUTH_CLIENTID netview-adapter-client-id
SETX NETVIEW_UAA_AUTH_CLIENTSECRET netview-adapter-client-secret
SETX NETVIEW_UAA_AUTH_PROJECTID {VALUE_of_Cluster_Project_ID}

# Get OMS Interface UAA Client ID and Secret
kubectl get secret oms-interface-service-secret -n oms -o jsonpath='{.data.clientId}' | base64
--decode
kubectl get secret oms-interface-service-secret -n oms -o jsonpath='{.data.clientSecret}' | base64
--decode
# Get NetView Adapter UAA Client ID and Secret
kubectl get secret netview-adapter-client-secret -n oms -o jsonpath='{.data.clientId}' | base64
--decode
kubectl get secret netview-adapter-client-secret -n oms -o jsonpath='{.data.clientSecret}' |
base64 --decode
```

7.5. Configuring Read-Only Servers

Important

Perform the following steps on each server where the referenced applications are running.

i Note

It is assumed that the ADMS Server and ADMS Interfaces kits are already deployed and the Outage Assist and Foundation ports are open in Windows firewall.

i Note

Backend services are still using naming convention as UAA but they all have been upgraded to use Zitadel token. Below configuration files still have 'UAA' but its actually using Zitadel token/secret.

Configuring the NetServerReadOnlyConfig.txt file

NetServerRO needs to communicate with Enterprise OMS services to get EOMS data for Outage Assist. NetServerRO also needs to access the UAA server to communicate with the AOR Manager application in the Outage Assist's Kubernetes cluster to get AOR data for Outage Assist.

To configure the %IDMS_ROOT%\config\server\NetServerReadOnlyConfig.txt file:

1. Modify configuration parameters as follows:

```
MAIN_SERVER_HOST = <Main_Server_IP>
MAIN_SERVER_PORT = 5025
DNAF_SERVER_HOST = <Main_Server_IP>
DNAF_SERVER_PORT = 5026

PERMIT_SERVER_IP = <Main_Server_IP>

WORKMODEL_WEBDB_ENABLED = 1
WORKMODEL_SERVICE_API = http://<EOMS_Server_IP>:8080/
WORKMODEL_EVENT_API = ws://<EOMS_Server_IP>:8083/oms/api/v1/events
WORKMODEL_GET_ORDERS_PAGE_SIZE = 1000

PERMISSION_UAA_END_POINT = https://zitadel.ingress.local/oauth/v2/token
PERMISSION_READ_VALUE    = Read
PERMISSION_WRITE_VALUE   = Write
PERMISSION_API_VERSION   = v3
PERMISSION_REFRESH_INTERVAL = 10

OMS_QUERY_THREADS = 64
```

i Note

The NetServerRO needs Permit and Permit API parameters to obtain permission data.

Configuring the NetViewerAdapterConfig.xml file (Read-Only Server, Query Mode)

To configure the %IDMS_ROOT%\config\interfaces\NetViewerAdapterConfig.xml file:

1. Enable the secure web service:

```
<SecureWS>true</SecureWS>
```

2. Set certificate subject matching O=GE of the ADMS CRT certificate (adms-server.crt/adms-server.pfx):

```
<CertificateSubject>GE</CertificateSubject>
```

3. Specify the listening port.

```
<Port>24468</Port>
```

4. Disable Common Notification Framework:

```
<DisableCnf>true</DisableCnf>
```

5. Disable pushing tiles to the Tile service.

```
<EnablePublishingToTileService>false</EnablePublishingToTileService>
```

6. Set a UAA endpoint:

```
<UaaTokenServer>https://zitadel.ingress.local/oauth/v2/token</UaaTokenServer>
```

7. Configure NetViewer Adapter running modes:

```
<ConfigMode>Disabled</ConfigMode>  
<CustomerMode>QueryOnly</CustomerMode>  
<DnomMode>QueryOnly</DnomMode>  
<DocumentMode>Disabled</DocumentMode>  
<LandbaseMode>Disabled</LandbaseMode>  
<OverviewMode>Disabled</OverviewMode>
```

8. Configure connection properties:

```
<DmsInterfaceServerUrl>http://<Main_Server_IP>:8801</DmsInterfaceServerUrl>  
<DmsServerIPs><Read_Only_Server_IP></DmsServerIPs>  
<DmsServerPort>5051</DmsServerPort>  
<HostName><Read_Only_Server_IP></HostName>  
<HostPort>5055</HostPort>
```

9. Disable DNAF result queries.

```
EnableDNAFQueries>false</EnableDNAFQueries>
```

10. Enable DNOM device searching and data querying.

```
DNOMQueriesInterfaceServer>true</DNOMQueriesInterfaceServer>
```

11. Save the file.

For more details, see appendix [NetViewer Adapter Configuration File](#).

Configuring the NetDataReplicatorReadOnlyConfig.xml file (Read-Only Server)

To configure the %IDMS_ROOT%\config\server\NetDataReplicatorReadOnlyConfig.xml file:

1. Disable using role from CfgProxy:

```
<UseRoleFromCfgProxy>false</UseRoleFromCfgProxy>
```

2. Make sure that the mode is set to Read Only:

```
<ReadOnlyMode>true</ReadOnlyMode>
```

3. Specify the connection properties:

```
<LocalListenerAddress>localhost:8201</LocalListenerAddress>
<RemoteListenerAddresses><Main_Server_IP>:8200</RemoteListenerAddresses>
<OmsServerHost>localhost</OmsServerHost>
<OmsServerPort>5051</OmsServerPort>
```

4. Make sure that the OMS data is replicated:

```
<ReplicationQueryIntervalSeconds>5</ReplicationQueryIntervalSeconds>
<ReplicateArchiveJournal>false</ReplicateArchiveJournal>
<ReplicateCallServerJournal>false</ReplicateCallServerJournal>
<ReplicateIncidentJournal>true</ReplicateIncidentJournal>
<ReplicateOrderJournal>true</ReplicateOrderJournal>
<ReplicateRealTimeIncidentJournal>true</ReplicateRealTimeIncidentJournal>
<ReplicateCrewJournal>true</ReplicateCrewJournal>
<ReplicateDailySnapshots>true</ReplicateDailySnapshots>
<ReplicationDailySnapshotDaysRequired>7</ReplicationDailySnapshotDaysRequired>
```

Configuring credentials to access the Kubernetes cluster

For NetServer Read-Only to access the UAA server in the Kubernetes cluster, add environment variables that define service connection credentials:

Execute the following commands to add environment variables for NetServer.

```
# Get NetServer UAA Client Readonly ID and Secret from kuberntes.
kubectl get secret net-server-readonly-secret -n oms -o jsonpath='{.data.clientId}' | base64 --decode
kubectl get secret net-server-readonly-secret -n oms -o jsonpath='{.data.clientSecret}' | base64
--decode
```

```
SETX NETSERVER_UAA_AUTH_CLIENTID net-server-readonly-client-id
SETX NETSERVER_UAA_AUTH_CLIENTSECRET net-server-readonly-secret
```

```
SETX NETSERVER_UAA_AUTH_PROJECTID {VALUE_of_Cluster_Project_ID}
```

7.5.1. Read-Only Data Load Balancing

Outage Assist supports scaling the system setup depending on the expected system load. A simple setup with one Netviewer Adapter and one NetServer Read Only is applicable to a non-production testing or development environment. A production or performance testing environment must use load balancing capabilities to route concurrent client connections to multiple NetServer Read Only instances.

The Apache HTTP Server application is a recommended tool for the load balancing setup. Apache HTTP Server must be configured to be an intermediate layer between the Netviewer Adapter Read-Only and Outage Assist's Kubernetes cluster External Gateway component. Apache HTTP Server must host HTTPS certificates to secure incoming connections from Outage Assist's Kubernetes cluster. Outage Assist uses the Incremental Protocol implementation. Therefore, for the load balancing setup, all Apache HTTP Server instances must use identical configuration files with all Netviewer Adapter Read-Only instances referenced in the route configuration.

One instance of NetServer Read Only with 4 instances of Netviewer Adapter Read-Only can serve about 300 concurrent Outage Assist clients. Depending on the available hosts and resources, the applications can run on the same host or on separate hosts.

It is recommended that you allocate resources as follows:

- One instance of Netviewer Adapter Read-Only requires 1.5 logical processors.
- One instance of Netviewer Adapter Read-Only with the demo Fantasy Island model requires 800-MB of RAM when running idle, with no load from connected Outage Assist clients.
- One instance of Netviewer Adapter Read-Only with the demo Fantasy Island model requires 2.2-GB of RAM when running under load from connected Outage Assist clients.
- One NetServer Read Only instance requires 2 logical processors.
- In the demo Fantasy Island model run with 300 concurrent Outage Assist clients, one NetServer Read-Only instance consumes 4-GB of RAM.
- One Apache HTTP Server instance requires 2 logical processors.

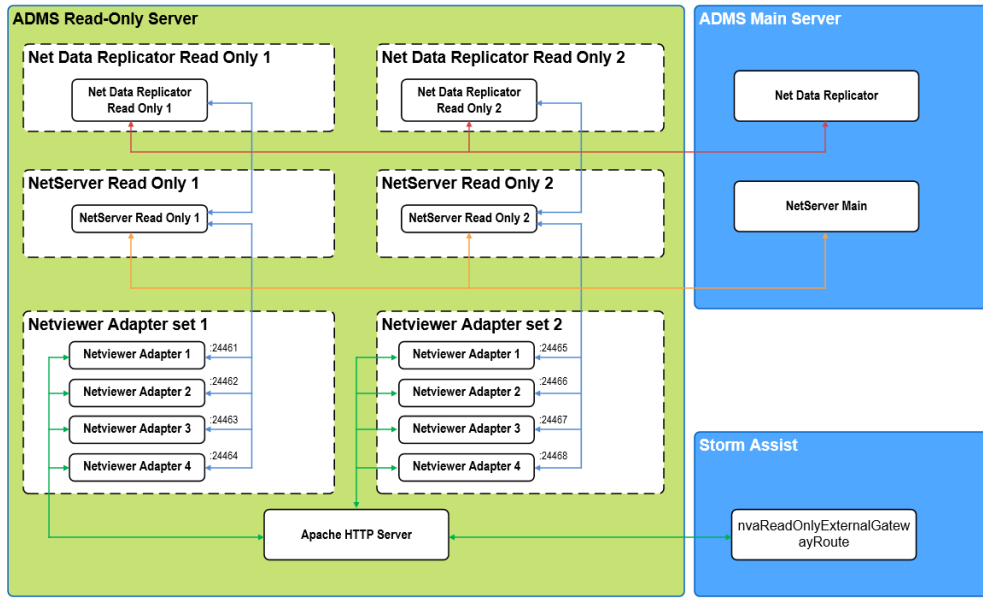


Figure 7. Read-Only Data Load Balancing Schema

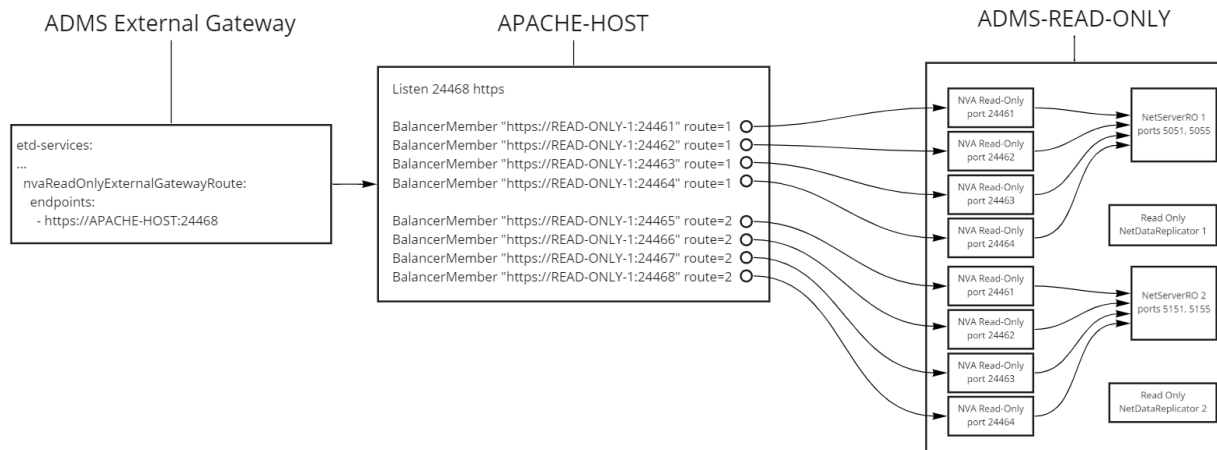


Figure 8. Example of Read-Only Data Load Balancing Running on One Apache Host and One Read-Only Host

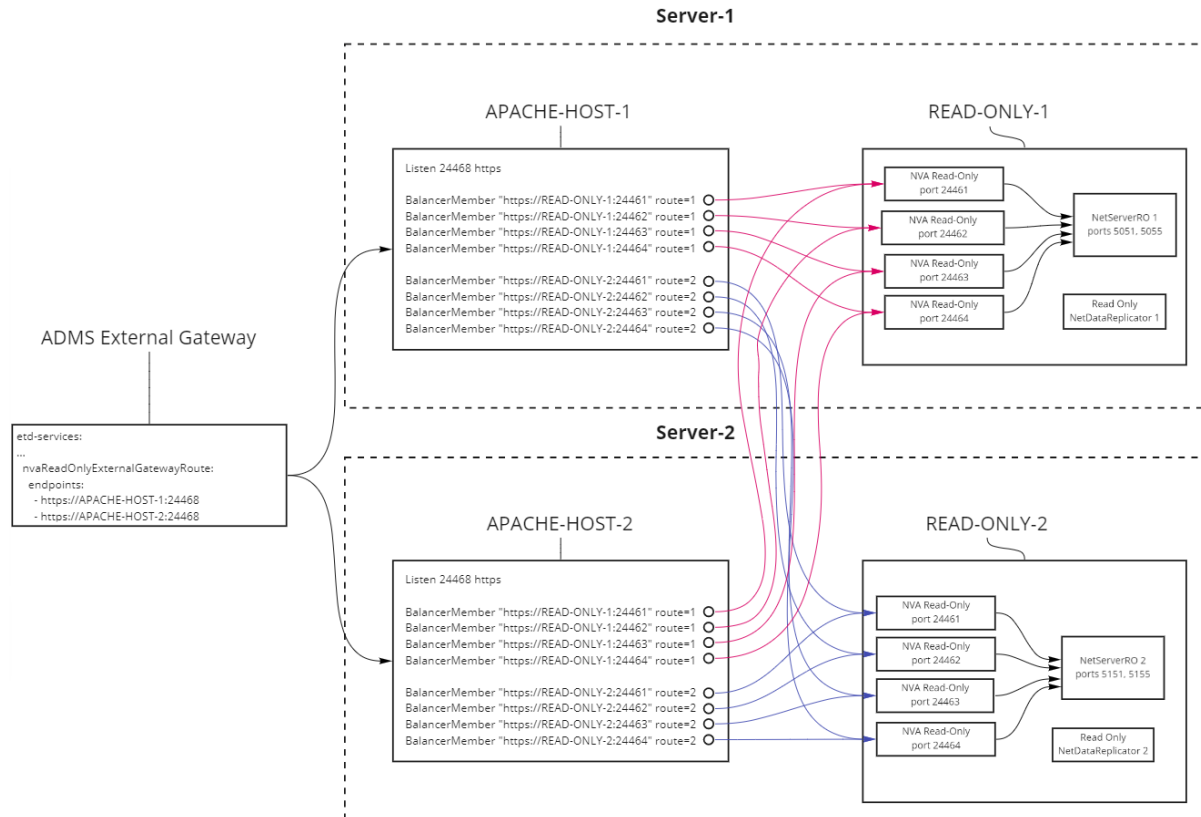


Figure 9. Example of Read-Only Data Load Balancing Running on Two Hosts with One Read-Only and One Apache Server

Example System Setup

- 2 hosts with 16-CPU threads and 1-Gbp/s network bandwidth capability each.
- 1 Apache HTTP Server on each host.
- 1 NetServer Read-Only instance on each host.
- 4 Netviewer Adapter Read-Only instances on each host. Each set of 4 Netviewer Adapter instances is configured to connect to a separate NetServer Read-Only instance.
- Net Data Replicator Read-Only instance on each host.

Pre-configuration

- Create 8 NetViewer Adapter configuration files (for example, NetViewerAdapterConfig-1.xml - NetViewerAdapterConfig-8.xml).
- Create 8 NetViewer Adapter shortcuts to use the above configuration files.
- Create folders for NetViewer Adapter dynamics (for example, %IDMS_ROOT%\dynamics\server\dmsaccess-24461 - %IDMS_ROOT%\dynamics\server\dmsaccess-24468)
- Create 2 NetServer Read Only configuration files (for example, NetServerReadOnlyConfig-1.txt and NetServerReadOnlyConfig-2.txt).

- Create 2 NetServer Read Only shortcuts to use the above configuration files.
- Create 2 Net Data Replicator Read Only configuration files (for example, NetDataReplicatorReadOnlyConfig-1.xml and NetDataReplicatorReadOnlyConfig-2.xml).
- Create 2 Net Data Replicator Read Only shortcuts to use the above configuration files.
- Create folders for the second NetServer Read Only and Net Data Replicator Read Only instances. For example:
 - %IDMS_ROOT%\dynamics\ServerReadOnly-node2\OMSServer
 - %IDMS_ROOT%\dynamics\ServerReadOnly-node2\WMArchiveAdapter
 - %IDMS_ROOT%\dynamics\ServerReadOnly-node2\SwitchOrderServer

Configuring ADMS applications

- Configure the NetViewerAdapterConfig-1.xml file:

```
<Port>24461</Port>
<DynamicsPath>%IDMS_ROOT%\dynamics\server\dmsaccess-24461</DynamicsPath>
<LogFile>%IDMS_ROOT%\log\NetviewerAdapter-24461.txt</LogFile>
```

- Configure the NetViewerAdapterConfig-2.xml file:

```
<Port>24462</Port>
<DynamicsPath>%IDMS_ROOT%\dynamics\server\dmsaccess-24462</DynamicsPath>
<LogFile>%IDMS_ROOT%\log\NetviewerAdapter-24462.txt</LogFile>
```

- Configure the NetViewerAdapterConfig-3.xml - NetViewerAdapterConfig-8.xml files similarly.
- Configure the NetViewerAdapterConfig-1.xml - NetViewerAdapterConfig-4.xml files to connect to NetServer Read Only 1:

```
<OmsConnectionConfig>
<OmsServerIPs><Read_Only_Server_IP></OmsServerIPs>
<OmsServerPort>5051</OmsServerPort>
</OmsConnectionConfig>
<WorkModelConfig>
<WorkModelWebDbEnabled>0</WorkModelWebDbEnabled>
</WorkModelConfig>
<DmsConnectionConfig>
<HostName><Read_Only_Server_IP></HostName>
<HostPort>5055</HostPort>
</DmsConnectionConfig>
```

- Configure the NetViewerAdapterConfig-5.xml - NetViewerAdapterConfig-8.xml files to connect to NetServer Read Only 2:

```
<OmsConnectionConfig>
<OmsServerIPs><Read_Only_Server_IP></OmsServerIPs>
<OmsServerPort>5151</OmsServerPort>
</OmsConnectionConfig>
```

```

<WorkModelConfig>
<WorkModelWebDbEnabled>0</WorkModelWebDbEnabled>
</WorkModelConfig>
<DmsConnectionConfig>
<HostName><Read_Only_Server_IP></HostName>
<HostPort>5155</HostPort>
</DmsConnectionConfig>

```

- Configure the NetServerReadOnlyConfig-1.txt file:

```

SERVER_PORT = 5055
OMS_SERVER_PORT = 5051
DYNAMIC_PATH = %IDMS_ROOT%\dynamics\ServerReadOnly
LOG_FILE_NAME = %IDMS_ROOT%\log\netsrv_read-only.log
ISD_LOG_FILE_NAME = %IDMS_ROOT%\log\isd.log
DNAF_SERVER_RESULTS_PATH = %IDMS_ROOT%\dynamics\ServerReadOnly\DNAF
OMS_CALL_JOURNAL_FILE_PATH = %IDMS_ROOT%\dynamics\ServerReadOnly\OMSServer\CallJournal\
OMS_ARCHIVE_JOURNAL_FILE_PATH = %IDMS_ROOT%\dynamics\ServerReadOnly\OMSServer\ArchiveJournal\
OMS_INCIDENT_JOURNAL_FILE_PATH = %IDMS_ROOT%\dynamics\ServerReadOnly\OMSServer\IncidentJournal\
OMS_CREW_JOURNAL_FILE_PATH = %IDMS_ROOT%\dynamics\ServerReadOnly\OMSServer\CrewJournal\
OMS_ORDER_JOURNAL_FILE_PATH = %IDMS_ROOT%\dynamics\ServerReadOnly\OMSServer\OrderJournal\
OMS_SNAPSHOT_FILE_PATH = %IDMS_ROOT%\dynamics\ServerReadOnly\OMSServer\Snapshot
OMS_SERVER_LOGFILENAME = %IDMS_ROOT%\log\NetRTServer_read-only.log
OMS_RETIREMENT_JOURNAL_AND_SNAPSHOT_FILE_PATH =
%IDMS_ROOT%\dynamics\ServerReadOnly\OMSServer\JournalWS
OMS_QUERY_THREADS = 300

```

- Configure the NetServerReadOnlyConfig-2.txt file:

```

SERVER_PORT = 5155
OMS_SERVER_PORT = 5151
DYNAMIC_PATH = %IDMS_ROOT%\dynamics\ServerReadOnly-node2
LOG_FILE_NAME = %IDMS_ROOT%\log\netsrv_read-only-node2.log
ISD_LOG_FILE_NAME = %IDMS_ROOT%\log\isd-node2.log
DNAF_SERVER_RESULTS_PATH = %IDMS_ROOT%\dynamics\ServerReadOnly-node2\DNAF
OMS_CALL_JOURNAL_FILE_PATH = %IDMS_ROOT%\dynamics\ServerReadOnly-node2\OMSServer\CallJournal\
OMS_ARCHIVE_JOURNAL_FILE_PATH = %IDMS_ROOT%\dynamics\ServerReadOnly-
node2\OMSServer\ArchiveJournal\
OMS_INCIDENT_JOURNAL_FILE_PATH = %IDMS_ROOT%\dynamics\ServerReadOnly-
node2\OMSServer\IncidentJournal\
OMS_CREW_JOURNAL_FILE_PATH = %IDMS_ROOT%\dynamics\ServerReadOnly-node2\OMSServer\CrewJournal\
OMS_ORDER_JOURNAL_FILE_PATH = %IDMS_ROOT%\dynamics\ServerReadOnly-node2\OMSServer\OrderJournal\
OMS_SNAPSHOT_FILE_PATH = %IDMS_ROOT%\dynamics\ServerReadOnly-node2\OMSServer\Snapshot
OMS_SERVER_LOGFILENAME = %IDMS_ROOT%\log\NetRTServer_read-only-node2.log
OMS_RETIREMENT_JOURNAL_AND_SNAPSHOT_FILE_PATH = %IDMS_ROOT%\dynamics\ServerReadOnly-
node2\OMSServer\JournalWS
OMS_QUERY_THREADS = 300

```

- Configure the NetDataReplicatorReadOnlyConfig-1.xml file:

```

<OmsJournalFilePath>%IDMS_ROOT%\dynamics\ServerReadOnly\OMSServer</OmsJournalFilePath>
<WorkModelJournalFilePath>%IDMS_ROOT%\dynamics\ServerReadOnly\WMArchiveAdapter</WorkModelJournalF

```



```

ilePath>
<SwitchOrderServerFilePath>%IDMS_ROOT%\dynamics\ServerReadOnly\SwitchOrderServer\</SwitchOrderServerFilePath>
<ServerLogFileName>%IDMS_ROOT%\log\NetDataReplicatorReadOnly.log</ServerLogFileName>
<OmsServerHost>localhost</OmsServerHost>
<OmsServerPort>5051</OmsServerPort>

```

- Configure the NetDataReplicatorReadOnlyConfig-2.xml file:

```

<OmsJournalFilePath>%IDMS_ROOT%\dynamics\ServerReadOnly-node2\OMSServer</OmsJournalFilePath>
<WorkModelJournalFilePath>%IDMS_ROOT%\dynamics\ServerReadOnly-node2\WMArchiveAdapter\</WorkModelJournalFilePath>
<SwitchOrderServerFilePath>%IDMS_ROOT%\dynamics\ServerReadOnly-node2\SwitchOrderServer\</SwitchOrderServerFilePath>
<ServerLogFileName>%IDMS_ROOT%\log\NetDataReplicatorReadOnly-node2.log</ServerLogFileName>
<OmsServerHost>localhost</OmsServerHost>
<OmsServerPort>5151</OmsServerPort>

```

Configuring sa-manifest.yaml

- Configure the sa-manifest.yaml file:

```

etd-services:
...
  nvaReadOnlyExternalGatewayRoute:
    endpoints:
      - https://<APACHE_HTTP_SERVER_HOST1>:24468
...
      - https://<APACHE_HTTP_SERVER_HOSTN>:24468

```

Configuring Apache HTTP Server

To install the Apache HTTP Server:

1. Download the Apache HTTP Server installer from <https://httpd.apache.org/download.cgi>. For example, download httpd-2.4.52-o111m-x64-vc15.zip from <https://www.apachehaus.com/cgi-bin/download.plx>, which is an official mirror of Apache builds for Windows.
2. Install the Apache HTTP Server to a local folder (for example, D:/Apache).
3. Add the SRVROOT user environment variable with the path "<Apache HTTP Server_Installation_Folder>". For example:

```
setx SRVROOT D:/Apache
```

To configure the Apache HTTP Server:

Edit the default httpd.conf file in D:\Apache\conf:

- Set the listening port to 24468 and enable the HTTPS mode:

```
Listen 24468 https
```

- Enable modules:

```
LoadModule actions_module modules/mod_actions.so
LoadModule allowmethods_module modules/mod_allowmethods.so
LoadModule asis_module modules/mod_asis.so
LoadModule auth_basic_module modules/mod_auth_basic.so
LoadModule authn_core_module modules/mod_authn_core.so
LoadModule authn_file_module modules/mod_authn_file.so
LoadModule authz_core_module modules/mod_authz_core.so
LoadModule authz_groupfile_module modules/mod_authz_groupfile.so
LoadModule authz_host_module modules/mod_authz_host.so
LoadModule authz_user_module modules/mod_authz_user.so
LoadModule autoindex_module modules/mod_autoindex.so
LoadModule cgi_module modules/mod_cgi.so
LoadModule deflate_module modules/mod_deflate.so
LoadModule dir_module modules/mod_dir.so
LoadModule env_module modules/mod_env.so
LoadModule http2_module modules/mod_http2.so
LoadModule headers_module modules/mod_headers.so
LoadModule include_module modules/mod_include.so
LoadModule isapi_module modules/mod_isapi.so
LoadModule lbmethod_byrequests_module modules/mod_lbmethod_byrequests.so
LoadModule lbmethod_bybusyness_module modules/mod_lbmethod_bybusyness.so
LoadModule log_config_module modules/mod_log_config.so
LoadModule mime_module modules/mod_mime.so
LoadModule negotiation_module modules/mod_negotiation.so
LoadModule proxy_module modules/mod_proxy.so
LoadModule proxy_ajp_module modules/mod_proxy_ajp.so
LoadModule proxy_balancer_module modules/mod_proxy_balancer.so
LoadModule proxy_connect_module modules/mod_proxy_connect.so
LoadModule proxy_html_module modules/mod_proxy_html.so
LoadModule proxy_http_module modules/mod_proxy_http.so
LoadModule proxy_wstunnel_module modules/mod_proxy_wstunnel.so
LoadModule rewrite_module modules/mod_rewrite.so
LoadModule setenvif_module modules/mod_setenvif.so
LoadModule slotmem_shm_module modules/mod_slotmem_shm.so
LoadModule socache_shmcb_module modules/mod_socache_shmcb.so
LoadModule ssl_module modules/mod_ssl.so
LoadModule xml2enc_module modules/mod_xml2enc.so
LoadModule proxy_hcheck_module modules/mod_proxy_hcheck.so
LoadModule watchdog_module modules/mod_watchdog.so
```

- Set HTTPS (SSL) settings as follows. (To run in HTTPS mode, Apache HTTP Server requires certificates to be referenced in the configuration file and recorded in the system certificate store. The certificates used for the Apache HTTP Server hosting are the same as the certificates used for the Netview Adapter HTTPS configuration, as described in [Importing Certificates to the ADMS Hosts](#).)

```
<VirtualHost _default_:24468>

    ServerName localhost
    SSLEngine on
```

```

    SSLCertificateFile "${SRVROOT}/conf/adms-server.crt"
    SSLCertificateKeyFile "${SRVROOT}/conf/adms-server.key"
    SSLCertificateChainFile "${SRVROOT}/conf/root-ca.pem"

</VirtualHost>

```

- Configure load balancing for services:

```

SSLProxyEngine on

Header add Set-Cookie "ROUTEID=backend.%(BALANCER_WORKER_ROUTE)e; path=/"
env=BALANCER_ROUTE_CHANGED

ProxyPass "/" "balancer://nva/" stickysession=ROUTEID
ProxyPassReverse "/" "balancer://nva/"

ProxyHCEXpr ok234 %{REQUEST_STATUS} =~ /^[234]/}

<Proxy balancer://nva>
    ProxySet lbmethod=bybusyness

    # Netview Adapter Read-Only instances connected to NetServer Read-Only 1
    BalancerMember "https://<Read_Only_Server_1_IP>:24461" route=1 hcmethod=HEAD hcexpr=ok234
    hcinterval=5 hcuri=/api/health
    BalancerMember "https://<Read_Only_Server_1_IP>:24462" route=2 hcmethod=HEAD hcexpr=ok234
    hcinterval=5 hcuri=/api/health
    BalancerMember "https://<Read_Only_Server_1_IP>:24463" route=3 hcmethod=HEAD hcexpr=ok234
    hcinterval=5 hcuri=/api/health
    BalancerMember "https://<Read_Only_Server_1_IP>:24464" route=4 hcmethod=HEAD hcexpr=ok234
    hcinterval=5 hcuri=/api/health

</Proxy>

<IfModule mpm_winnt_module>
    ThreadsPerChild 600
    MaxRequestsPerChild 0
</IfModule>

```

Where:

- The ThreadsPerChild value must be equal to the number on NetServer Read Only instances multiplied by the OMS_QUERY_THREADS value in NetServerReadOnlyConfig.txt (for example, 2x300 600).

This is required to avoid overloading NetServer Read Only by requests from Outage Assist's Kubernetes cluster. The Apache HTTP Server is limiting the number of concurrent requests, but keeps being buffered and processed once the queue is resolved.

- The BalancerMember "route" value identifies a back-end Netviewer Adapter Read-Only instance. Each BalancerMember entry must have a unique "route" value.

This is required for the Incremental Protocol functionality used in Outage Assist.

- The route value can be any alphanumeric value without spaces.
- The cookie name (here, "ROUTEID") can be any plain text value.
- The cookie value must have "." as a route separator. The cookie value can also be as follows (ignoring the first part):

```
ROUTEID=.{BALANCER_WORKER_ROUTE}e;
```

Important

For the load balancing setup, all Apache HTTP Server instances must use identical configuration files. In the configuration files, BalancerMember "route" values must reference all Netviewer Adapter Read-Only instances on all hosts. This allows the browser to route requests correctly via different Apache HTTP Server instances when storing the route values in cookies.

7.6. Creating Navigation Files

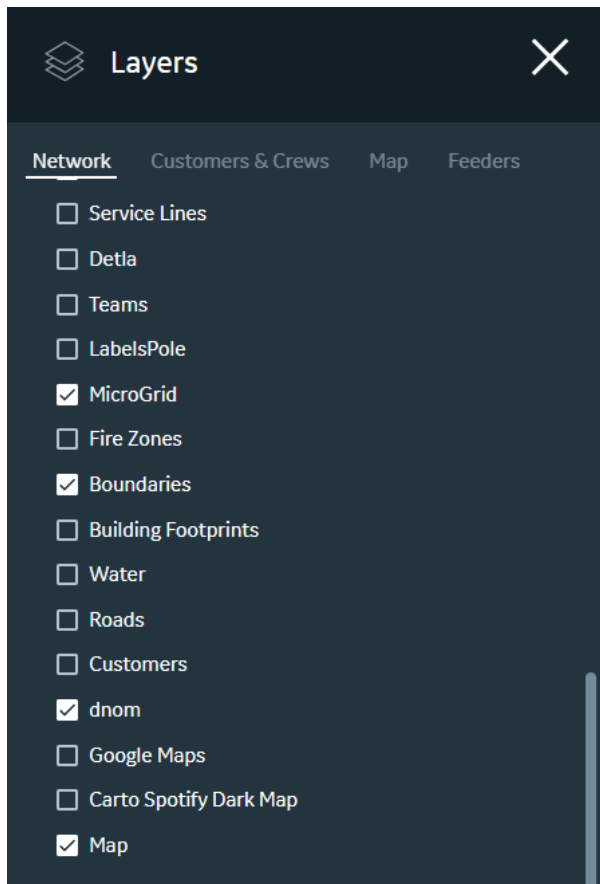
Use the Navigation Builder to create a new Overview.nvw file. This process also generates a new tilesetlist.xml file. For more information, refer to the *ADMS Model Manager User Guide*.

The Common Network Viewer uses PNG files instead of BMP files. Use the Viewer Configuration Editor to generate any missing PNG files from existing BMP images. For more information, refer to the *Viewer Configuration Editor User Guide*.

Note

The Viewer Configuration Editor only creates PNG images for BMP images that are used on the Bitmaps tab and included in ViewerConfig.xml.

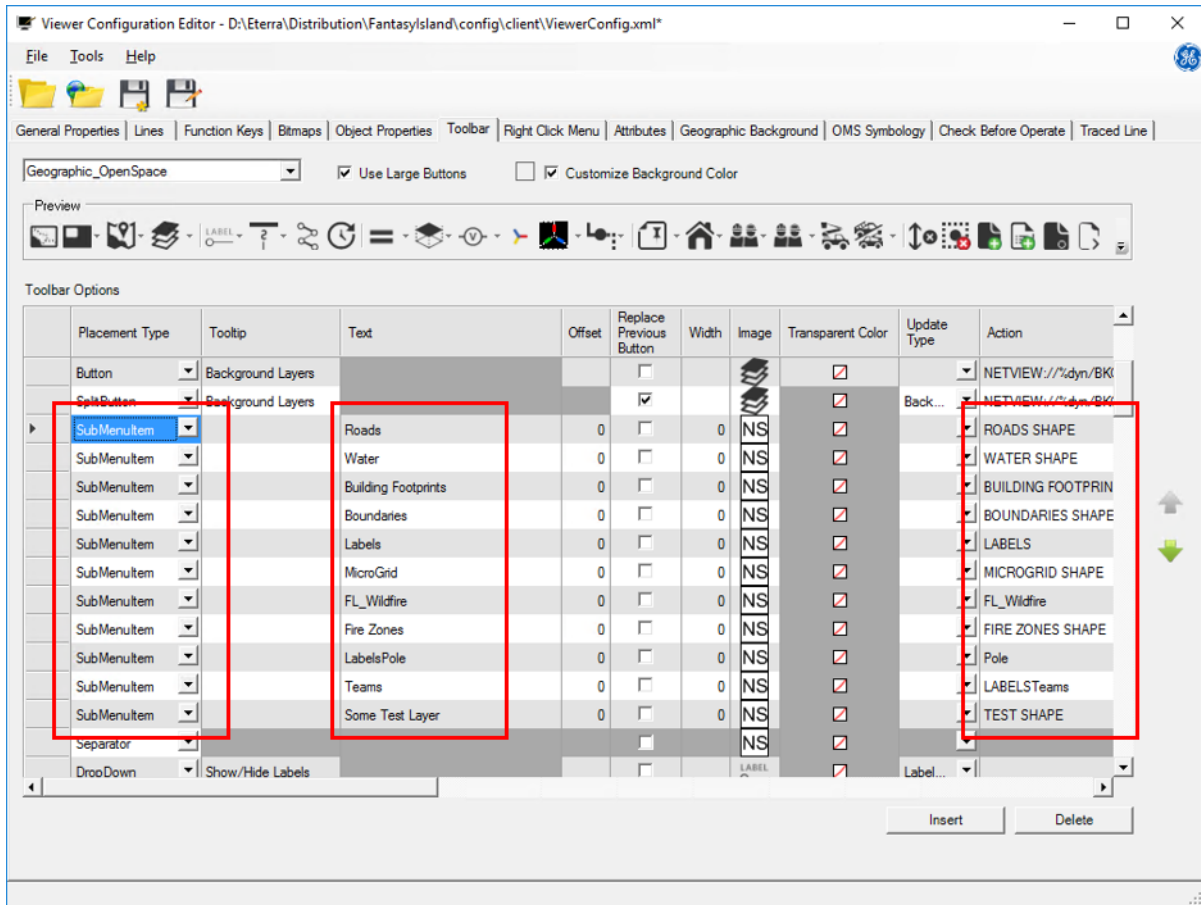
7.7. Configuring Background Layer Names



To configure the background layer names in Outage Assist that appear on the Landbase tab in the Layers panel:

1. Open the Viewer Configuration Editor.
2. Navigate to the Geographic Background tab.
3. Click the Modify Layer List button.
4. Modify the layer list, as appropriate. For more information, refer to the "Adding a New Background Layer to the Background Layers List" section in the *Viewer Configuration Editor User Guide*.

In Outage Assist, all defined background layers will appear as options on the Landbase tab in the Layers panel.



10. In the Text column, specify the layer label to use on the Landbase tab in the Layers panel.
11. Save the file. 12. Restart the NetViewer Adapter to apply the changes.

Note

For all background layers not listed in the Action column or without a Text value, the default layer names will be used for background option labels on the Landbase tab in the Layers panel in Outage Assist.

8. Validating the Outage Assist Setup

To validate that the Outage Assist system is operating correctly:

1. Run Outage Assist pods and services in Kubernetes.
2. Start the core ADMS servers (DMS server, OMS server, Call server, Crew server, etc.).
3. Start the ADMS NetViewer Adapter (or restart the application if it is already running to apply changes).
4. Start the ADMS OMS Interface Server.
5. Start the ADMS Enterprise OMS servers (Work Order Adapter, Notification service, Event Handler service, and Preference service server).
6. Create an incident in ADMS.
7. Navigate to <https://service.ingress.local/deui-storm-assist>.
8. Log in to Outage Assist.
9. In Outage Assist, check whether the Orders tab shows the correct order details.
10. Similarly, update an incident via the Outage Assist UI and verify that the details are updated in ADMS.
11. Check whether Outage Assist's Crews tab shows the correct crew details.
12. Check whether Outage Assist shows the correct geographic network view, layers, components, etc.

9. Maintenance and Troubleshooting

9.1. Using the Developer Console

Use the developer console in a web browser to debug the application.

To show the developer console in Google Chrome, press Ctrl+Shift+J.

9.2. Maps and Map Decorations are Not Shown

Problem: Maps and map "decorations" (incident pins, crew pins, tags, etc.) are not shown.

Solution: Verify that ADMS servers that communicate with Outage Assist are configured correctly, running, and connected to the Outage Assist Kubernetes cluster.

- Verify that OMS Interface Server, DMS Interface Server, and NetViewer Adapter are running and that their log files do not contain errors.
- Verify that the "TileServiceClient websocket connection has been OPENED." message appears in the NetViewer Adapter log.

9.3. Cannot Confirm Device Status

Problem: Cannot confirm device status (toggle a device). An error appears: "Device Update Failed, User has no permission to access the AOR".

Solution: Verify that the DMS Interface Server is configured correctly, running, and connected to the OMS Interface Server.

9.4. Outage Assist UI Hangs

Problem: The Outage Assist UI hangs.

Solution: Restart the Outage Assist UI pod using the Kubernetes dashboard (delete the pod and wait until it re-appears and initializes).

9.5. Reviewing the GraphQL Log

Problem: Need to review the GraphQL log.

Solution: Review the GraphQL log from the Kubernetes dashboard.

9.6. Outage Assist Does Not Show Data

Problem: Outage Assist does not show data.

Solution 1: Review the GraphQL log from the Kubernetes dashboard.

Solution 2: Check the ADMS configuration.

Solution 3: Check the Area of Responsibility (AOR) configuration in Foundation. For more details, refer to your GE project team.

9.7. Error: HTTP Could Not Register URL

Problem: The following OMS Interface Server exception appears:

```
HTTP could not register URL http://+:8812/oms/services/. Another application has already registered this URL+++
```

Explanation: In the default configuration, the OMS Interface Server listens at `http://+:8812/oms`. Listening at a particular HTTP address requires administrator privileges or account permissions to reserve the URL.

The error could occur due to one of the following reasons:

- There are no proper rights.
- The port is registered by another application.
- The port is registered with https (not http).

Solution 1:

Run the OMS Interface server as an administrator.

Solution 2:

Use Netsh.exe to give the account permissions to reserve the URL.

To check and update the system HTTP URL namespace reservations:

1. Run Windows PowerShell as Administrator.
2. Run the "netsh" command.
`netsh`
3. Run the "http" command. `http`
4. Run one of the following commands to check the current URL assignments in the system:
 - `show urlacl`
 - `show urlacl | findstr /I 880`

The list may show whether another URL has http or https binding at the same port number

(8812).

5. Run the following command to remove an unwanted URL assignment:

```
delete urlacl URL=<URL>
```

For example:

```
http delete urlacl URL=http://+:8812/
```

Or:

```
http delete urlacl URL=https://+:8812/
```

6. Run the following command to re-add a new URL assignment for the OMS Interface Server:

```
add urlacl URL=http://+:8812/ user="NT Network Service"
```

9.8. Chrome Exits when the System is under Load

Problem: When the system is under high load, a message saying "Aw Snap, something went wrong" appears and Chrome exits.

Reason: Chrome may exit when using a 32-bit version.

Solution: Use the 64-bit version of Chrome.

10. ADMS Configuration Files

10.1. NetViewer Adapter Configuration File

The NetViewer Adapter reads the NetViewerAdapterConfig.xml configuration file, which is located in the %IDMS_ROOT%\config\interfaces folder. This section provides details for NetViewerAdapterConfig.xml parameters.

General The general parameters are as follows:

- **SecureWS:** Set to True to run NetViewer Adapter in HTTPS mode. Set to False to run NetViewer Adapter in HTTP mode.
- **CertificateSubject:** The certificate subject (name) for HTTPS mode. The certificate must be placed locally in the Certificates > Trusted Root Certification Authorities > Certificates folder.

To use different certificate subjects for different hosts in a multiple standby host configuration, you can define subjects using the "certsubject" system environment variable and set the CertificateSubject value to %certsubject%.

- **Port:** Port number that is used by the NetViewer Adapter.
- **ConfigPath:** Path to the folder with ADMS client configurations. The NetViewer Adapter reads client configuration files for styles, symbols, colors, zoom levels, and other user interface data so that the network view in Outage Assist looks similar to the network view in the ADMS client.
- **DynamicsPath:** Location of the dynamics.sav file for Outage Assist. Specify a unique folder, so that Outage Assist does not conflict with another server writing its dynamics.
- **NetConfigFileName:** Name of the Net Dynamics configuration file to be read by the NetViewer Adapter. The file is stored in the folder with ADMS client configurations.
- **DmsInterfaceServerUrl:** DMS Interface Server URL. DMS Interface Server is used to query for DNAF results, specifically, potential fault locations.
- **EnableDNAFQueries:** Specifies whether DNAF result querying is enabled (specifically, queries for potential fault locations). Set to true if NetViewer Adapter is used for generating tiles. Set to false if the NetViewer Adapter is used for DNOM device searching and data querying in DMS-only environment.
- **DNOMQueriesInterfaceServer:** Specifies whether the DNOM device searching and data querying is enabled. Set to true if the NetViewer Adapter is used for DNOM device searching and data querying in DMS-only environment and needs device data for GraphQL queries.
- **CustomSecretStore:** The NetViewer Adapter can retrieve UAA secret values from the CyberArk Enterprise Password Vault so that you do not need to provide the value in the configuration file (as a plain text or encrypted string). For more information, refer to the *ADMS Software Installation and Maintenance Guide*.
 - **AssemblyName:** Name of the assembly (DLL) containing a class that implements the CyberArk secret store.

- **ClassName:** Name of the class that implements the CyberArk secret store. This class is implemented based on the CyberArk software development kit (SDK) and supports connecting to CyberArk API.
- **OtherAssemblyNames:** Name of the assembly (DLL) that implements the .NET Standard Password SDK.
- **UaaIdObjectKey:** CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the NetViewer Adapter UAA ID defined in the Enterprise Vault Server. The format is
`<DatabaseMemberUserName>,<AccountSafe>,root,Application-<AccountPlatformID>-<AccountAddress>-<AccountUsername>.`
- **UaaSecretObjectKey:** CyberArk object key, which contains CyberArk application ID, safe, database (folder), and object definitions of the NetViewer Adapter UAA secret defined in the Enterprise Vault Server.
- **DisableCnf:** Common Notification Framework (CNF), also known as Notification Service, sends notifications about tile and dynamics updates to clients. This option specifies whether the NetViewer Adapter sends notifications to CNF. Typically, the NetViewer Adapter is running with the Tile Service, which itself sends notifications to CNF. Therefore, disable CNF (set this parameter to True) if the NetViewer Adapter is running with the Tile Service and EnablePublishingToTileService is set to True.
- **CnfUrl:** Notification service URL. By default, `http://localhost:9090/notification-service/notify`.
- **EnablePublishingToTileService:** If set to True, the adapter publishes DNOM and customer tiles to the Tile Service.
- **TileServiceUrl:** The URL of the web socket Tile Service endpoint. By default, `ws://localhost:24467/tilecacheservice`.
- **TileServiceCompression:** The compression type to apply to published tiles and configurations. Can be one of the following: "gzip" (Gzip compression), "br" (Brotli compression), or "none" (no compression).

Note

Using Brotli compression is recommended for better client performance.

- **UaaTokenServer:** The name of the server that hosts UAA services. For security reasons, the NetViewer Adapter does not directly connect to a service in the Kubernetes cluster. The NetViewer Adapter must connect to a UAA server to get a token for authorizing connection with a service in the cluster.
- **DnomDynamicLoopTime:** Time period, in milliseconds, between checks for dynamic updates to DNOM data. The default value is 1000.
- **CustomerDynamicLoopTime:** Time period, in milliseconds, between checks for dynamic updates to customer data. The default value is 1000.
- **DocumentDynamicLoopTime:** Time period, in milliseconds, between checks for dynamic updates to document/annotation data. The default value is 1000.

- **CustomerQueryRequestTime:** Time period, in milliseconds, between queries for customer data for updates. The NetViewer Adapter uses this time period to wait for more customer data changes after a change occurs, to avoid sending a number of notifications and updating a number of tiles. (For example, a number of successive customer events are expected after a device outage, so the NetViewer Adapter waits for those to settle). The default value is 5000.
- **FormatJsonOutput:** If set to True, the JSON output is formatted to human-readable text. Formatted JSON files are larger than unformatted files.
- **DisableSearchService:** If set to True, the search service is disabled and returns empty results. Use for debugging purposes.
- **DisableQueryService:** If set to True, the query service is disabled and returns empty results. Use for debugging purposes.
- **DisableIntrospection:** If set to True, the schema for GraphQL queries are disabled.

Modes

This section describes parameters in the Modes section, which define how the NetViewer Adapter processes the network view tile and configuration data.

A mode parameter value can be one of the following:

- **Normal:** Get data from ADMS and publish the data to the Tile Service. If all modules are loaded in the Normal mode, the NetViewer Adapter consumes around 3 GB of RAM.
- **File:** Read data from files.
- **Write:** Get data from ADMS, save the data to files, and publish the data to the Tile Service).
- **Disabled:** Disable the mode. This option only supports GraphQL queries. If all modules are disabled, the NetViewer Adapter consumes around 70 MB of RAM.
- **QueryOnly:** Support DNOM, customer, and GraphQL queries. If the ConfigMode, CustomerMode, and DnomMode modules are loaded in the QueryOnly mode, the NetViewer Adapter consumes around 1.7 GB of RAM.

The Modes section includes the following parameters:

- **ConfigMode:** Indicates whether to generate or read a display configuration for the Common Network Viewer.
- **CustomerMode:** Indicates whether to generate or read customer data for the Common Network Viewer.
- **DnomMode:** Indicates whether to generate or read DNOM data for the Common Network Viewer.
- **DocumentMode:** Indicates whether to generate or read documents data, such as annotations and safety docs for the Common Network Viewer.
- **LandbaseMode:** Indicates whether to generate or read landbase data for the Common Network Viewer.
- **OverviewMode:** Indicates whether to generate or read overview data for the Common Network

Viewer.

Log File

The log file parameters are as follows:

- **EnableDebugLog:** Write a debug log file.
- **LogFile:** The name and location of the debug log file
- **MaxMessagesInLogFile:** Maximum number of log entries to retain.

Selection Configuration (SelectionConfig)

The selection parameters define the appearance of selected objects in the client display.

- **SelectionDrawStyle:** The appearance of the line selection indicator. Can be one of the following: PlainColorBorder, Pattern, BlueWhiteDiagonalChecks5, BlueWhiteDiagonalChecks7, BlueWhiteDiagonalChecks9, BlueWhiteDiagonalChecks11, BlueWhiteDiagonalChecks13, BlueWhiteDiagonalChecks15, BlueWhiteTriangle5, BlueWhiteTriangle7, BlueWhiteTriangle9, or BlueWhiteTriangle11.

Note

PlainColorBorder draws a solid color border around the selection; use SelectionColor to set the color. Pattern specifies a custom image to use for a pattern with the SelectionPattern property.

- **SelectionLineWidth:** The width of the line selection indicator.
- **SelectionAnimationStyle:** The animation for the line selection indicator: Can be one of the following:
 - **None:** No animation.
 - **Flash:** Line selection flashes.
 - **Expand:** Line selection expands/contracts on a half-second interval.
 - **ExpandSlow:** Line selection expands/contracts on a one-second interval.
 - **PatternCrawl:** When used with a pattern, the pattern crawls to draw attention to the selection.
 - **SelectionColor:** The color of the line selection in hex format, such as "#0088AA". Applies to the PlainColorBorder draw style only.
 - **SelectionPattern:** A custom image to be used for the line selection pattern. (For the Pattern draw style only).

The SelectionPattern animation style contains the following sub elements:

- **name:** Can include any name that is unique in the configuration (CustomSelectionPattern is recommended).
- **imageData:** Image data, in the data URL format.

For more information, see https://developer.mozilla.org/en-US/docs/Web/HTTP/Basics_of_HTTP/Data_URIs.

- **SelectionPatternReplacementColors:** Use this option to programmatically replace colors in the SelectionPattern image. This option should include a list of ReplacementColors elements for replacing several colors. Each ReplacementColors element includes the following sub-elements:
 - **orig:** The color to replace in the source image, in hex format.
 - **target:** The color to use to replace the "orig" color in the source image, in hex format.

Drawing Array

The drawing array configuration parameters are as follows:

- **EnableDrawingArrayExport:** Specifies whether the NetViewer Adapter automatically exports drawing array files to xml files. Drawing arrays are an intermediate format. Data goes from in-memory DNOM to drawing arrays, from drawing arrays to tiles, and then from tiles to JSON string representing tiles. Exporting drawing array files to xml files is used for debugging purposes.
- **DrawingArrayPath:** Path to the folder with drawing array files.
- **LandbaseBoundary:** Boundaries for landbase geo files. The boundaries are defined using the X1, Y1, X2, and Y2 coordinates. Any geo file out of boundaries is ignored.

Tiles

The tile configuration parameters are as follows:

- **LoadTilesDuringNavigation:** Specifies whether tiles are loaded in the NetViewer Adapter during navigation (panning). If set to false, the tile loading is delayed till panning is completed.
- **Boundary:** Limits the tile output area. Any customer and landbase tiles outside of the configured area will not be processed.
- **RotationType:** Indicates whether rotation angles in display tiles are interpreted clockwise or counter-clockwise (cw, ccw).
- **InitialScale:** Initial OpenLayers scaling factor for displays.
- **MinZoom:** Minimum zoom level for the displays.
- **MaxZoom:** Maximum zoom level for the displays.
- **SmallDisplaySize:** Size (width or height) under which a display is considered "small". This setting triggers using the SmallDisplayInitialScale setting.
- **SmallDisplayInitialScale:** Initial scaling factor for small displays.
- **DocTileWidth:** Width of document (annotation) tiles, in display units.
- **DocTileHeight:** Height of document (annotation) tiles, in display units.
- **CustomerTileWidth:** Width of customer tiles, in display units.
- **CustomerTileHeight:** Height of customer tiles, in display units.

- **DnomPatchWidth:** Width of station model (DNOM) tiles, in display units.
- **DnomPatchHeight:** Height of station model (DNOM) tile patches, in display units.

Note

For better performance, it is recommended to adjust tile sizes on both the server and client. Tiles that are too large take longer to generate and therefore perform poorly in the client. Tiles that are too small result in a large number of requests for tiles to the server, which may result in slower response times. To disable the tile patching functionality, remove these parameters or set their values to 0.

Also note that tile sizes are in the units of measure of the geographic display; therefore, if the coordinates are in meters, the width/height sizes are also in meters.

- **LandbaseTileLabelMinZoom:** Minimum zoom level for landbase labels. Use this parameter to declutter labels.
- **DisableIndividualDynamicUpdate:** If set to True, customer tiles are not updated for individual customer dynamics updates; customer tiles are only updated at the timeout specified by the TileStaleTimeout parameter.
- **TileStaleTimeout:** Timeout, in seconds, for recreating a tile affected by dynamics. When the last dynamic that affects a tile is older than this timeout, the tile is recreated.

Failover

The failover configuration parameters are as follows:

- **EnableFailoverRoleAssignment:** If set to True, the adapter queries the Failover Broker for its role, ENABLED or STANDBY.
- **FailOverBrokerUrl:** A comma-separated list of Failover Broker base URLs. For example: `http://hostname_a:24470/api/role,http://hostname_b:24479/api/role,http://hostname_c:27832/api/role`.
- **HealthCheckInterval:** Time period, in seconds, for sending health check messages to the Failover Broker. This interval should be less than the health check interval that is applied to the Failover Broker application.

Geographic Background

The geographic configuration parameters are as follows:

- **MapSource:** The source URL template for the background base map. The source URL template must include placeholders for x, y, and z coordinates, although their placement in the URL is up to the service provider.

Examples:

- `http://{1-4}.basemaps.cartocdn.com/dark_all/{z}/{x}/{y}.png`
- `http://mt0.google.com/vt/lyrs=y&hl=en&x={x}&y={y}&z={z}&s=Ga`



Note

You can also define a URL for an internally-hosted resource.

Note

The MapSource configuration is a legacy configuration, which supports only one map source. To use multiple map sources for background and overlay map layers, use the MapBackgroundLayerSources parameter group.

- **Proj4Def:** Projection definition in "PROJ4" format. Use the same projection that is used to translate DNOM/customer data into x/y coordinates in which the data is expressed. In particular, the "units" setting must match the units of measure of the data. For more information, see <https://proj4.org/>.

For example:

```
proj=tmerc +lat_0=24.33333333333333 +lon_0=-81 +k=0.999941177 +x_0=200000.0001016002 +y_0=0  
+ellps=GRS80 +towgs84=0,0,0,0,0,0 +units=us-ft +no_defs
```

- **Proj4Name:** Spatial reference name used internally by OpenLayers. You can choose any value, not necessarily a standard name. For more information, see <http://spatialreference.org/ref/?search=oregon&srtext=Search>.
- **Proj4Extent1:** Minimum longitude value for geographic data, expressed in decimal degrees.
- **Proj4Extent2:** Minimum latitude value for geographic data, expressed in decimal degrees.
- **Proj4Extent3:** Maximum longitude value for geographic data, expressed in decimal degrees.
- **Proj4Extent4:** Maximum latitude value for geographic data, expressed in decimal degrees.

Note

Proj4Extent settings are optional. They indicate the maximum bounds at which the projection string is reliable. If bounds are specified, data beyond the bounds may be ignored by OpenLayers; this may or may not be desirable.

- **StartCenterX:** X-coordinate for the initial map position.
- **StartCenterY:** Y-coordinate for the initial map position.
- **MapBackgroundLayerSources/MapOverlayLayerSources:** Configurations for multiple background and overlay map sources.

Note

Ampersand characters ("&") must be escaped with ' to pass the XML parser validation.

Multiple map sources can be defined using multiple MapTileSource parameter groups. Each MapTileSource parameter group includes the following parameters:

- **name:** Map source unique identifier.

- **type:** Map source type. Only the "xyz" and "tilewms" types are supported.

XYZ tiles come from compliant web data sources that expect the x, y, and z position of the display to be embedded in the URL when fetching tiles. WMS (web map service) tiles are returned as images or tiled data.
- **declutter:** The name of the associated declutter layer. This parameter is optional. It should be used for layers that need to be visible only within certain zoom ranges.
- **label:** Map source layer label to be shown on the user interface.
- **url:** Map source URL.
- **opacity:** Layer opacity. This parameter is optional. The allowed range is 0-1.

For example, 0.5 is 50% opacity so that the layer does not completely overlay the background.
- **MapSourceParameters:** Map source parameters only apply to the "tilewms" map source types. This is an "any" object allowing the specifying of params understood for this map type.
 - **ParamName:** Parameter name. For example, LAYERS.
 - **ParamValue:** Parameter value. For example, 1.
- **defaultVisible:** Specifies whether the map source layer is visible by default.

OMS Server Connection

The NetViewer Adapter sends query and search requests that cannot be resolved locally on the OMS Server. The OMS server configuration parameters are as follows:

- **OmsServerIPs:** OMS server IPs.
- **OmsServerPort:** OMS server port number.
- **DmsQueryUserId:** DMS query user ID. The default value is {8599AE31-2409-4875-9ECF-0698FAE23EDA}.
- **RequestTimeoutPeriod:** Request timeout period, in seconds.
- **UseBinarySerialization:** Specifies whether binary serialization is used.
- **SendCompressedMessage:** Specifies whether OMS message data is compressed.

Work Model Configuration

The Work Model (Enterprise OMS) configuration parameters are as follows:

- **WorkModelWebDbEnabled:** If enabled (=1), NetViewer Adapter queries the Enterprise OMS REST API for the order data.
- **WorkModelServiceApi:** Work Model Event Handler Service endpoint.
- **WorkModelEventApi:** Work Model Notification Service events endpoint.
- **WorkModelTlsEnabled:** Specifies whether the HTTPS TLS security is enabled for the Work

Model Event Handler Service connection for web clients and web socket clients.

- **WorkModelTlsSubjectName:** Specifies the certificate subject name that is used when searching for client certificates to send to the Work Model Event Handler Service.
- **WorkModelTlsIgnoreRemoteNameMismatch:** Specifies whether the remote server certificate name mismatch is ignored. If enabled (=1), the remote server certificate name can be different from the URL host name that is used to connect to the server.

DMS Server Connection

The DMS server configuration parameters are as follows:

- **HostName:** DMS server name.
- **HostPort:** DMS server port number.
- **Options:** Hello period and connection retry period. For example:

```
HELLO_PERIOD:45;CONNECT_RETRY_PERIOD:45
```

Tags

The tag configuration parameters are as follows:

- **Name:** Name of the tag image.
- **ImageData:** Tag image, in data URL format. For more information, see https://developer.mozilla.org/en-US/docs/Web/HTTP/Basics_of_HTTP/Data_URIs.
- **Width:** Tag image width.
- **Height:** Tag image height.
- **InsertionPoint:** X and Y tag offset relative to the associated device.
- **ColorToReplace:** Tag image color to replace to indicate the tag type.
- **TagPriority:** Collection of tag type colors, where the first entry is the highest priority.

Customer Search

The customer search configuration parameters are as follows:

- **SearchableFields:** List of customer fields that will be searched. Can include the following: FIRSTNAME, LASTNAME, MIDDLENAME, CUSTOMERNAME, ACCOUNTNUMBER, METERNUMBER, SERVICEPOINT, PHONEPREFIX, PHONESUFFIX, PHONENUMBER, STREETNUMBER, STREETNAME, STREETSUFFIX, APARTMENTNUMBER, CITY, ZIPCODE, AREACODE, OPERATINGCENTER, and SERVICECENTER.
- **ResultFields:** Customer fields to be returned as the result of a customer query/search.

Device Search

The device search configuration parameters are as follows:

- **ResultFields:** A comma-separated list of device fields to be returned as the result of a device query/search.

- **SearchType:** Type of search. Can be one of the following: CONTAINS, EXACTMATCH, BEGINSWITH, ENDSWITH, or DYNAMIC. If not specified, the CONTAINS type is used.
- **ResultFields:** A comma-separated list of device fields to be returned as the result of a device query/search. The default value is "DeviceId,HasGeometryInfo,Dyn,SubType,HasMeasurement,DnomType,SubGeoName,EmsName,Id,StationName,Name,AlternateName,Address,GeometryName,IsGeographic,FeederMain,HasPlacement,Instation,InstationMeasurment,Underground,Future".

Display Parameters

The display configuration parameters define which filters and layers apply to primary (geographic) and non-primary (non-geographic) displays:

- **Zoom:** Default zoom level for the map.
- **Filters:** Default filters for the map.
- **Layers:** Default layers to be shown on the map.

10.2. OMS Interface Server Configuration

The OMS Interface server reads the OmsInterfaceServerConfig.xml configuration file, which is located in the %IDMS_ROOT%\config\interfaces folder.

The OmsInterfaceServerConfig.xml configuration file includes the following parameters:

- **OmsServerIPs:** Name or IP address of the OMS server (NetServer) host machine. The default value is localhost.
- **OmsServerPort:** TCP/IP port number on which the listener host is listening for client connections. The default value is 5050.
- **OmsServerUseStreams:** If set to true, the OMS Server uses the streams OmsConnection interface. (This setting must be set to the same value in all configuration files.)
- **OmsServerUseTLS:** If set to true, the OMS Server uses the TLS security protocol. (This setting must be set to the same value in all configuration files.)
- **OmsServerCheckMessageSecurity:** If set to true, the OMS Server checks for security token authenticity/expiration. (This setting must be set to the same value in all configuration files.)
- **OmsServerCheckMessageSequence:** If set to true, the OMS Server checks message sequence numbers. (This setting must be set to the same value in all configuration files.)
- **OmsServerUseCredentialFile:** If set to true, the OMS Server uses a credential file for defining the private key and user SIDs. (This setting must be set to the same value in all configuration files.)
- **OmsServerCredentialInfoFile:** Path to the credential file.
- **OmsServerTimeToLive:** Time to live for OMS Server connections in seconds.
- **RequestTimeOutPeriod:** Timeout period for a request, in seconds.

- **UseBinarySerialization:** Specifies whether to use binary serialization.
- **SendCompressedMessage:** Specifies whether to send compressed messages.
- **SendBufferSize:** Buffer size for sending messages, in KB.
- **ReceiveBufferSize:** Buffer size for receiving messages, in KB.
- **EnableMessageLog:** Specifies whether to log all incoming web service requests and their responses. To use EnableMessageLog in HTTP mode, set WSSecurityEnabled to False.
- **EnableDebugLog:** Specifies whether to log detailed information for debugging purposes.
- **LogFile:** Specifies the full path to the file that is used for OMS Interface server logging purposes.
- **MaxMessagesInLogFile:** The maximum number of records in the log file. When this number is reached, another log file is created. The default value is 10,000.
- **EnableWebApi:** If set to true, the OMS Interface server web service is enabled.
- **WritePermissionValue:** The AOR write permission value. To enable the write permission checking, use the oms.Write value. To disable the write permission checking, use the blank value.
- **WebServiceBaseURL:** Configures the base URL for trusted applications. Note that the trailing "/" is needed at the end. The default value is http://localhost:8812/Oms/.

Each web service application adds a specific ending to this base. For example, the URL base for the OMS Services web service is http://localhost:8812/Oms/Services.

Note

Use the "http" value in the URL to support both http and https modes, as appropriate. To enable secure connection, set SecureWebService to True.

- **DMSInterfaceServerURL:** Configures the endpoint to the DMS Interface Server's DMS Control service that the OMS Interface Server uses to update DMS data and request services from the DMS server. The default value is http://localhost:8801/dms/services/.
- **AorApiVersion:** The AOR Manager API version. By default, version 3 is used.
- **UseUAAToken:** Specifies whether UAA authentication is enabled. Enables retrieving a valid UAA authentication token from the UAA server.
- **UaaEndpoints:** List of endpoints for communicating with the UAA server and retrieving a valid UAA token, separated by semicolon.

Use multiple UAA token endpoints to enable the OMS Interface Server to support connection to multiple servers (Kubernetes clusters).

- **UaaEndpointCheckingInSeconds:** Time period (in seconds) for checking the UAA endpoint.
- **IncidentAorSyncInSeconds:** Time period (in seconds) for requesting AOR data for all incidents from NetServer.
- **DisableServiceAuthCheck:** If this is set to true, authorization header checking for service calls

is disabled. This is added for ease of testing and should **not** be set to true in a production system.

- **WSSecurityEnabled:** Set to false to test the OMS Interface Server calls using SoapUI or to use the EnableMessageLog option in HTTP mode. Make sure that WSSecurityEnabled value in OmsInterfaceServerConfig.xml is the same.
- **SecureWebService:** Configures whether a secure web service connection is used (the web service runs in HTTPS mode). The default value is false.
- **DmsQueryUserId:** Configures the trusted user ID for the DMS Interface Server. Required for implementing DMS commands.

 Note

Use the same setting for this parameter and the TRUSTED_USER_ID_DIS parameter in the NetServerMainConfig.txt configuration file.

- **TrustedUserId:** Configures the trusted user ID for the OMS Interface Server. Required for implementing OMS commands.

 Note

Use the same setting for this parameter and the TRUSTED_USER_ID_OIS parameter in the NetServerMainConfig.txt configuration file.

- **OMS Interface Server Web Service Plugin Parameters:** This group includes OMS Interface Server web service plugin parameters.
 - **Enabled:** If set to true, the OMS Interface Server web service is enabled.
 - **Type:** The type that implements the OMS Interface Server web service plugin.
 - **Name:** OMS Interface Server web service plugin name.
 - **BasePath:** OMS Interface Server web service root URL.
 - **BaseUri:** The web service root URL. If populated, BaseUri takes precedence over the URL constructed by the UriScheme, UriHost, UriPort, UriPath, BasePath, and Version parameters.
 - **Name:** OMS Interface Server web service name to be shown on the user interface.
 - **Version:** OMS Interface Server web service message version.
 - **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
 - **UseHeaderAuthorization:** If set to true, "authorization" header checking is enabled for incoming web service calls.
 - **QueryService:** The name of the OMS Interface query service.
 - **PW:** Password value that is used to hash and place in the request authorization header. Use the Encryption tool for encrypting the password.

- **UseUAAToken:** Specifies whether UAA authentication is enabled. Enables validating a UAA authentication token (for servers and web services) or enables retrieving a valid UAA authentication token (for clients).
- **UaaIntrospectEndPoint:** Endpoint for validating a UAA authentication token.
- **Plugin URI Parameters:** The URI parameters in this group apply to all OMS Interface Server web service plugins.
 - **UriScheme:** URI scheme: HTTP or HTTPS.
 - **UriHost:** URI host.
 - **UriPort:** URI port.

Note

Use different ports for different web services.

- **UriPath:** URI path.

10.3. Configure DMS API Server Communication with OMS Interface Server

To enable the DMS API Server to communicate with the OMS Interface Server, perform the following steps:

Step 1: Modify the NetServerDnafConfig_localhost.txt file

- Locate the file:
%IDMS_ROOT%config/server/NetServerDnafConfig_localhost.txt file.
- Uncomment the following lines under the DMS API server-related services section:

```
GRPC_SERVICE_PORT = 50055

DMS_ENABLE_GRPC_API_SERVICE = 1

DMS_ENABLE_GRPC_MSG_SERVICE = 1
```

```

'-----
'      DMS API server related services options
'-----
GRPC_SERVICE_PORT    = 50055
'The Grpc service binding IP, default will be ANY(0.0.0.0) if not specified
'GRPC_SERVICE_IP     = 0.0.0.0
'GRPC_SERVICE_SSL_CERT_FILE = %IDMS_ROOT%\config\server\certs\adms-server.cert
'GRPC_SERVICE_SSL_CERT_KEY_FILE = %IDMS_ROOT%\config\server\certs\adms-server.key

' Enable DMS API server service, default is disabled
DMS_ENABLE_GRPC_API_SERVICE = 1

' Enable GRPC message streaming service, default is disabled
DMS_ENABLE_GRPC_MSG_SERVICE = 1

```

Step 2: Update the OMS Interface Server configuration files

- Open the file:

%IDMS_ROOT%\config\client\OmsInterfaceServer.xml

or

C:\eterra\Distribution\FantasyIsland\config\interfaces

- Add the following parameter to the OmsInterfaceServerConfig_EC2AMAZ-J9Q0BHI.xml and OmsInterfaceServerConfig_UAA_EC2AMAZ-J9Q0BHI.xml

```

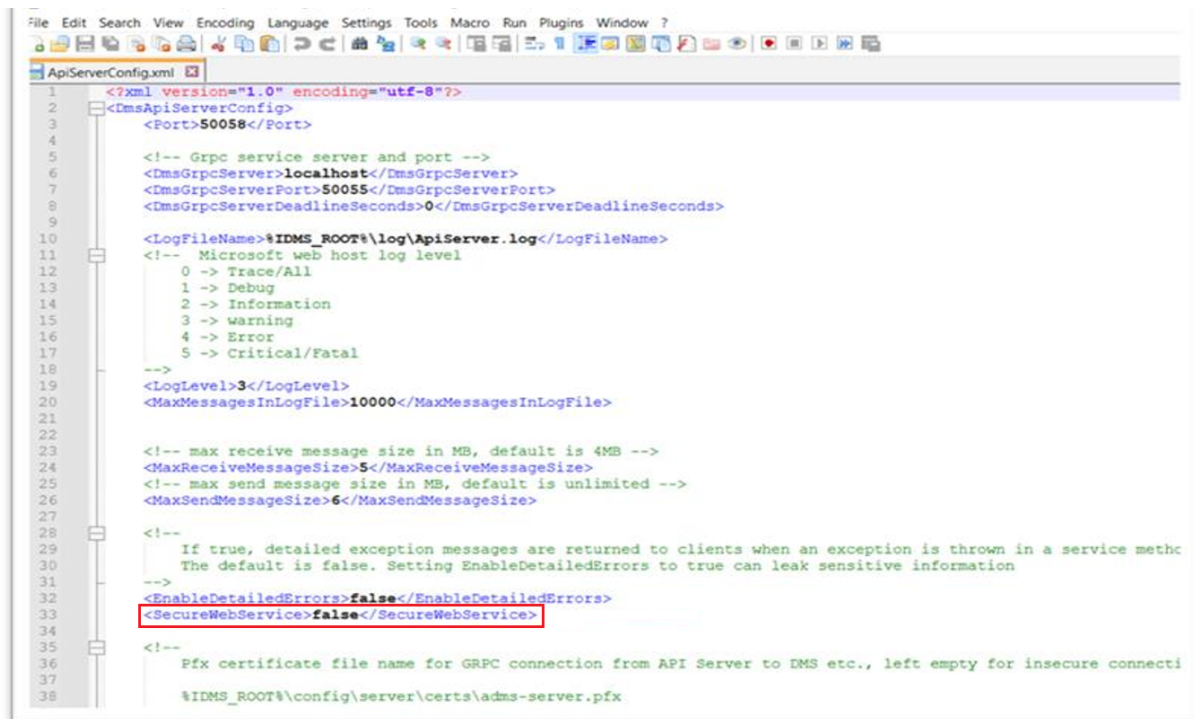
<code>&lt;DMSApiServerURL&gt;<a
href="http://localhost:50058/api/v1/&lt;/DMSApiServerURL&gt;"
class="bare">http://localhost:50058/api/v1/&lt;/DMSApiServerURL&gt;</a></code>

```

Use http instead of https

Step 3: Update the ApiServerConfig.xml file

- Navigate to %IDMS_ROOT%\config\server/
- Open ApiServerConfig.xml file with Admin privileges.
- Update SecureWebService value to false.



- Optionally, enable detailed logging:

```
<AlsoLogToConsole>true</AlsoLogToConsole>
<LogLevel>0</LogLevel>
```

Step 4: Update the apisettings.json file

- Navigate to C:\eterra\Distribution\Server\bincore .
- Open appsettings.json file with Admin privileges. If not, add it to configure for http and https:

```
{
  "Kestrel": {
    "Endpoints": {
      "Http": {
        "Url": "http://*:50058"
      }
    }
  }
}
```

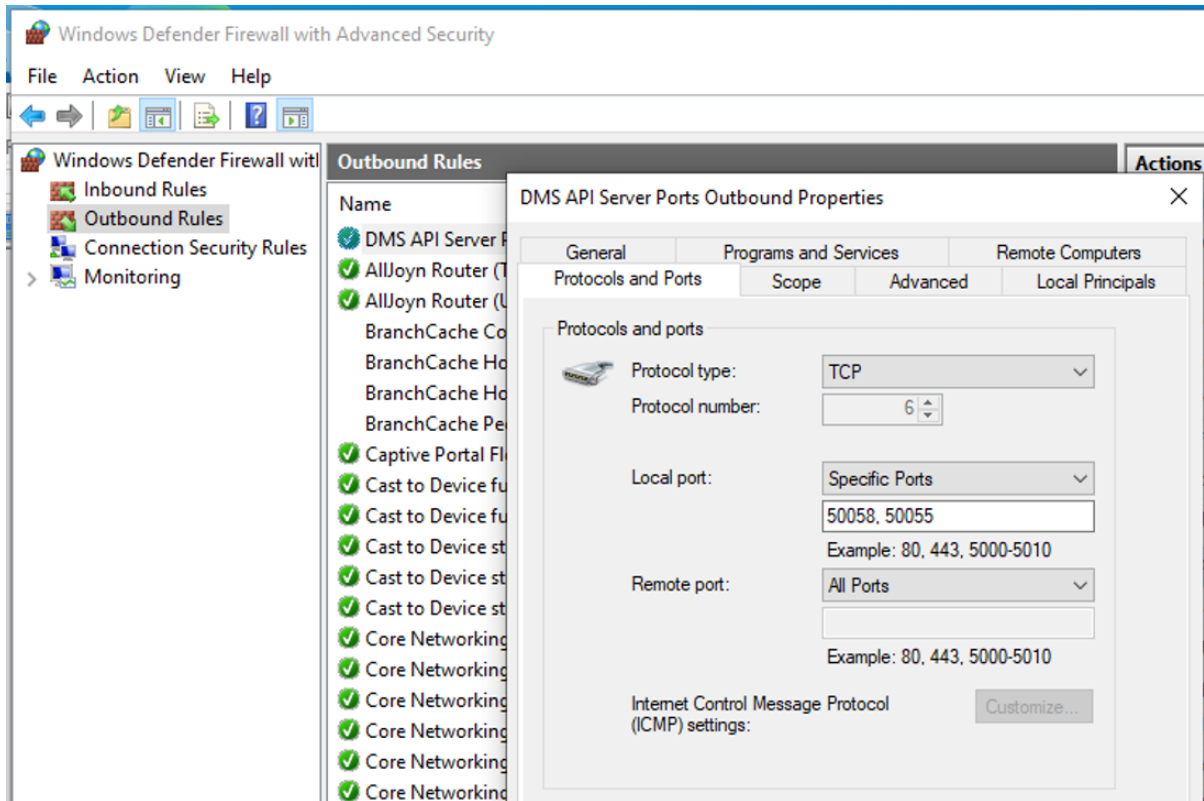
Eg:

```
{
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft": "Warning",
      "Microsoft.Hosting.Lifetime": "Information"
    }
  },
  "AllowedHosts": "*"
}
```

```
"Kestrel": {
  "Endpoints": {
    "Http": {
      "Url": "http://*:50058"
    }
  }
}
```

Step 5: Update Firewall Settings

- Add the DMS API Server port to the outbound properties in the firewall.



Step 6: Restart Services

- Restart the server and ADMS services.
- Start the services from the Task Manager.

10.4. Configure Quick Close settings

Quick Close requires configuration on both the server and client.

For details, see the Configuration Files section in the *ADMS Installation Guide*.